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Global Financials

Digital Banks: From Niche to Mainstream

Mobile banking will become the new norm as technology evolves and product offerings expand. We find that policy paths have been key for success but even under tough conditions, leading digital banks have shown their mettle. The next step is to go mainstream.

Why look at digital banks at this juncture? In a span of three years, from having no digital banks under our coverage we now have seven amongst an expanding global industry.

Similar rules apply for digital banks; policy paths matter more: We find that how supportive policy direction is for digital banks will be more of a key factor than rules, as most countries have similar rules for digital banks and incumbents. We find Latam/Brazil, ASEAN and South Africa have the most supportive policies. In the case of the US regulatory barriers exist for digital banks, while for Europe and Korea we expect increased scrutiny on digital banks.

Retail banking is where the competition is most fierce and digital banks remain secondary/tertiary banks in most markets while **some banks are moving into SME businesses**. Digital banks are expanding their product breadth to boost the revenue base, which should be key to reaching growth and ROE targets.

Incumbents holding their own in DM regions while new players taking share in Asia/EM: For DM regions including the US, Europe and Japan, in some cases we are seeing incumbent banks trying to maintain their strong positions by advancing their mobile capabilities to be on par or better than digital banks. We find that Europe is in need of consolidation of digital banks.

The story is somewhat different for EM digital banks as the possibility of digital banks going from niche players to the mainstream is much higher in markets such as Latam/Brazil, ASEAN, and South Africa, considering their higher under-banked customer base and a younger demographic. Nubank is gaining market share not only in Brazil but also in Mexico and Colombia, which also have large populations and high margins. In China, tightened regulation has seen digital banking functions returning to the traditional banks, with BigTech players more focused on payments and some retail lending.

For Korea and Japan, the story is about how the leading digital banks, including KKB, Rakuten Bank and SBI Sumishin Net Bank, will utilize their ecosystem effectively and continuing to push innovation to take market share and monetize.

Leading digital banks should provide positive long-term returns and we believe valuation premiums are deserved for leading digital banks over traditional banks assuming growth dynamics are stronger, cost efficiencies are attained, technology is

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better utilized, and they evolve into mobile financial platforms.

Key risks are policy incoherence, macro downturn and increased competition.

We are **Overweight Bank Jago (ARTO.JK)**, **KKB (323410.KS)**, **Nu Holdings (NU.N)** and **Equal-weight on Rakuten Bank (5838.T)**, **SBI Sumishin Net Bank (7163.T)** and **SoFi (SOFI.O)** while **Inter & Co (INTR.O)** is **Underweight**.

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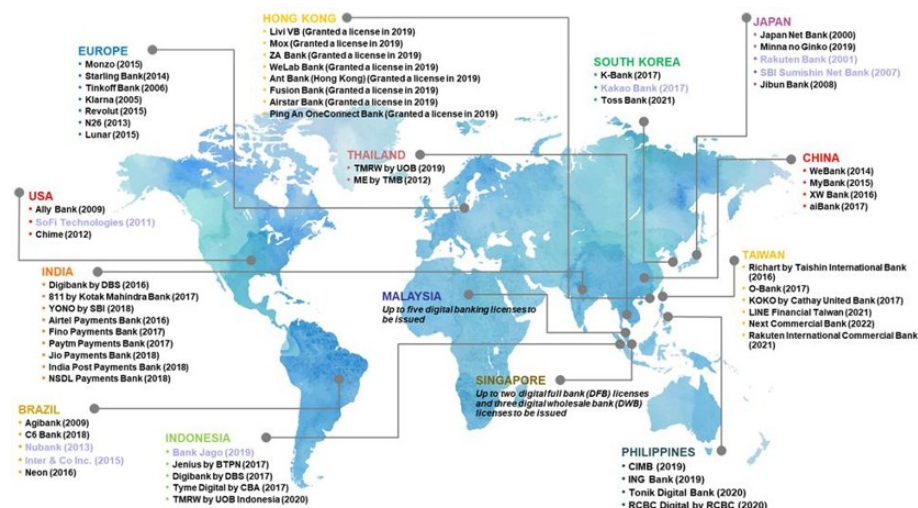
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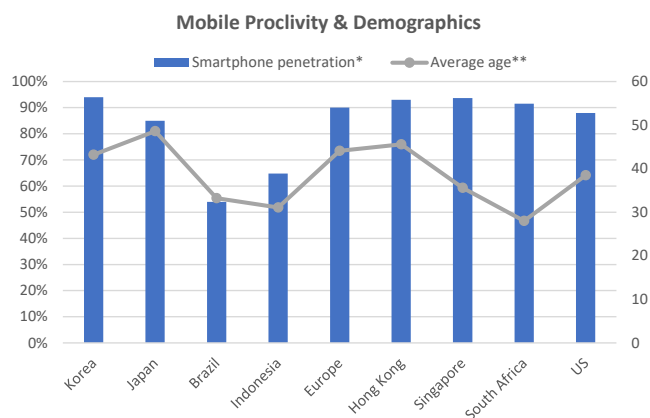
Just Pictures

Exhibit 1: Numerous digital banks have been launched globally but there are a limited number of listed ones



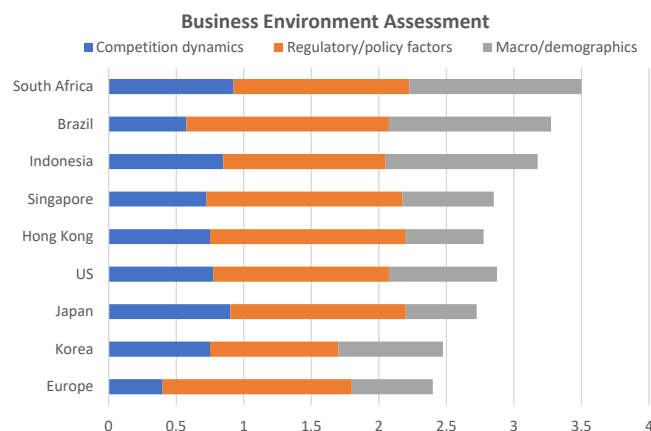
Source: Company Data, Morgan Stanley Research. Purple coloured banks are covered by Morgan Stanley.

Exhibit 2: Higher smartphone penetration and a younger demographic are favorable for digital banks and fintech players



Source: Statista, WorldData.info

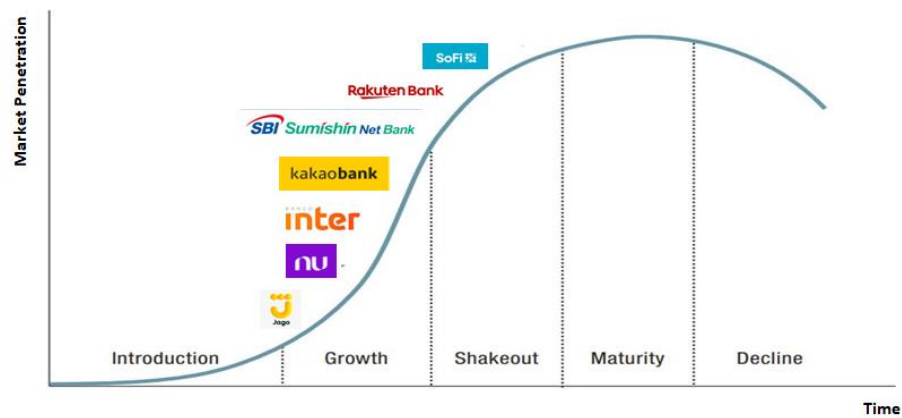
Exhibit 3: Our assessment of macro, policy and competition dynamics shows EM markets more favorable than DM



Source: Morgan Stanley Research

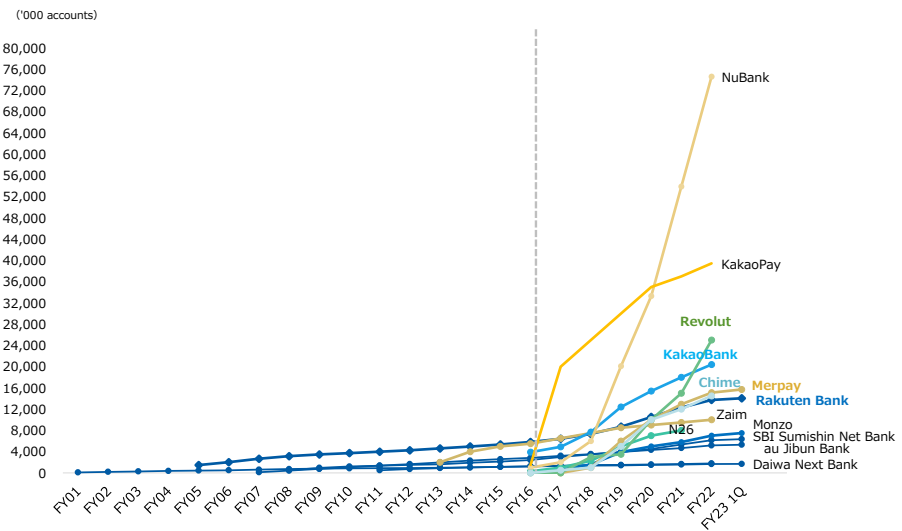
Exhibit 4:

Digital banks under Morgan Stanley coverage: most digital banks are in the growth stage



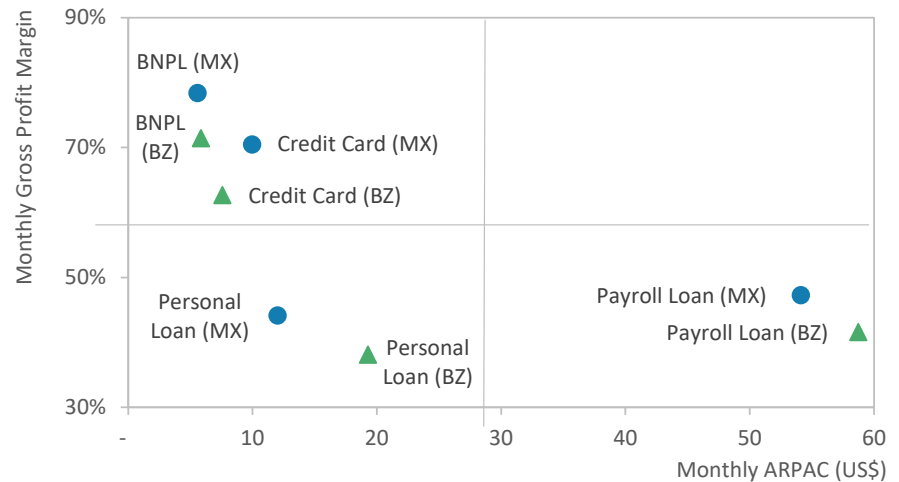
Source: Morgan Stanley Research

Exhibit 5: Customer acquisition is key to growth and digital banks that are innovative and have strong ecosystems show quicker customer base growth



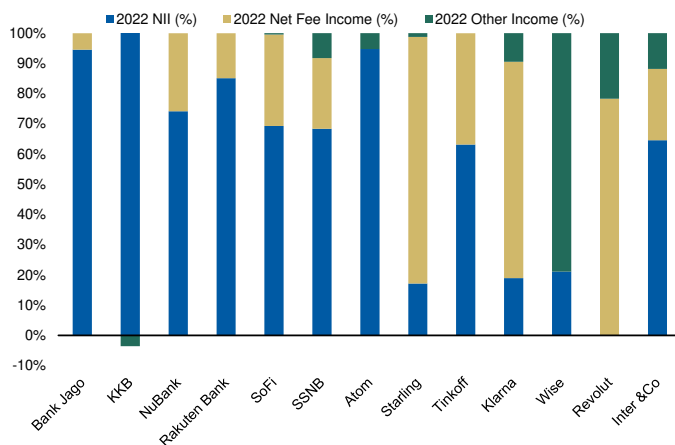
Source: Company Reports, Morgan Stanley Research

Exhibit 6: Overseas expansion is another avenue for growth but available only for a select few digital banks: Nubank operates in Mexico and Mexico's credit offering offers greater profitability vs. Brazil



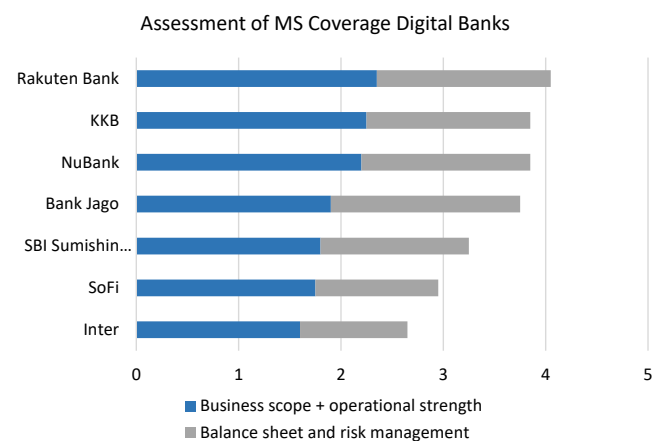
Source: Morgan Stanley Research estimates (e). BNPL = buy now, pay later; ARPAC = average revenue per active customer

Exhibit 7: Many digital banks are dependent on their credit business but we are seeing increased expansion of product breadth for WM and other peripheral services



Source: Company reports, Morgan Stanley Research

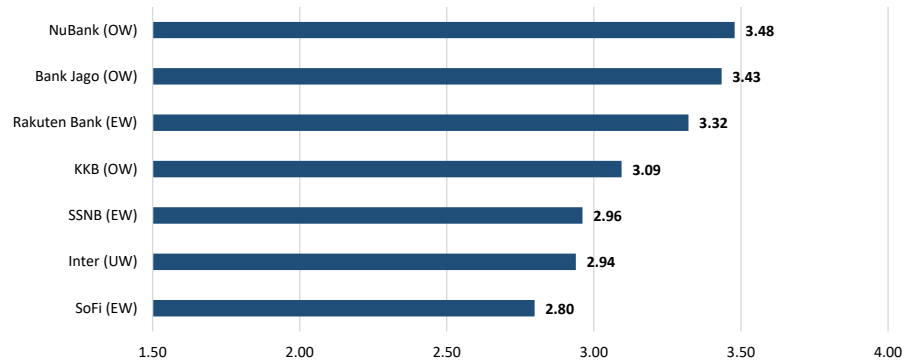
Exhibit 8: Our assessment of MS coverage digital banks shows that rather than business conditions, innovative capacity and an effective ecosystem matter more



Source: Morgan Stanley Research

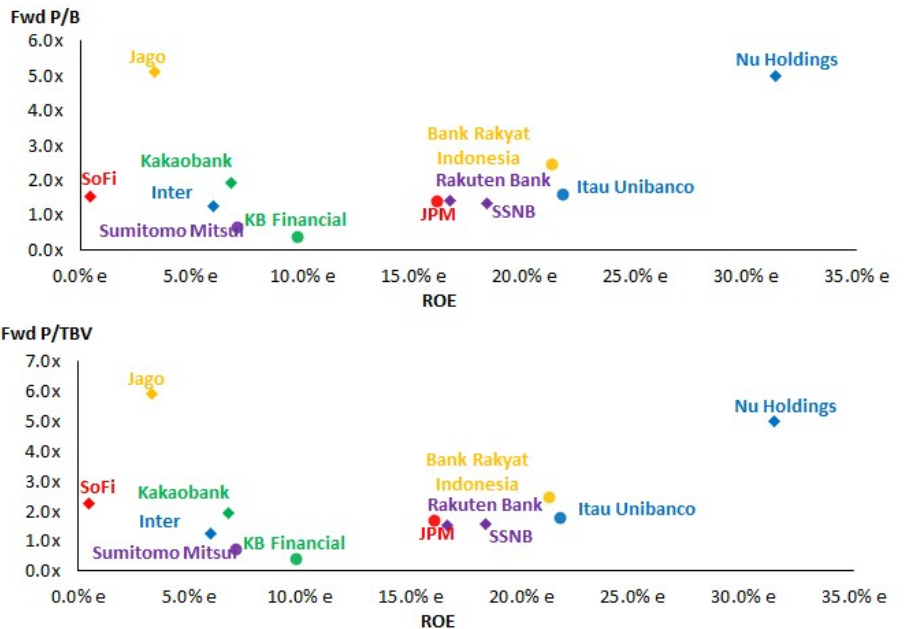
Exhibit 9:

Our overall assessment of the digital banks under MS coverage shows that Nubank and Bank Jago look to be the best positioned to maintain their leading digital bank positions in their respective markets and grow from being niche players to mainstream players



Source: Morgan Stanley Research

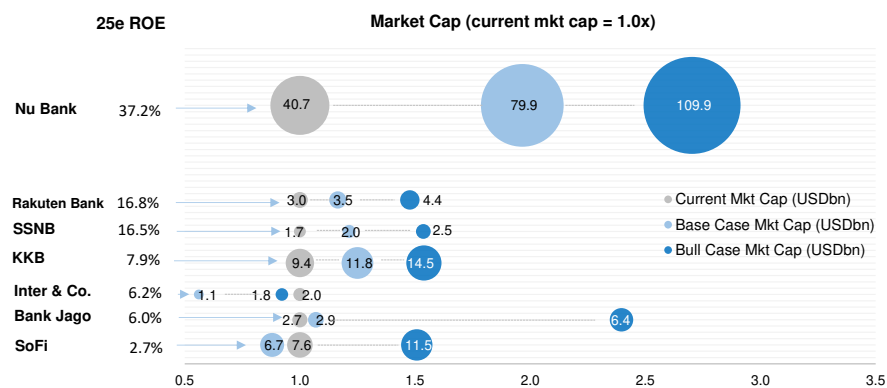
Exhibit 10: Digital banks vs. traditional banks: Fwd P/B, Fwd P/TBV vs. ROE – valuation gap between leading digital banks and traditional banks is deserved, assuming sustained strong long-term prospects



Note: Series with Diamond markers represents digital banks and Circle markers represents traditional banks. Also banks are color coded by region.

Source: Morgan Stanley Research

Exhibit 11: Leading digital banks showing considerable upside for market cap expansion using our base and bull cases



Note: Current market caps as of Dec 8, 2023. Base case and bull case market caps are calculated using our analysts' target prices and bull case valuations. Source: Morgan Stanley Research

Investment Summary: Going from Niche to Mainstream

Why are we writing on digital banks now?

In the past few years, digital banking has proliferated at a fast pace, fueled by growing usage of smartphones, more accessible mobile banking apps and increasing number of financial products available through mobile platforms. Digital banks have been central to the mobile banking revolution and in the years to come, penetration by digital banks in the financial services realm will only increase.

We have seen a number of digital banks grow rapidly, and meaningfully alter the competition dynamics of their respective markets including banks such as KakaoBank, Nubank, Rakuten Bank and SBI Sumishin Net Bank. More importantly, these digital banks have become publicly traded stocks and there seems to be more digital banks waiting in the wings, which will increase the pool of digital banks available for investment.

In past reports, we have focused on financial digitization and digital banking more holistically with there being no listed digital banks, but today Morgan Stanley Research currently covers seven digital banks: Bank Jago, Inter & Co, Kakao Bank, Nu Holdings, Rakuten Bank, SoFi and SBI Sumishin Net Bank.

[Europe: Neobanks: how does the path to profitability look? \(20 Dec 2022\)](#)

[Europe: Neobanks: Exploring disruptive potential \(31 May 2022\)](#)

[Latam Financials: The Rise of Digital Banks \(3 Jan 2022\)](#)

[Asia Financials: Asia Banking 2025: Digitalisation to Redefine the New Normal \(13 Oct 2020\)](#)

[Asia Primer: Digital/Mobile Push; Incumbents vs. BigTech \(12 Dec 2019\)](#)

[Banks & Software / IT: Digitisation in Banking: on the cusp of operational revolution? \(13 Mar 2018\)](#)

[Global Banks & Diversified Financials: Banking at the Speed of Light \(8 Jan 2018\)](#)

Our definition of a digital bank

A bank established with a separate digital banking license or an existing banking license that offers financial products/services using a digital platform or electronic channels with no or limited utilization of physical branches or offline channels. These banks tend to be cloud-native and have no mainframe-based infrastructure but not all jurisdictions allow cloud-native core banking systems (i.e. Korea).

Where do digital banks stand?

We are seeing digital banks in very different phases of operations and levels of penetration with there being no set formula for success. In many countries, the digital banking agenda was formulated by the government in order to set up a new competitive

landscape and improve customer convenience, while some were more private sector driven.

The level of embracement by customers of digital banks is dependent on smartphone penetration, banking penetration level, and accessibility but also the level of satisfaction for incumbent players. The markets where we are seeing stronger deposit and loan market share grab by digital banks are Korea, Japan and Brazil, while Europe looks to be in a consolidation mode. In the US, focus on digital banking tends to be more of an extension of existing banking offerings via the online/mobile channel, with Fintech players generally more niche or focused on payments.

Globally, we believe there are largely four types of digital banks:

1. BigTech-linked multi-faceted platforms (e.g., KakaoBank, Rakuten Bank);
2. Fintech start-up platforms generally targeting niche segments (e.g., Nubank, Revolut, Chime, SoFi);
3. Those established by a financial company as a platform for channel diversification (e.g., Ally Bank, SBI Sumishin Net Bank, Inter & Co); and
4. Consumer-related company-linked platforms (e.g., Sony Bank, KBank).

Exhibit 12: MS Digital Banks Coverage: Comps Table

Company	Rating	Currency	TP	Bull/Basis/Bear			Current Price	Mktcap (USD bn)	P/B		P/E		ROE		EPS Growth	
				Bull	Basis	Bear			2023E	2024E	2023E	2024E	2023e	2024e	2023e	2024e
Bank Jago	OW	IDR	3,200	7,189	3,972	1,231	5,100	2.9	5.2	5.1	45.1	158.9	1.2%	3.3%	50.5%	108.5%
KakaoBank	OW	KRW	32,500	40,000	32,500	17,500	26,000	9.5	2.1	1.9	36.2	30.2	6.0%	6.8%	29.9%	19.6%
Nu Holdings	OW	USD	16	22	16	5	8.14	38.8	6.5	5.0	33.0	17.5	21.0%	31.4%	na	88.4%
Rakuten Bank	EW	JPY	2,830	3,590	2,830	1,820	2,430	3.0	1.7	1.4	11.9	9.8	16.7%	16.6%	20.9%	21.7%
SoFi	EW	USD	7	12.0	7.0	2.5	7.96	7.7	1.5	1.5	na	331.8	-2.4%	0.5%	na	na
SBI Sumishin Net Bank	EW	JPY	1,920	2,430	1,920	1,180	1,582	1.7	1.6	1.3	10.3	8.6	17.5%	18.4%	14.3%	19.0%
Inter & Co Inc.	UW	USD	2.8	4.6	2.8	1.2	5.0	2.0	1.3	1.3	35.2	22.4	4.0%	6.0%	na	57.2%

Note: Data as of Dec 8, 2023. Source: Company Data, Factset, Refinitiv, Morgan Stanley Research Estimates (E)

What matters for digital banks in future

Globally, we see digital banks having a diffusion effect on their respective markets rather than disruption considering incumbents will continue to play a role in mobile banking. Meanwhile financial authorities are focussed more on digital banks from a regulatory standpoint as they grow in size.

Macro factors and mobile banking characteristics point to a structural trajectory for the proliferation of digital banking but... Countries with a high level of smartphone penetration, where customers are tech-savvy and have a lower level of satisfaction for traditional bank offerings, have a higher tendency for digital banks to penetrate the market quickly. Hence, we find that digital banking providers with better UI/UX platforms and connectivity to existing mobile BigTech platforms have been able to grow faster and monetize sooner.

...Government policy direction towards digital banks now more neutral compared to early development cycle for digital banks as managing risks gains importance + calls for a fairer playing field by incumbents: We are seeing different levels of policy support between countries with the most important policy factors being: (1) regulatory arbitrage vs. incumbents; (2) level of regulatory supervision; and (3) competition-related policy. Generally, markets that need to increase financial services penetration or want to change competition dynamics have seen digital banks flourishing as policy support provides a foundation for growth.

Incumbent banks are fighting back harder with improved mobile banking capabilities

and more personalized services being offered to customers: Retail customers are becoming more familiar with digital banks and are finding them to have lower transaction costs, easier accessibility and more attractive conditions for core products/services. However, the breadth of financial services offered by digital banks is much narrower than traditional banks and an omni-channel platform provides more flexibility for customers especially ones that are more affluent.

Digital banks not the primary bank for customers but open banking and innovative service platform swaying customers to increase digital bank utilization: Digital banks are still largely used by customers as a secondary bank rather than their primary bank but the utilization of secondary accounts looks to be rising.

We expect leading digital banks to be able to monetize via cost efficiencies – but more importantly, expansion of product breadth and deepening relationships with customers look to be key: Most digital banks are still in the growth phase, needing to reach critical mass in order for the lower operational cost factor to really kick in.

Not all digital banks are the same; Differentiation to be more pronounced between digital banks

Growing is not an easy task but realizing profitable growth is even more difficult especially for fintech and digital banks considering the high fixed costs. The story for digital banks will gradually transition away from growth to monetization and sustainability of operations. In many markets, we are seeing the widening gap between the stronger digital banks and the weaker ones with this trend expected to be even more discernible in the coming years. So what will differentiate the winners from the laggards?: We approach this question taking into account the '3 Ms', which we define as:

1. **Mechanics:** Capital strength combined with risk management ability and funding franchise;
2. **Motor:** Innovative capabilities combined with a strong ecosystem; and
3. **Mettle:** Strategic flexibility and operational efficacy.

Capital strength and committed stakeholders go hand-in-hand and this will be a key factor in the near-term for digital banks that are endeavoring to reach critical mass:

Digital banks generally have high capital adequacy ratios especially the listed digital banks as they were able to raise sizable capital to position themselves for growth. We expect to see disparate levels of capital strength going forward with key differentiating factors being stakeholder commitment.

Normalization of rates at higher levels, and increased competition from incumbents and new players, will require digital banks to possess core operational strength namely a funding franchise, which will not be the case for all digital banks:

When interest rates are low funding franchise difference is not so discernible but when rates are higher, funding cost disparity between the strong funding franchises and the weaker ones widens. KKB and Rakuten Bank have funding costs that are lower than their incumbents and this should enable them to be more competitive in the lending business.

- KKB developed a deposit product called a group deposit that allows a group of

customers to use a single account to manage joint funds efficiently and the deposit rate at 0.1% is very low. Other banks have tried to emulate this product but none of them have been able to do this. This product, together with other innovative deposit products, enables KKB to have the lowest funding cost among all Korean banks.

Breadth of financial services and innovative products will be key to monetization as digital banks will need to establish a business model that consumes less capital in the long-term:

A low loan-to-deposit ratio (LDR) at digital banks implies that incremental growth should be positive for NIM but the general trend for digital banks is a move to safer loans such as mortgages, which points to limited NIM expansion. In order to boost sustainable ROE, digital banks will need to expand their breadth of financial services into areas such as WM (Wealth Management)/PFM (Personal Financial Management) that consume less capital.

- Innovation is certainly a key factor but is also an effective business strategy such as the one used by Nubank, which is a no-frills cost-efficient business model, and has seen success.

Digital banks with effective captive ecosystems like Rakuten Bank are advantaged and ones that do not have such a setup will need to strengthen collaborative partnerships:

Digital banks have to be more than lenders and sellers of financial products in order to differentiate themselves from each other and incumbent players. We are seeing digital banks with effective ecosystems like Rakuten Bank and KakaoBank able to expand their customer base more efficiently and be more innovative owing to their ecosystems.

- Leading digital banks target synergies with powerful ecosystems. For example, Rakuten Bank (a consolidated subsidiary of Rakuten Group) is successfully reaping synergies within the Rakuten ecosystem drawing on the group's powerful customer base of approximately 100mn ID holders, while SBI Sumishin Net Bank is in a position to unleash synergies with the SBI Holdings group and its subsidiaries.
- Bank Jago has strategic partnerships with one of the largest digital ecosystems in Indonesia – GoTo – and through this linkage the bank has been able to build up its customer base and increase revenue.

Most digital banks have not experienced a full through-cycle for credit risk and with strong growth comes growing pains, which will impart more importance on risk management capability:

There are growing concerns from investors on credit risk for digital banks due to high interest rates and also the possibility of adverse selection of riskier customers as digital banks have grown assets at a breakneck speed.

- Nubank presents a valuable case study of one digital bank that has successfully provided strong growth while maintaining sound asset quality throughout cycles — including periods of sharp GDP contraction in Brazil (2015, 2016, and 2020) and most recently a sharp hiking cycle which took rates in that country from 2.0% in March 2021 to 13.75% in August 2022.

Most digital banks will be focusing mainly on the domestic business but a select few with strategic flexibility will be able to expand into overseas markets:

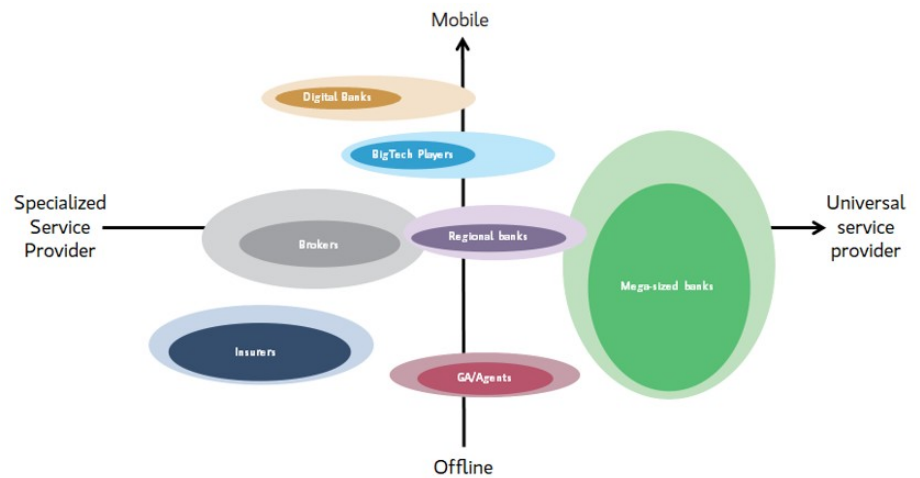
Not a strategy

that is open to all digital banks but this is currently being successfully carried out by Nubank with the expansion into Mexico and Colombia.

- KKB is also planning on entering overseas markets through joint ventures including into Thailand and Indonesia.
- Revolut has tried to be global from the very beginning, and is currently present in 36 countries. N26 is present across the Eurozone, and tried to expand to the US, although quickly realized that it didn't have enough resources and decided to exit that market.

The mobile financials business will be populated by many different types of business entities including digital banks, commercial banks, BigTech platforms, brokerage firms and fintech players.

Exhibit 13: Financial industry – competitive landscape



Source: Morgan Stanley Research

Our Assessment of Key Markets and Players

Our assessment of digital banking markets

As a way of assessing the business environment of digital banks in each of the countries/regions, we have designed a framework to evaluate what we believe are key factors that matter for the success of digital banks in respect to macro, regulatory/policy and competition dynamics. In the below table, we provide an explanation of what factors we evaluated and why we selected them.

Exhibit 14: Business Environment Assessment Framework

Category	Key Factors	Weighting	Description
Macro/ demographics	Financial penetration (High: 1 vs. Low: 5)	10.0%	We assess the level of financial services penetration for the population with lower financial penetration generally being more favorable for new players
	Tech savviness + smartphone penetration (High: 5 vs. Low: 1)	7.5%	High smartphone penetration and tech savviness generally means that level of mobile app accessibility is high and proclivity to use mobile-based services
	Demographics (Low avg. age: 5 vs. high avg. age: 1)	7.5%	We find that the younger generation are more inclined to use digital banks with higher stickiness for the younger customers
Regulatory	Regulatory advantages/leeway (High: 5 vs. Low: 1)	5%	If it is ruled that digital banks have more lenient regulatory requirements compared to the traditional banks (i.e. lower CET1 minimum requirement)
	Ownership/capital rules (Less restrictive: 5 vs. Very restrictive: 1)	10%	Ownership restrictions for non-financial entities or other entities when taking stake holdings in digital banks and/or high initial capital requirements
	Level of regulatory intervention (High: 1 vs. Low: 5)	12.5%	Do the financial authorities try to control the banks directly or indirectly in terms of operations or other aspects
	Overall government policy direction (Positive: 5 vs. Negative: 1)	15%	The level of supportiveness for the digital bank industry and on the core businesses that digital banks are involved in
Competition dynamics	Number of competing digital banks (Low: 5 vs. High: 1)	10%	The level of competition within the digital bank industry based on number of digital banks and skew of market share to the top player(s)
	Potential for new entrants (Low: 5 vs. High: 1)	7.5%	Potential for the government giving out new licenses and/or new digital banks emerging through existing players in the market (i.e. traditional banks establishing a digital bank affiliate)
	Mobile banking strength of incumbent players (Low: 5 vs. High: 1)	15%	How well incumbent banks are positioned in mobile banking services and ability to prevent customer attrition to new players

Source: Morgan Stanley Research

The results of our assessment find that the markets that show relatively more favorable business conditions for digital banks are South Africa, Brazil, and Indonesia – mainly due to favorable macro conditions and policy backdrop for digital banks that are quite favorable. On the other hand, Korea and Europe look to be less favorable markets due to high level of financial penetration in both markets. For Korea the more pressing issue is the unfavorable regulatory/policy backdrop while for Europe it is more about competition dynamics.

- In **Brazil**, we think certain digital challengers will continue to succeed and compete against incumbents for market share. Cross-selling represents a notable opportunity for these challengers, as many have amassed large client bases and are now looking to ramp up new products, like personal loans, payroll loans, investments, and more.
- **Indonesia** is one of the largest unbanked/underbanked markets in Asia, with retail loan penetration of 10% vs. ASEAN peers of 4-6%, making it an attractive market for digital banks. Similarly, SME is also an underbanked segment and we expect SME lending market to grow at least three times. We see a strong push from governments towards financial inclusion, including implementation of financial infrastructure, such as standardised QR codes and BI Fast, creating a supportive environment for digital banks to grow. There is room for a few digital banks of significant scale to serve the expanding market, along with incumbent banks.
- In **Japan**, digital banks and other "neobanks" have been expanding since around 2000 as operational guidelines for "new forms of banking" were established along with revisions to the Banks Act. Essentially, under Japanese law and regulations, there is no special framework for digital banks, and the names are treated under

local business practices. Still in recent years, the banking industry, especially in the retail area, has seen a rapid increase in the accounts and deposit balances at new specialized online banks. By 2030, we think digital banks should reach 35-40mn accounts and their share of total deposit value should expand from about 2-3% at this point to 6-7% in our base case and about 13-14% in our bull case. We think a handful of digital banks will be driving share growth.

- **Korea** has a very well financially serviced population with over-banked being the case rather than underbanked. However, the level of satisfaction of incumbent banks is not that high and mobile banking is becoming the preferred servicing channel. Policy direction was initially supportive of digital banks but in recent years the policy backdrop is not favorable with retail lending being tightened by regulators. Incumbent banks have become more active in upgrading digital channels to prevent attrition but customers are now actively using digital banks as secondary banks especially due to competitive pricing.
- In **Europe** we expect consolidation to happen within the Digital banks space – we think it's not sustainable to have more than 10 digital banks operating in each market when Europe is already overbanked to start with, but we do expect some leaders to emerge, in particular those able to get to scale and profitability faster. Meanwhile, we think digital banks will continue pushing incumbents to evolve and innovate.
- Both **Hong Kong** and **Singapore** have set up a limited number of digital banks, with specific digital banking licenses, with the aim of filling underserved parts of the market, and driving product innovation. Thus far, the digital banks have remained small players, despite supportive policy and high digital penetration and awareness in both markets. The main issue the digital banks face is that both HK and Singapore already have well developed and highly penetrated banking markets and the incumbent banks are large regional, or even global banks, with both deep pockets and the propensity to invest in their own digital banking capabilities. We therefore see the outlook for digital banks in both markets as uncertain. Based on current data, we believe ZA Bank in HK, plus Standard Chartered's sponsored banks Mox (HK) and Trust (SG), are most likely to break out and become mainstream banks.
- In **South Africa**, limited scale and narrow consumer-focused product suites without meaningful physical points of presence, leave digital banks challenged for profitability, despite encouraging deposit and client growth in a concentrated market (six banks hold 92% of assets). In time, we see digital banks broadening their product offerings (into SME, merchant acquiring, etc) to improve relevance, client primacy and data gathering capabilities to unlock more developed credit origination strategies needed to improve profitability. Meanwhile, we see digital banks continuing exerting competitive pressure on banking fees and innovation levels, although as a cohort, the digital banks are not yet leaders in digital user-experience
- In the **US**, we expect incumbents to continue to dominate market share driven by significant regulatory barriers and scale. Non-bank winners in the US digital banking space tend to be niche, focused on payments, or operating in partnership with incumbent banks.
- The **Indian** Government has initiated measures to boost financial inclusion (Jan Dhan – no frills account), built a universal biometric identification system

(Aadhaar), and implemented real-time payment systems (Unified Payments Interface – UPI and Bharat QR) with further efforts being made to further accelerate growth in digital lending and/or other financial services. That said, India is yet to come up with a full stack digital banking regime. NITI Aayog (a public policy think tank of the Government) did propose licensing and regulatory regime for digital banks in India but we are yet to see any progress on this front. The report ([link here](#)) highlights that full stack digital banks are the potential solution for the persistent policy challenge of credit deepening and this is the next stage of financial inclusion.

- In **China**, both digital payment and online digital banking have experienced rapid growth in the past 10 years. The rapid innovation by some of the payment firms, such as Tenpay and Alipay, have revolutionized digital payment in China and push banks to further digitalize not only payment but also retail client service, credit approval and risk management as well as wealth management services. In addition, with the policy focus shift to industrial sectors, the government and financial system further digitalized SME and trade finance related financial services. However, the government has also tightened regulations to ensure all financial activities are properly regulated under the same standards. This has pushed the key digital bank functions back to the traditional banks with tech firms and payment firms choosing to focus more on the capital light payment business (more similar to the function of Visa and Master) and loan facilitation business rather than the full bank functions.

Exhibit 15: Business environment assessment by country/region in the perspective of digital banks

	Weighting	Korea	Japan	Brazil	Indonesia	Europe	Hong Kong	Saudi Arabia	Singapore	South Africa	US
Macro/demographics	Financial penetration (High: 1 vs. Low: 5)	10%	2	2	4	4	2	1	3	1	4
	Tech savviness + smartphone penetration (High: 5 vs. Low: 1)	7.5%	5	3	4	3	4	5	4	5	4
	Demographics (Low avg. age: 5 vs. high avg. age: 1)	10.0%	2	1	5	5	1	1	5	2	5
	Weighted average sub-total	28%	0.78	0.53	1.20	1.13	0.89	0.58	1.10	0.88	1.28
Regulatory	Regulatory advantages/leeway (High: 5 vs. Low: 1)	5%	3	3	4	3	3	3	4	3	3
	Ownership/capital rules (Less restrictive: 5 vs. Very restrictive: 1)	10%	3	3	4	3	4	3	3	4	4
	Level of regulatory intervention (High: 1 vs. Low: 5)	10%	2	4	3	3	4	4	3	3	4
	Overall government policy direction (Positive: 5 vs. Negative: 1)	15%	2	3	4	3	3	4	4	4	3
	Weighted average sub-total	40.0%	0.95	1.30	1.50	1.20	1.40	1.45	1.40	1.45	1.30
Competition dynamics	Number of competing digital banks (Low: 5 vs. High: 1)	10%	3	3	2	4	1	3	3	2	4
	Potential for new entrants (Low: 5 vs. High: 1)	7.5%	4	4	3	2	2	4	4	5	3
	Penetration level of incumbent players (Low: 5 vs. High: 1)	15%	1	2	1	2	1	1	3	1	2
	Weighted average sub-total	33%	0.75	0.90	0.58	0.85	0.40	0.75	1.05	0.73	0.93
Weighted average total		100%	2.48	2.73	3.28	3.18	2.40	2.78	3.55	2.85	3.50

Source: Morgan Stanley Research

Short takes on key factors

1) Regulatory framework for digital banks generally not so different to incumbent traditional banks, but the approaches to digital bank licensing differ

The objectives for giving out new digital bank licenses are quite similar across the globe with key objectives looking to be: (1) changing the competitive dynamics of the existing banking sector; (2) boosting innovation in the financial system; and (3) improving financial inclusion. Based on this fundamental backdrop, the financial authorities have adopted slightly different digital bank licensing approaches with there being countries that set up specific licensing and regulatory structures while other countries used the existing traditional banking licensing and regulatory framework.

- Key differences between specific regulatory approach to licensing and tradition licensing one is that there has been some leeway(s) provided to digital banks compared to traditional banks together with the regulatory underpinning of

supervising digital banks disparately to traditional banks.

- Singapore is an outlier on the standpoint of having two different types of licenses – one for a full digital bank license and one for a digital wholesale bank, with the capital requirements being disparate.
- For example in Europe some type of digital 'banks' are in fact operating under an e-money licence (like Revolut and Tide in the UK).

For markets using the traditional regulatory framework, the digital bank ownership rules are the same as for traditional banks. There are countries that have stricter restrictions on digital bank ownership namely Korea and Taiwan, where traditional ownership for non-financial entities is limited to 10%.

- In the case of Korea and Taiwan, there have been changes made to ownership rules with a new digital bank by-law allowing non-financial entities to hold up to 34% stake in digital banks (limited to 10% for banks) while in Taiwan non-financial entities with capabilities in financial technology, e-commerce or telecoms, allowed to own more than 10% stake in a digital bank.

Exhibit 16: Licensing and regulatory framework by country

Specific licensing and regulatory framework
Hong Kong, Korea, Malaysia, Philippines, Singapore, Taiwan
Based on existing licensing and regulatory framework
Australia, Brazil, China, EU, Indonesia* Japan, Saudi Arabia, UAE, UK, US,

Source: Morgan Stanley Research

2) Capital rules are key with capital adequacy rules not so different for digital banks and traditional banks except for DSIB (Domestic Systemically Important Banks)/GSIB (Global Systemically Important Banks)

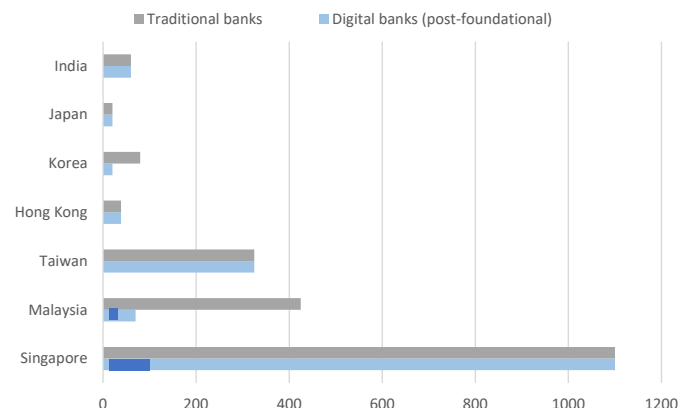
The opportunity for regulatory arbitrage for digital banks seems to be quite limited contrary to market views as existing banking rule frameworks or similar on-going business requirements are mostly being utilized to govern digital banks.

Capital initial requirements are mostly the same for digital banks and traditional banks across the countries/regions we looked at except for Korea and Malaysia. However, this advantage is not so important considering that once a digital bank is firmly on a growth trajectory more capital should be needed in order to meeting capital adequacy requirements.

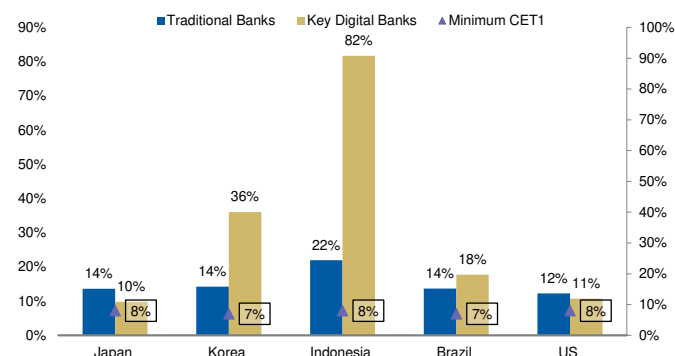
- The capital requirements for some countries have a phasing in approach, which provides both the digital bank owner(s) and the regulators time to determine the viability of the endeavor.
- In most cases, digital banks have been able to raise capital to grow but there have been cases, namely KBank a Korean digital bank, had difficulty raising capital due to ownership restrictions – non-financial entities were limited to 10% ownership before rule changes were made in 2020.

Capital adequacy and liquidity requirements matter more on an on-going business standpoint. In most countries, the capital adequacy and liquidity rules are the same for

traditional banks and digital banks

Exhibit 17: Minimum paid-in-capital requirements by key countries

Source: Morgan Stanley Research

Exhibit 18: Capital adequacy minimum requirement by country

Source: Morgan Stanley Research, BIS

3) Overall government policy stance towards digital banks and core businesses gaining importance especially for digital banks that are reasonably large

When we compare the various countries that currently have an active digital banks industry, we find that government policy had a strong influence in pushing the digital banking initiative. Early on, most governments had been supportive in the digital bank proliferation journey but along the way there have been policy path changes:

- The launch of new digital banks was quite active from 2000-2015 but this trend has slowed down since mid-2010s due to various reasons including difficulty in monetizing the business, less regulatory arbitrage opportunities and growing risks related to digital security.

One important point is whether there tends to be lighter or heavier regulatory intervention in the banking industry in general. Another one is how the financial authorities view digital banks in the context of the competitive landscape, meaning whether digital banks are seen more as needed fintech disruptors or just banks with digital platforms.

- Compared to emerging market countries, developed markets tend to have less policy intervention and the regulatory guidelines are more transparent. For most of the markets our analysts had assessed, the policy environment is neutral or favorable.
- For example, in the UK where government policy has promoted the entry of newcomers to financial business to address a monopoly of large financial institutions in the financial industry; or Brazil where digital banks are growing rapidly amid efforts to promote financial inclusion; and
- Still others along the lines of Japan, Korea and South Africa amid moves to design financial businesses that deliver, speedily and at low cost, personalized services not available at conventional banks.
- However, the Korean digital bank industry at this juncture is experiencing a

somewhat difficult policy environment as household debt control is being endeavored while digital banks have to meet strict mid-prime lending targets.

4) Competition, co-existence and collaboration may all go hand-in-hand between digital banks and incumbent players. Servicing of customers at different life cycles and also with disparate cost and convenience preferences to matter for digital banks.

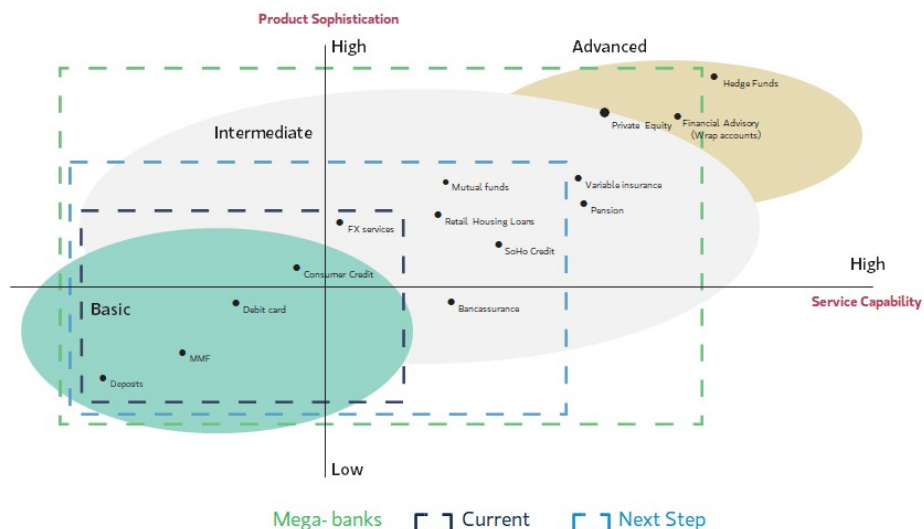
The business domain of existing financial institutions is their existing commercial banking businesses, but this is not to say that they are merely sitting by and watching the rise of digital banks, and instead they too are pursuing digitization, whether through working to improve financial services by digital means while reining in resources allocated to real branches, building their own new digital bank organizations, or tying up with fintech companies, for example.

Digital banks seem likely to continue increasing share, but the probability of them dealing a serious blow to the share of conventional banks in a short space of time seems low, and the more likely scenario may be that the two sides work together while also competing with each other.

- Many users of digital banks may have both conventional bank accounts and digital bank accounts, and the likelihood seems that the share of conventional banks will remain larger. Nevertheless, in Japan and Brazil for example, digital banks are gaining ground with loans and unsecured loans, and settlement solutions, particularly in the retail arena.
- A feature in countries where digital banks are scoring success is that they are managing to appeal to customers centered on younger age groups, have effective eco systems, and are able to offer key operational advantages (i.e. lower funding costs), aspects in which a handful of players are succeeding.

5) Breadth of financial services offered by digital banks is narrower compared to incumbents but digital banks tend to perform better in areas of business focus namely retail lending – moving to SME lending

Most digital banks are catering to retail customers providing mostly basic services including time deposits, unsecured retail loans, payment services and with some intermediate services such as FX services. This still does not compare to the breadth of services provided by the traditional banks but we believe what is more important is how effectively digital banks penetrate their core businesses.

Exhibit 19: Retail product mapping by sophistication and service capacity

Source: Morgan Stanley Research

Most digital banks seem to focus on their retail businesses and are taking discernible market share from the incumbent players. On the lending side, digital banks tend to be focused more on unsecured retail credit products, and digital banks that are in more mature markets or need to expand into other products are generally moving to mortgage lending.

- KKB has been particularly aggressive in building up its mortgage book since launching the product in 4Q22 and it has already reached W5tn mortgage balance as of 3Q23.
- Inter in Brazil has also been active in the mortgage market – in part because of its origins as a mortgage origination arm of a real estate development conglomerate.

There are a few digital banks now moving into the corporate lending space with focus on SME and SoHo (small business) lending. We believe this will become another avenue for growth for digital banks and there are digital banks that are focused only on the SME segment.

Exhibit 20: Lending business scope for key countries

Country	Service Provider	Retail Lending		
		Retail unsecured	Mortgage/Home equity	Mid/sub-prime
Brazil	NuBank	●	◐	○
Brazil	Inter	●	●	○
Indonesia	Bank Jago	◐	○	○
Japan	Rakuten Bank	●	◐	◐
Japan	SBI Sumishin Net Bank	●	◐	○
Korea	Kakao Bank	●	●	◐
Korea	Kbank	●	◐	◐
Korea	Toss Bank	●	○	◐
US	SoFi	●	●	◐

Source: Morgan Stanley Research

Our assessment of digital banks under Morgan Stanley coverage from a medium-/long-term perspective

Within our global coverage, we currently cover seven digital banks and we mostly have positive medium-/long-term outlooks for these digital banks. We think the fact that they were able to get listed puts these banks on another level to their peers and their stories are appealing.

Together with our assessment of the business environment for each of the respective markets, we also performed an evaluation of the digital banks under MS coverage on how they look to be positioned on a mid/long-term perspective. In the table below, we provide an explanation of the key competencies that we believe are important for mid/long-term profitable growth.

Exhibit 21: Key Competencies for digital banks

Category	Key Factor	Weighting	Description
Business Dynamics for Individual Banks			
Business scope + strength	Funding franchise vs. all competitors (Much stronger: 5 vs. Much weaker: 1)	15%	Funding franchise being a core competency for any bank, we evaluated how funding costs compare with peers and traditional banks
	Loan growth vs. all competitors (Much higher: 5 vs. Much lower: 1)	10%	Expected loan growth in the mid/long-term vs. traditional banks and also digital banking peers
	Product breadth/cross-selling vs. incumbents (Much higher: 5 vs. Much lower: 1)	10%	Breadth of product offerings vs. traditional banks with a broader set of products/services presenting more cross-selling opportunities
	Ecosystem + partnerships (Very strong: 5 vs. Very weak: 1)	15%	We find that an effective ecosystem especially one that is captive is a great advantage, but an ecosystem set up with strong partners could be equally as powerful. The comparison was made within the digital bank industry
	Overseas expansion (Overseas ops growing: 5 vs. No overseas exposure: 1)	10%	Most digital banks will be domestically oriented but a select few would have opportunities to expand into neighboring markets providing additional earnings growth
	LDR (Very low: 5 vs. Very high: 1)	10%	Conventional wisdom points to low loan-to-deposit ratio (LDR) which would enable higher loan growth while also having a positive impact on NIM
Balance sheet	Capital strength (Very strong: 5 vs. Very Weak: 1)	15%	Similar to LDR, stronger capital structure and the ability to raise capital would enable higher asset growth and more room to maneuver on acquisitive growth also
	Credit risk profile and capabilities (Higher risk: 5 vs. Lower risk: 1)	15%	One of the key foundations for sustainability of profitable growth is risks management and the capacity to maintain stable through-cycle credit costs is important

Source: Morgan Stanley Research

The digital banks that have lower business condition scores mainly tend to have higher core competency scores and we believe this is attributed to three factors: (1) funding franchise + cost efficiency; (2) effective ecosystems.

- Many of the global digital banks are spearheaded by Internet companies, online platforms, e-commerce players and fintech startups. The sources of competitive advantage for digital banks are: 1) collaboration within and outside the group; 2) UI/UX capabilities; and 3) data analysis capabilities. For digital banks, ties with online services within the group are a source of competitiveness, and for Rakuten Bank, the linkage of service usage within the Rakuten Group and loyalty program points are strengths. Kakao has been successful in growing its business via its eco system and is likely to be significantly larger and more profitable than its peers.
- The services provided by digital banks are fast, seamless and highly transparent, in ways that are not apparent at traditional banks. In particular, many digital banks have their own system development, operation, and maintenance systems, which facilitates high system cost competitiveness, and in many cases they can provide almost all services within a single app in a smartphone with a superior UI/UX. The ability to create in-house-developed services can drive growth, and unlike traditional banks, the ability to maintain and operate in-house without being dependent on vendors is a major strength.

Exhibit 22: Core Competency Assessment for MS-covered Digital Banks

Macro/Policy/Competition Dynamics		Weighting	KKB	Rakuten Bank	SSNB	NuBank	Inter & co.	Bank Jago	SoFi
Macro/demographics									
Population with high tech savviness (High: 5 vs. Low: 1)		10%	2	2	2	3	3	4	2
Smartphone penetration (High: 5 vs. Low: 1)		8%	5	3	3	4	4	3	4
Demographics (Low avg. age: 5 vs. high avg. age: 1)		10%	2	1	1	5	5	5	1
Weighted average sub-total		28%	0.78	0.53	0.53	1.10	1.10	1.13	0.60
Regulatory									
Regulatory advantages/leeways (High: 5 vs. Low: 1)		5%	3	3	3	4	4	3	1
Ownership and capital rules (Less restrictive: 5 vs. Very restrictive: 1)		10%	3	3	3	4	4	3	4
Level of regulatory intervention (High: 1 vs. Low: 5)		10%	2	4	4	3	3	3	4
Overall government policy direction (Positive: 5 vs. Negative: 1)		15%	2	3	3	4	4	3	3
Weighted average sub-total		40%	0.95	1.30	1.30	1.50	1.50	1.20	1.30
Competition dynamics									
Number of competing digital banks (Low: 5 vs. High: 1)		10%	3	3	3	2	2	4	4
Potential for new entrants (Low: 5 vs. High: 1)		8%	4	4	4	3	3	2	3
Penetration level of incumbent players (Low: 5 vs. High: 1)		15%	1	2	2	1	1	2	1
Weighted average sub-total		33%	0.75	0.9	0.9	0.575	0.575	0.85	0.775
Business Dynamics for Individual Banks		100%	2.48	2.73	2.73	3.18	3.18	3.18	2.68
Business scope + strength									
Funding franchise vs. all competitors (Much stronger: 5 vs. Much weaker: 1)		15%	5	4	3	3	3	3	2
Loan growth vs. all competitors (Much higher: 5 vs. Much lower: 1)		10%	3	4	4	5	3	4	4
Product breadth/cross-selling vs. incumbents (Much higher: 5 vs. Much lower: 1)		10%	3	4	3	3	2	2	4
Ecosystem + partnerships (Very strong: 5 vs. Very weak: 1)		15%	4	5	3	3	3	5	3
Overseas expansion (Overseas opp. growing: 5 vs. No overseas exposure: 1)		10%	3	2	2	5	2	1	2
Weighted average sub-total		2.25	2.35	1.8	2.2	1.6	1.9	1.75	
Balance sheet									
Loan-to-deposit ratio (Very low: 5 vs. Very high: 1)		10%	4	5	4	3	3	5	3
Capital generation capacity and strength (Very strong: 5 vs. Very Weak: 1)		15%	5	4	4	5	3	5	3
Credit risk profile and capabilities (Lower risk: 5 vs. Higher risk: 1)		15%	3	4	3	4	2	4	3
Weighted average sub-total		1.6	1.7	1.45	1.65	1.05	1.85	1.2	
Weighted Score			3.85	4.68	3.25	3.85	2.65	3.75	2.85
Weighted average grand total points			3.09	3.32	2.96	3.48	2.94	3.43	2.80

Source: Morgan Stanley Research

Bank Jago (Overweight): Digital bank pioneer in an untapped market

Bank Jago is Indonesia's digital bank pioneer. Its first-mover advantage, seasoned management team and leverage to its shareholder GoTo's online ecosystems, put it in a strong position to lead the digital bank path in Indonesia. Bank Jago focuses on millennials and the younger generation for its funding franchise and to date it has successfully established a respectable funding franchise for a digital bank with CASA ratio of 73% as of end-September 2023. On the lending side, Bank Jago has adopted partnership lending which allows it to manage credit quality prudently. Its primary lending target market is the SME market which is underbanked in Indonesia.

Kakao Bank (Overweight): Has the mettle but policy path a swing factor

KakaoBank is the largest digital bank in Korea, despite not being first, with more than 22mn customers and deposits of total loan assets of ₩37tn as of 3Q23, which corresponds to a 25.5% CAGR since 3Q17. We believe that among the digital banks in Korea, KKB has the mechanics, motor and mettle to maintain its position as the top digital bank in Korea and to be more profitable than commercial banks within 2-3 years. The key strengths possessed by KKB are its banking industry lowest funding cost, connectivity to KakaoTalk, brand recognition, and product innovation. The bank has become one of the most active lenders in the housing loan related market and plans to expand to the SoHo market. KKB's platform business provides cross-selling opportunities for co-branded credit cards, brokerage accounts and soon the bank will start selling investment products (4Q23 launch expected).

Nubank (Overweight): Large runway ahead given cross-sell in Brazil and expansion into Mexico and Colombia

Nubank is one of the world's largest and fastest growing digital banking platforms with 90 million clients across Brazil, Mexico, and Colombia. And we think it is well-positioned to build one of the most valuable banking franchises in Latam with superior technology, best-in-class customer satisfaction, strong cohort performance and unit economics. At the core of the product offering is a distinctive no-frills, no-fee credit card with a fully-digital app experience. However, the company is further monetizing through the cross-sell of other products, including personal loans, payroll loans, Buy Now, Pay Later (BNPL), investments, insurance, and more. We see adjusted ROE reaching ~35% in 2028 and think the company could reach US\$100 billion valuation by 2026. All told, we are Overweight

and Nu is our Top Pick in the Latam Financials space.

Rakuten Bank (Equal-weight): Well positioned among Japan's digital banks with a strong ecosystem

Rakuten Bank was established on January 14, 2000 and was listed in April 2023. Rakuten Bank is efficiently acquiring new customers by leveraging the robust customer base (around 100 million user IDs) covered by Rakuten Group, Inc., whose members are generally known to be highly loyal to Rakuten's various services, and Rakuten Bank has been able to gain the trust of these members to quickly attract a vast number of new customers. Links with the financial businesses within the group are strong, and cross-selling is advancing. As an earnings structure, around 60% of ordinary income in F3/23 was net interest income (NII) and around 40% non-NII. The company is expanding interest income by building up high-earning assets, and the strategy for non-interest income is to grow earnings by increasing the ratio of main accounts.

SBI Sumishin Net Bank (Equal-weight): BaaS holds key for growth

SBI Sumishin Net Bank was the first Internet specialty bank in Japan to list on the TSE Standard market when it listed on March 29, 2023. SBI Sumishin Net Bank has digital bank, BaaS, and THEMIX businesses, though the main income source is digital bank business that primarily handles mortgage loans (95.5% of ordinary profit at end F3/23). The bank provides financing, payment, deposit, and other services to individuals and corporations and aims to deliver products with high added value utilizing AI reviews, biometric authentication, and other technologies. The KPIs that demonstrate business scale are ¥7.98trn in deposit value at end-March 2023, 6.146mn deposit accounts, and ¥1,485.2bn in mortgage loan execution value. In shareholder composition, while Sumitomo Mitsui Trust Bank and SBI Holdings each owned 50% prior to the listing, ownership ratios of these two firms dropped to 34% each at listing.

SoFi (Equal-weight): Risk Reward Balanced; Watchful of Capital, Loan Sale Risks

SoFi Technologies, established in August 2011, is a challenger consumer lender largely focused on refinancing student loans and offering unsecured personal loans (also offers mortgages and credit cards); as an online digital bank, SOFI also offers high-yield deposits and brokerage/investment solutions. SoFi has been one of the strongest growth stories in consumer finance in recent years, including 2022 revenue growth of +50% y/y. However, we do expect top-line growth will continue slowing from here, and we remain below consensus expectations for revenue growth in 2024/25 as fair value marks/gain on sale economics continue to slow. While we are Equal-weight on the stock, we remain watchful of the company's ability to execute on loan sales from here; SOFI did take some steps that should help offset its shrinking capital base going forward, namely the announcement of a US\$2bn flow agreement with a new loan buyer.

Inter (Underweight): Challenges remain; we are skeptical about long-term profitability

Inter is one of the largest and most prominent digital banks in Brazil with over 30 million total clients and a "super app" that provides access to a wide range of products, from

consumer and SME loans to an e-commerce marketplace. The company has been lending in Brazil since 1994 but started a complete transformation in 2015 to go fully digital. Today, the transformation story continues, and it could result in faster growth and higher profitability. We think, however, that there are several key challenges that may hamper the bank's aggressive growth plans, including: difficulties around cross selling and client monetization, rollout of new products, low profitability, and a tough competitive environment. Valuation seems high for a bank with long-term ROE below cost of equity. We are Underweight.

Key risks to our view

- (1) Regulatory and policy environment becomes less favorable for digital banks limiting them from growing assets, reaching targeted cost efficiency, and/or expanding business scope.
- (2) Macro conditions weaken to a level where loan demand declines sharply and/or credit costs rise significantly as asset quality deteriorates.
- (3) Competition becomes more severe as more digital banks enter the market and/or incumbent banks aggressively enhance their mobile capabilities and product offerings that satiate customer demands enough to keep customer attrition at bay. In addition, Fintech/BigTech mobile financial platforms may emerge to become competitive with digital banks in certain areas of business (i.e. WM business).
- (4) Digital banks are not able to differentiate themselves from incumbent players due to strategic and tactical missteps made by management. Expansion into new businesses for cross-selling does not achieve the desired results indicating limitations in becoming a diversified mobile financial platform.

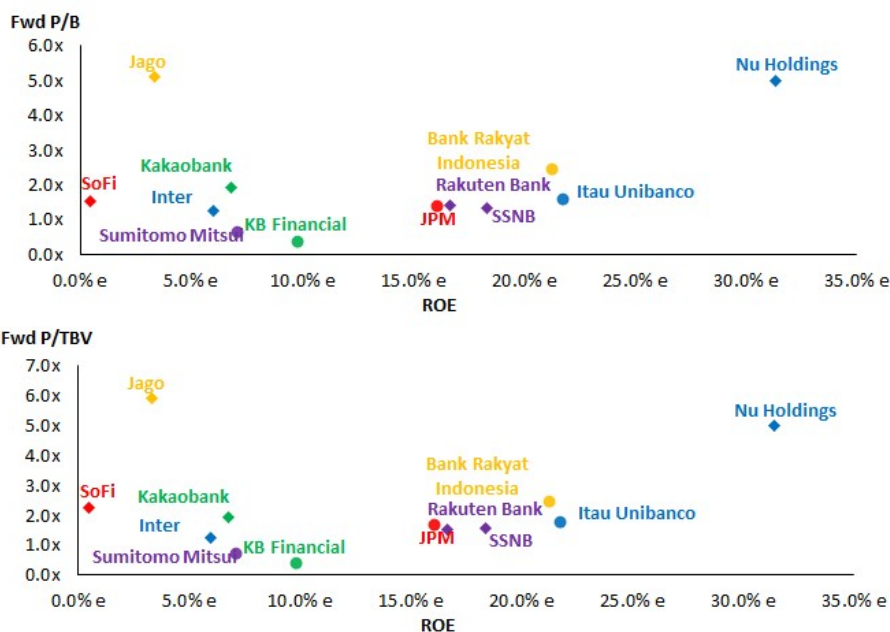
How to Value Digital Banks

MS-covered digital banks: Different valuation methods but similar assumptions

We are often asked whether digital banks deserve their lofty valuations, especially compared to the traditional banks. The market showing the widest gap between traditional banks vs. digital banks on P/B perspective is Indonesia with Bank Jago trading at 4.6x 1yr Fwd P/B and Bank Negara Indonesia at 1.2x 1yr Fwd P/B followed by Brazil with Nu Holdings trading at 5.0x 1yr Fwd P/B vs. Itau Unibanco at 1.2x 1yr Fwd P/B.

If we focus on P/TBV (tangible book value), in the US, the premium gap is narrower, with SoFi trading at a 1yr Fwd P/TBV of 2.3x vs JPMorgan at 1.7x (note: for SoFi, we consider P/TBV a more relevant valuation metric than P/BV, as SOFI has ~US\$1.8bn intangible assets/goodwill largely related to its Tech Platform business, which are not stripped out of book value but are removed from tangible book value).

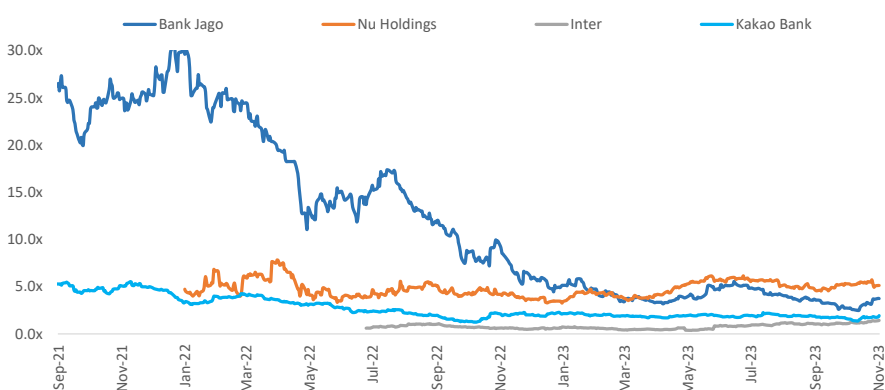
Exhibit 23: Digital banks vs. traditional banks: 1yr Fwd P/B, 1yr Fwd P/TBV vs. ROE



Note: Series with Diamond markers represent digital banks and Circle markers represent traditional banks. Also banks are color coded by region.

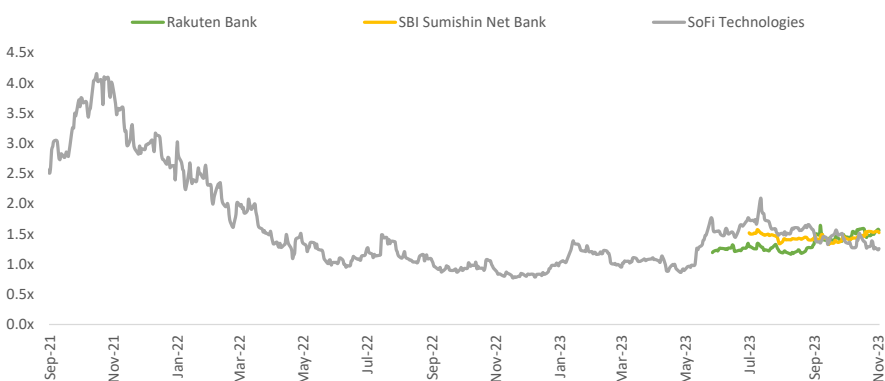
Source: Morgan Stanley Research

For the EM markets, we have seen a de-rating of KKB and Bank Jago in the past two years while Nubank post-listing has shown more steady share price performance. Nu Holdings has overtaken Bank Jago on a Fwd P/B standpoint – both are still trading above KKB's forward P/B of 2.0x.

Exhibit 24: Digital banking peers 12m Fwd P/B: Emerging markets

Source: Bloomberg, Morgan Stanley Research

In the case of the DM markets, Japan has only had listed digital banks starting this year and share price performance has generally been positive for both Rakuten Bank and SBI Sumishin Net Bank. SoFi in the US was trading at 4.2x Fwd P/B at the peak in November 2021 but is now at around 1.5x Fwd P/B.

Exhibit 25: Digital banking peers 12m Fwd P/B: Developed markets

Source: Datastream, Morgan Stanley Research

Do digital banks deserve much higher valuations than traditional banks? Yes, for the leading digital banks

We are almost always asked the question whether digital banks deserve their lofty valuations especially compared to commercial banks. In order to try to address this question, we need to see if digital banks are different to the traditional banks and whether they can monetize their business platforms.

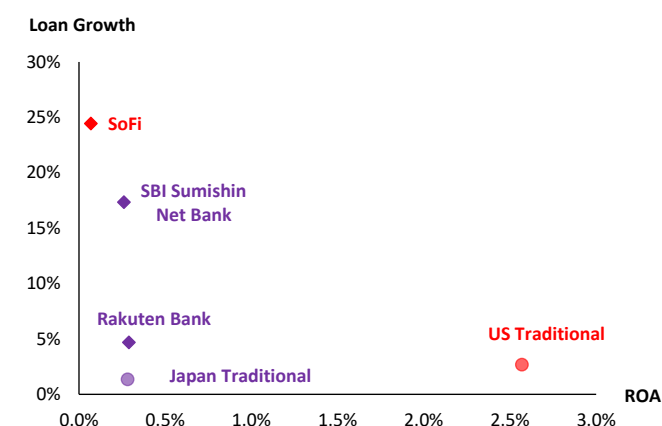
We believe that to evaluate the performance of digital banks, key metrics at software-as-a-service companies (LTV, CAC, DAU/MAU) give an informative perspective. This is because digital banks operate as stock-based businesses with a long lifespan and investment is prone to increase in the early stages of startups, a characteristic that they have in common with SaaS companies. Looking ahead, disclosure of such indicators, not just by challenger banks but by traditional financial institutions too, could invite calls from the market for improvements in earnings and cost efficiency.

We believe the key prerequisites for digital banks being valued as more of a technology or

Internet company depend on the following: 1) top-line growth is significantly higher than for traditional banks; 2) cost efficiencies are attained; 3) technology is better utilized (i.e. cloud, AI); and 4) evolves into more than just a lender but a mobile financial platform.

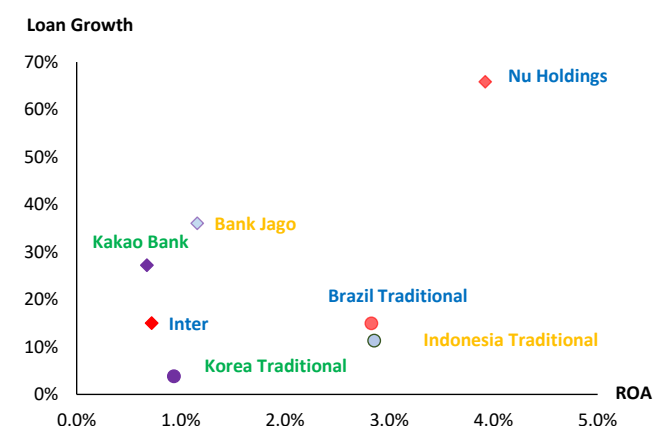
- Loan growth for digital banks based on our analysts' forecasts looks to be significantly higher than traditional banks and the loan growth gap is sizable.
- Aside from Nu Holdings, we find that the ROA for digital banks compared to traditional banks are still quite low as monetization is still mostly in the early stage with cost advantages not being fully realized.
- Cost-to-income ratios for digital banks are expected to come under traditional banks when critical mass is exceeded and revenue sources become more diverse leading to higher non-interest income levels.
- As previously discussed, we believe leading digital banks will be more than just lenders as they expand into other lending segments and/or expand product breadth to increase per customer product holdings, which are provided to customers more efficiently and proficiently using technology.

Exhibit 26: DM digital banks vs. traditional banks: loan growth vs. ROA (24e)



Source: Morgan Stanley Research

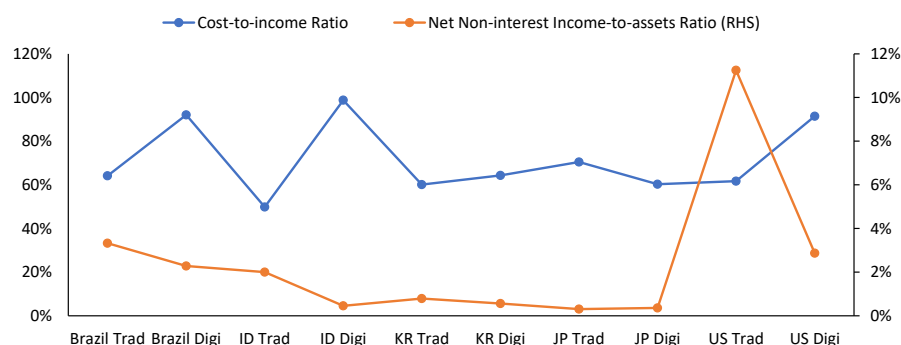
Exhibit 27: EM digital banks vs. traditional banks: loan growth vs. ROA (24e)



Source: Morgan Stanley Research

Exhibit 28:

Digital banks vs. traditional banks: Cost-to-income ratio vs. net non-interest income-to-assets ratio (FY22)



Source: Morgan Stanley Research

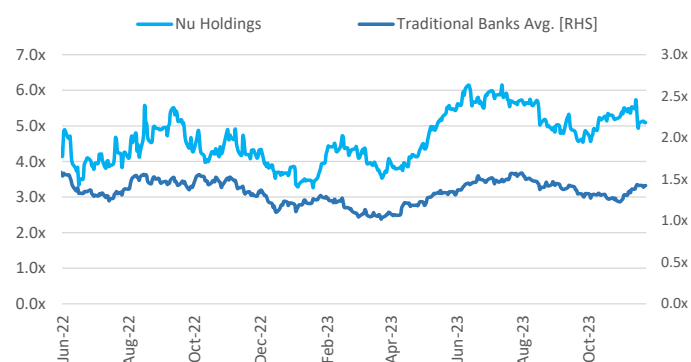
Our analysts apply different valuation methodologies but there are some common elements;

- Valuation horizon tends to be more long-term similar to high growth new world companies;
- Digital banks are seen to be different to traditional banks from the perspective of higher growth potential and higher long-term sustainable ROE than traditional banks; and
- Cost efficiency advantage to be realized after reaching beyond critical mass and business scope expansion to be achieved, which should enable higher ROE than traditional banks to

Nu Holdings Ltd. (NU.N, OW):

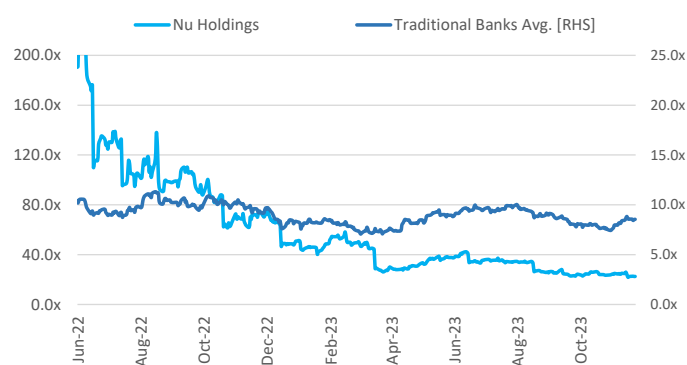
Jorge Kuri values Nubank using residual income analysis. The price target of US\$16 is derived assuming that the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). The valuation model uses a discount rate of 14.2% and long-term ROE of 35%. Jorge thinks Nubank is well-positioned owing to its superior technology, best-in-class customer satisfaction, and strong cohort performance and unit economics. He expects strong revenue growth and operating leverage to support profitability over the next few years, leading to an adjusted ROE of ~35% in 2028.

Exhibit 29: Nu Holdings vs. traditional banks 12m Fwd P/B trend



Source: Datastream, Morgan Stanley Research

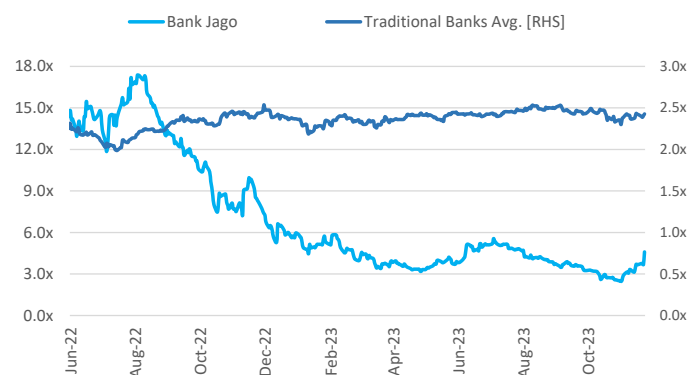
Exhibit 30: Nu Holdings vs. traditional banks 12m Fwd P/E trend



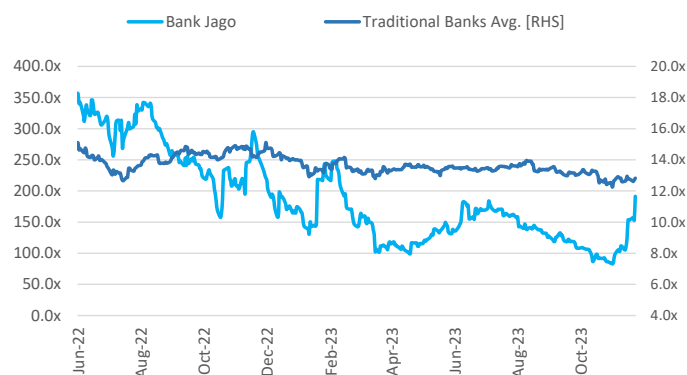
Source: Datastream, Morgan Stanley Research

Bank Jago Tbk Pt (ARTO.JK, OW):

Selvie Jusman, our Morgan Stanley ASEAN Financials analyst, values Bank Jago using a probability-weighted (10% bull, 50% base and 40% bear) Gordon Growth model over a 20-year horizon of financial forecasts. The skew here in the bear case reflects the impact of higher rates on this growth stock. The price target of IDR3,208 is derived assuming the base case cost of equity of 13.4% (risk premium 4.4%, risk-free rate 6.5%, beta 1.6), sustainable ROE of 18.0% and sustainable growth of 12.6%. Selvie thinks Bank Jago's healthy operating metrics are supported by its high CASA ratio, prudent lending policies, and strong partners. She expects it to remain profitable with a net profit CAGR of 219% for 2022-2025e despite initial adjustment to credit underwriting.

Exhibit 31: Jago vs. traditional banks 12m Fwd P/B trend

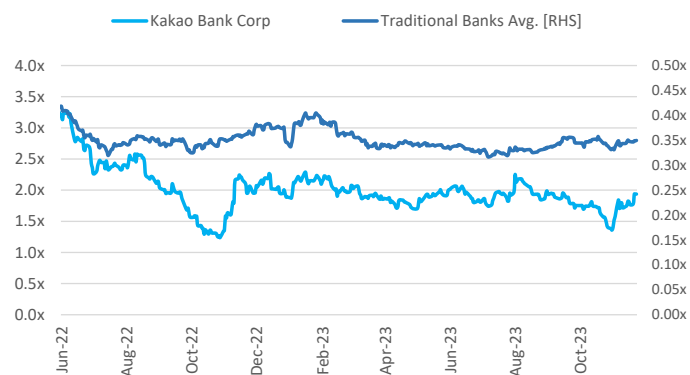
Source: Datastream, Morgan Stanley Research

Exhibit 32: Jago vs. traditional banks 12m Fwd P/E trend

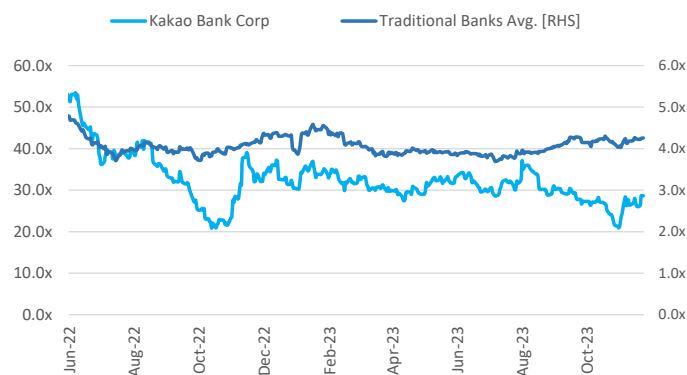
Source: Datastream, Morgan Stanley Research

Kakao Bank Corp (323410.KS, OW):

Joon Seok values Kakao Bank using an economic profit valuation methodology. The price target of KRW32,500 is based on our base case earnings estimates reflecting 2023e-2024e ROE reaching 5.8% and 6.6% respectively. The valuation model uses a cost of equity assumption of 10.1%, sustainable ROE: 14.8% (2030e) and terminal growth rate of 8.25%. Joon thinks the bank has been adversely affected by macro, regulatory and competitive factors. However, he thinks the negative factors such as regulation and competition are now becoming more favorable for the bank, which should enable it to rekindle asset growth.

Exhibit 33: Kakao Bank vs. traditional banks 12m Fwd P/B trend

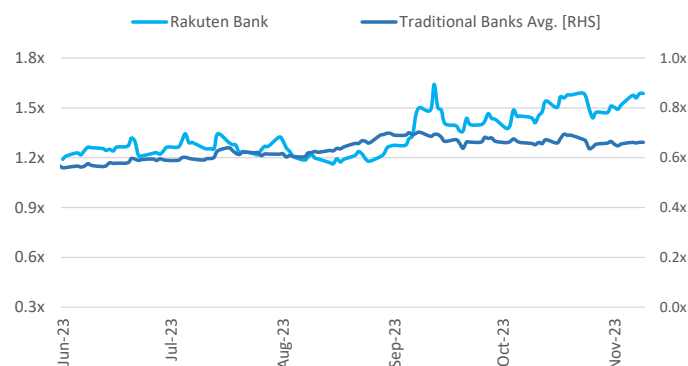
Source: Datastream, Morgan Stanley Research

Exhibit 34: Kakao Bank vs. traditional banks 12m Fwd P/E trend

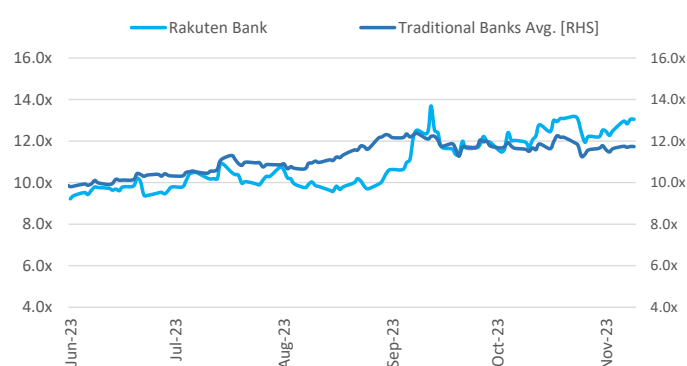
Source: Datastream, Morgan Stanley Research

Rakuten Bank (5838.T, EW):

Our Morgan Stanley Japan Banks analyst, Mia Nagasaka, values Rakuten Bank using a P/B-based valuation methodology. The price target of JPY2,830 is based on a 1.6x P/B multiple on our F3/25 BPS estimate. Leveraging its access to the Rakuten ecosphere might boost the bank's strategic value and scarcity value to levels other banks cannot achieve that is why the target multiple also includes a premium to the estimated F3/25e P/B (1.5x). Mia expects the company to achieve solid growth from rising transactions as more clients use Rakuten Bank as their main account and also thinks that the firm can achieve high ROE through an earnings structure with high operating leverage.

Exhibit 35: Rakuten Bank vs. traditional banks 12m Fwd P/B trend

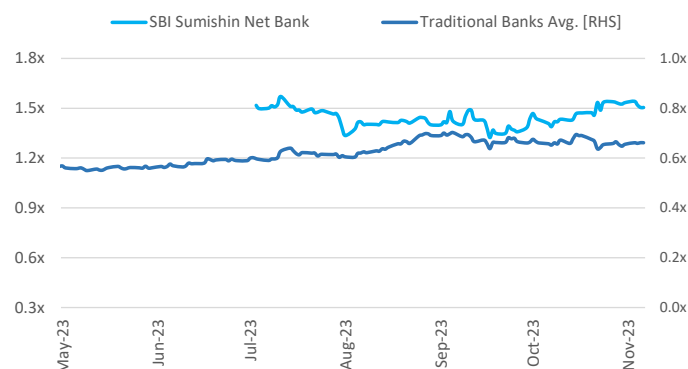
Source: Datastream, Morgan Stanley Research

Exhibit 36: Rakuten Bank vs. traditional banks 12m Fwd P/E trend

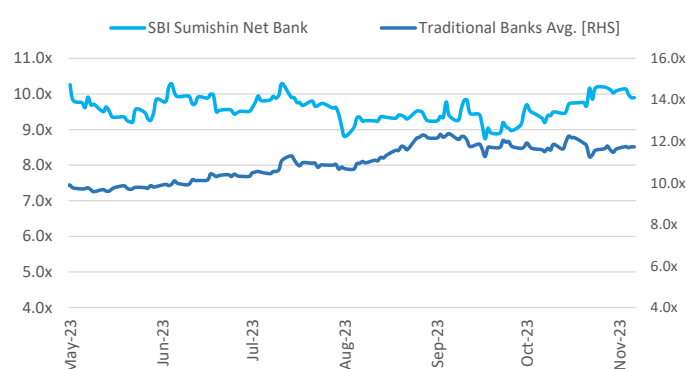
Source: Datastream, Morgan Stanley Research

SBI Sumishin Net Bank (7163.T, EW):

Mia Nagasaka uses a P/B-based valuation methodology to value SBI Sumishin Net Bank. The price target of JPY1,920 is based on a fair-value P/B multiple of 1.5x applied to our F3/25e BPS estimate. Mia expects an average growth of 13.8% for consolidated NP over the forecast period F3/24-26e as the market recognizes its high operating leverage compared to conventional banks and also due to the firm's ability to advance its business model. She thinks further share price gains will hinge on valuation on the basis of SaaS-type indicators via rising profit exposure to BaaS that draws a line between SBI Sumishin and conventional banks.

Exhibit 37: SBI Sumishin Net Bank vs. traditional banks 12m Fwd P/B trend

Source: Datastream, Morgan Stanley Research

Exhibit 38: SBI Sumishin Net Bank vs. traditional banks 12m Fwd P/E trend

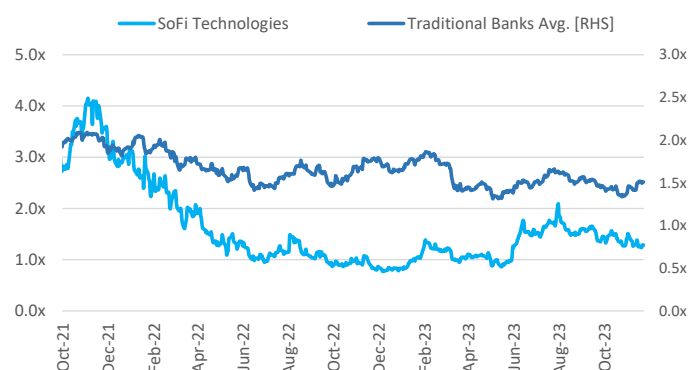
Source: Datastream, Morgan Stanley Research

SoFi Technologies, Inc. (SOFI.O, EW):

Jeff Adelson values SoFi Technologies using an SOTP valuation framework, focusing on 1) the banking business and 2) Technology Platform business. The price target of US\$7 is derived by layering on the value of bank at 1.5x P/TBV + the value of tech platform at 5x P/Sales. This price target implies a 21x P/E on 20226e EPS. Jeff views the underlying bank business (ex-Tech Platform) as capable of driving to a low-teens ROTCE level of profitability by 2026, which along with faster revenue growth than most banks, justifies a 1.5x P/TBV for the banking business. He also views capital raising risk as lower in the near-

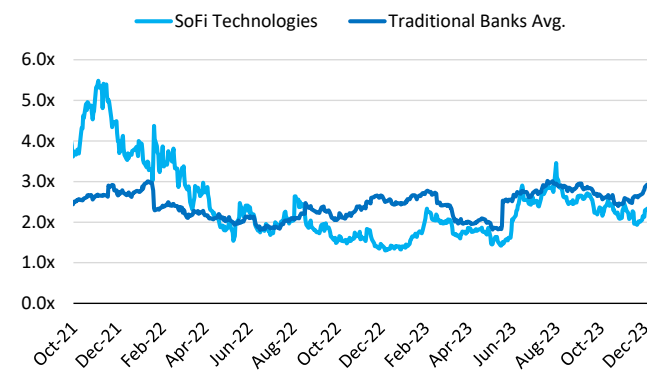
term following the announcement on 3Q earnings of US\$475mn in 4Q loan sales expected as well as the new US\$2bn 12-month flow agreement with a new loan buyer, but is watchful of the company's ability to execute from here.

Exhibit 39: SoFi vs. Traditional Banks: 12m Fwd P/BV trend



Source: Datastream, Morgan Stanley Research

Exhibit 40: SoFi vs. Traditional Banks: 12m Fwd P/TBV trend

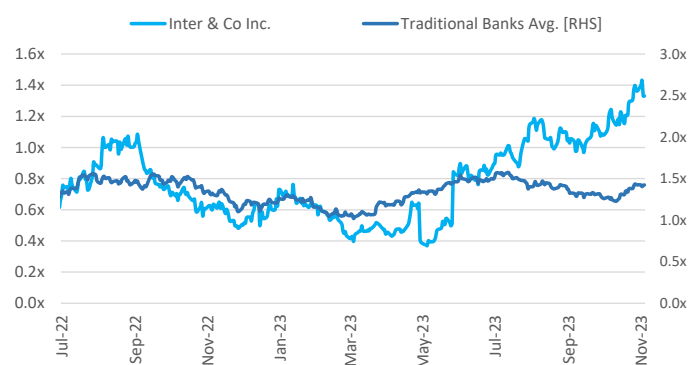


Source: Datastream, Morgan Stanley Research

Inter & Co Inc. (INTR.O, UW):

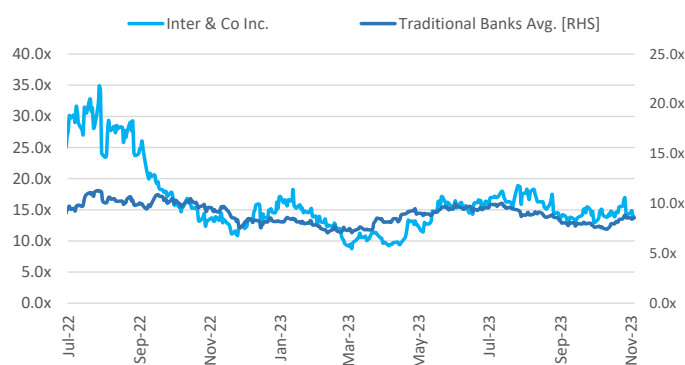
Jorge Kuri values Inter using residual income analysis. The price target of US\$2.80 is derived assuming that the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). The valuation model uses a discount rate of 14.3% and long-term ROE of 6.8%. Jorge believes Inter is currently undergoing a transformation that could result in faster growth and higher profitability. However, he thinks Inter's valuation seems high for a bank with long-term ROE below cost of equity.

Exhibit 41: Inter vs. traditional banks 12m Fwd P/B trend



Source: Datastream, Morgan Stanley Research

Exhibit 42: Inter vs. traditional banks 12m Fwd P/E trend



Source: Datastream, Morgan Stanley Research

Unlisted private digital banks: Slower fund raising and lower valuations

There are many digital banks currently in operation globally, with most of them being unlisted entities. These banks have continued to receive private fundraising, with assumed valuations having shown a sharp upward trajectory until 2022 but recent funding rounds have been flat or down.

For us the key debate around how to value a digital bank is whether this should be looked at as a tech company, which is more the case if the company doesn't take credit risk and provides only payments services, or like a bank, which in our view is the case if the company provides some form of lending (or wishes to do so in the future). The next key question is what's the competitive differentiating advantage vs traditional banks? Digital banks usually argue that tech allows them to be much more profitable, once at scale, and have a better customer experience. While we think the superior tech infrastructure may be a point in favor, and should lead to higher valuations, traditional banks on the flip side usually enjoy cheaper funding (because they have a deposit franchise), and are already at scale. In a nutshell, we think the valuation premium vis-à-vis the incumbent banks would be justified just to the extent that the neobanks can show faster growth, better customer experience, and on a path to achieve higher returns.

Based on recent private fundraising rounds, the digital bank with the highest *EV per customer ratio*, is Chime (US) at the US\$1,700 level followed by Revolut (UK) and N26 both at around the US\$1,100 level.

Exhibit 43: Comparison with Private Digital Banks

Bank Description					Key Business Stats		Valuation (recent fund raising)		
Name	Established	Country	Target customers	Key market	Customer Base	Deposits (US\$m, as of '22)	Recent Funding Timing	Assumed Valuation	Est. Value per Customer (US\$)
Chime	2013	US	Retail	US	14.5mn		Jul 2023	US\$25bn	1,724
N26	Feb. 2013	Germany	Retail	25 countries including US, Europe	8mn	2021: 7,215	Oct 2021	US\$9bn	1,125
Starling Bank	2014	UK	Retail & Business	UK	3.4mn	12,406	Apr 2022	US\$1.8bn	529
Revolut	July 2015	UK	Retail & Business	UK, Europe	25mn	2021: 10,127	Aug 2021	US\$19.8bn	792
Monzo	2015	UK	Retail & Business	UK	7.1mn	6,935	Feb 2022	US\$4.5bn	634
Klarna	2005	Sweden	Retail	UK, Europe, US	150mn	8,011	Jul 2022	US\$6.7bn	45
C6	2018	Brazil	Retail	Brazil	NA	NA	Aug 23	US\$2.1bn*	84
Agibank	1999	Brazil	Retail	Brazil	NA	NA	Sep 20	NA	NA
Neon	2016	Brazil	Retail	Brazil	NA	NA	Feb 22	US\$1.6bn	62

Source: Media (Bloomberg), Morgan Stanley Research

Details on Key Markets

Japan

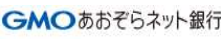
Mia Nagasaka

Competitive landscape

Digital banks in operation

The big names in digital banking in Japan include Rakuten Bank, SBI Sumishin Net Bank, PayPay Bank (formerly Japan Net Bank), Minna Bank, and GMO Aozora Net Bank. There are two that are independently listed in Japan: Rakuten Bank (listed April 2023) and SBI Sumishin Net Bank (listed March 2023).

Exhibit 44: Digital bank benchmarking

	 NEOBANK 住信SBIネット銀行	 Rakuten 楽天銀行	 PayPay 銀行	 Minna no Ginko	 GMO あおぞらネット銀行
	SBI Sumishin Net Bank	Rakuten Bank	PayPay Bank	Minna Bank	GMO Aozora Net Bank
Established	2007/9	2001/7	2000/10	2021/5	2018/7
Specialization	Deposits, loans (housing, consumer)	Smartphone apps, deposits, loans (housing, unsecured), joint services with Rakuten Card etc.	Online settlement, debit card, loans (personal unsecured loans, business loans)	Settlement, savings, consumer loans	Foreign currency, deposits, joint services with GMO Click Securities
No. of accounts (thousands)					
Sep 23	6,570	14,370	---	---	---
Jun 23	6,350	14,047	7,180	710	---
Mar 23	6,146	13,739	6,960	600	---
Deposits (JPY bn)					
Sep 23	8,643	9,550	---	---	---
Jun 23	8,587	9,488	1,689	24	---
Mar 23	7,978	9,130	1,667	23	413

Source: Company data, Morgan Stanley Research

Competitive dynamics amongst digital banks

In most markets, there are a small number of digital banks that have made a splash. We find them to generally have advantages in attracting younger clients, effective ecosystems, and key operational advantages (i.e., lower funding costs). Digital banks don't operate physical branches or their own ATMs. However, these online banks generally can't make good on the cost competitiveness of their business models without a certain scale. Rakuten Bank and SBI Sumishin Net Bank already are at the scale where they can enjoy such competitiveness, and thus can benefit from operating leverage. Accordingly, the banks now generate a higher ROE than the traditional banks in Japan, as well as other online competitors.

Incumbents are rushing digitalization: Behind the rush in megabank digital strategy is the entry into the financial industry of firms from other industries and emergence of digital banks. As a case study, SMFG, one of the mega banks, has been proceeding with branch

reforms and a super app concept.

- SMFG's Branch reform:** Sumitomo Mitsui Bank, under the SMFG umbrella, plans to expand its new specialized “stores” from 8 current locations to 250 locations in 2025. The concept for the “stores” is to extricate itself from large full-service branches located in prime locations and pursue a new model for channels. The new locations are to be in commercial facilities and other places with easy access, and are to operate also in evenings and on weekends with unattended reception machines and staff. Of 400 locations nationwide, Sumitomo Mitsui Bank plans to gradually convert 250 locations servicing only individual customers to stores specialized in consultation business. The group expects the conversion to stores to cut ¥28bn annually by FY25. Branch costs are exemplified by rent and so-called CRE cost, which includes branch maintenance costs. Previously, branches opened usually had an area of 1,000-1,650m² on multiple floors, but the new stores are to be run with about a 90% reduction in size, to 100-165m², greatly reducing branch maintenance costs. In addition, with automation of operations and such, the stores can be run with few people, and including staff efficiency, the group expects to operate with about a 70% reduction in running costs. SMFG has been working on branch reform and enhancing digitalization for the past 10 years. Needing branches to make progress on going cashless, it greatly increased the number of non-branch ATMs as general cash infrastructure, while at the same time pursuing more efficient operations, ultimately succeeding in shrinking large ATM-related losses to the breakeven point.
- SMFG's Super app:** In March 2023, SMFG announced Olive, a digital account at the center of group retail business strategy. SMFG announced that the number of Olive accounts had surpassed 850,000 in the five months since its release and 1mn in September. Accounts of workers are growing successfully, especially among the young, due to the merchantability of point increases at salary direct transfer branches. There has been an increase in customers using Olive as their main account combining salary direct transfer with accumulating deposits. An SBI tie-up and credit card accumulation have increased credit usage rates and usage prices, leading greater earnings from cashless payments.

Overall regulatory and policy landscape

Financial regulation moves toward digital banking in Japan

The digital banks that are pioneering services in Internet banking in particular in Japan are categorized as “new forms of banking” by the Financial Services Agency, and generally included as “others” in the agency’s summary list of licensed financial institutions. Essentially, under Japanese law and regulations, there is no special framework for digital banks, and the names are treated under local business practices.

1. Bank registration: Rakuten Bank, SBI Sumishin Net Bank, PayPay Bank, and Minna Bank are qualified banks, having registered as banks in the normal fashion.

2. Supervision guidelines: The FSA’s Comprehensive Guidelines for Small and Medium-sized and Regional Financial Institutions are applied to Japanese digital banks, but that term (digital banks) is not used under these guidelines, and there is no special framework

for them.

3. Capital regulations: Japan's digital banks are basically obligated to maintain the mandatory shareholders' equity requirement as domestic standard banks (at least 4%). For example, Rakuten Bank is required to maintain a shareholders' equity ratio of 4% or more as a domestic registered bank; if its shareholders' equity ratio falls below 4%, the FSA may issue directives, including orders to suspend all or part of operations. SBI Sumishin Net Bank goes further, adding to the 4% regulatory shareholders' equity requirement an appropriate capital buffer to be maintained.

Regulatory environment

Japan's digital banks and other "neobanks" have been expanding since around 2000. Amid a wave of financial deregulation, enterprises from other segments of the financial industry clarified strategies to enter banking business from around 1999, and in 2000 the Financial Reconstruction Commission and the Financial Service Agency established operational guidelines for "new forms of banking," followed in 2001 by the full-scale promulgation of revisions to the Banks Act. The strength of digital banks is low-cost operations with no physical outlets required, and compared with traditional banks, they have been able to quickly build up customer bases with low interest and fees on home loans. Around 2020, they became able to sell investment trusts with no commissions, and made progress in developing ways to win and maintain customers. The no-load (zero commission) investment trust products introduced online by Internet brokers also spread to Internet banks. Subsequently, a succession of major financial institutions and IT firms among others took steps to establish digital banks, and in recent years regional financial institutions have also entered the field of online banking. We recognize that the use of digital banking has been picking up since around 2013, but we think this evolution was given added momentum in recent years by Covid-19.

A new point that came to the fore in 2021 has been competition in lowering fees for transfers to other banks. Up until then digital banks had increased client numbers by, for example, charging no fees for up to a certain number of transfers depending on frequency of use, but when interbank charges in the interbank data communication system (the Zengin System) were reduced to nearly half previous levels (a uniform ¥62) from October 2021, moves to reduce fees themselves made progress in 2021. This was considered to be an effect of a report by the Fair Trade Commission in 2020, which said for the first time since the system went into operation in 1973 fees had become static at ¥117 for transfers of less than ¥30,000 and ¥162 for those upward of ¥30,000.

Loosening of financial regulations

Deposit-handling financial institutions, including banks, have been subject to strict constraints on the scope of their business. Meanwhile, moves to loosen restrictions relating to banking have gradually been making headway, in response to the argument that some of the rules that govern finance have not adapted to the changing environment. For example, amendment of the Banking Act implemented in 2017 made it possible for banks and bank holding companies to convert fintech entities into subsidiaries, and further amendment of the Banking Act in 2019 reviewed the "5% rule" limiting the acquisition of voting rights related to venture investments. In the case of Japan, financial groups founded on a bank cannot enter business operations other than those specifically stipulated. However, if a general business company is moving into banking, it does not face

constraints on its stipulated business operations under the Banking Act provided that eligibility as a major shareholder and certain regulatory hurdles are cleared. This is what has allowed the likes of Rakuten (banking, credit card business, etc.), Sony (Sony Financial), and Seven & I Holdings (Seven Bank) to move into banking. The Banking Act amendments in 2019 make it possible to obtain FSA approval to invest in companies engaged in business that is expected to enhance the sophistication and user convenience of banking business, and permits more latitude than the previous limitation to stipulated operations. In August 2022 as well, certain “advanced banking service companies” (i.e., fintech companies, etc.) were allowed to have voting rights in certain situations under relaxed regulatory standards.

Banking service scope and ecosystem

Other operations handled by digital banks in addition to ordinary deposits and time deposits include insurance and credit card issuance. Some digital banks also handle housing loans and investment trusts and offer foreign currency deposit services. Online banks are able to operate with low costs in the absence of physical branch networks, and can therefore offer more favorable deposit rates and hold remittance fees and mortgage loan rates lower than at in-person banks, which provides competitive strength. With incorporation of the latest technology, usage costs are low, and high convenience is a feature. Many online banks offer mortgage lending, and with megabanks no longer pushing these aggressively now that their mortgage profitability has diminished, online banks are capitalizing on their low costs and expanding their client base, especially among working generations. The leading digital banks target synergies with powerful ecosystems. For example, Rakuten Bank (a consolidated subsidiary of Rakuten Group) is successfully reaping synergies with the Rakuten ecosystem, drawing on the group’s powerful customer base of approximately 100mn ID holders, while SBI Sumishin Net Bank is in a position to unleash synergies with the SBI Holdings group and its subsidiaries.

The online banking realm includes banks that have found customers by forming alliances with stores in other industries. Alliances with other industries can be effective as a means of addressing the reservations of users who hesitate to conduct procedures exclusively online. Amendments to Japan’s Banking Act in 2006 opened up bank agency business to general business corporations. Bank agency branches can undertake banking business on a consigned basis (taking deposits, lending, foreign currency services, etc.).

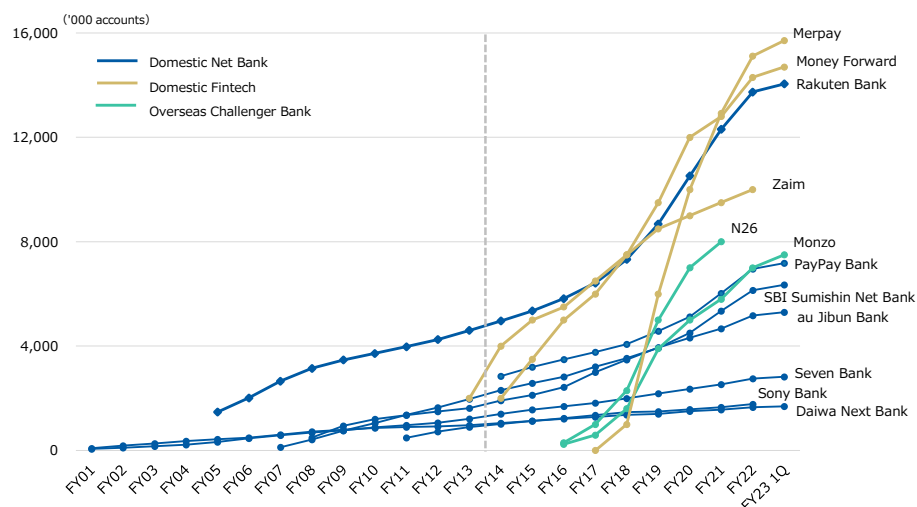
Advances in embedded finance BaaS (Banking as a Service)

BaaS, the delivery of banking infrastructure functions (such as payments, deposits, and loans) using API and other formats to business operators, has picked up. For example, SBI Sumishin Net Bank started providing service in the BaaS business in 2020 and is expanding it by consigning bank distributor work to partner companies in other industries. Under this arrangement, partner companies can deliver banking services to their customers by acquiring a license for bank distributor business, while SBI Sumishin Net Bank increases deposits by supplying the customer contact portion in the partner company’s specifications and connecting this portion to its own system. Though differences exist in the business models, Rakuten Bank, GMO Aozora Net Bank, Minna Bank, and others have already started delivery or are considering delivery of these services too. Fintech companies also run businesses that operate financial services as assistance to other

industries.

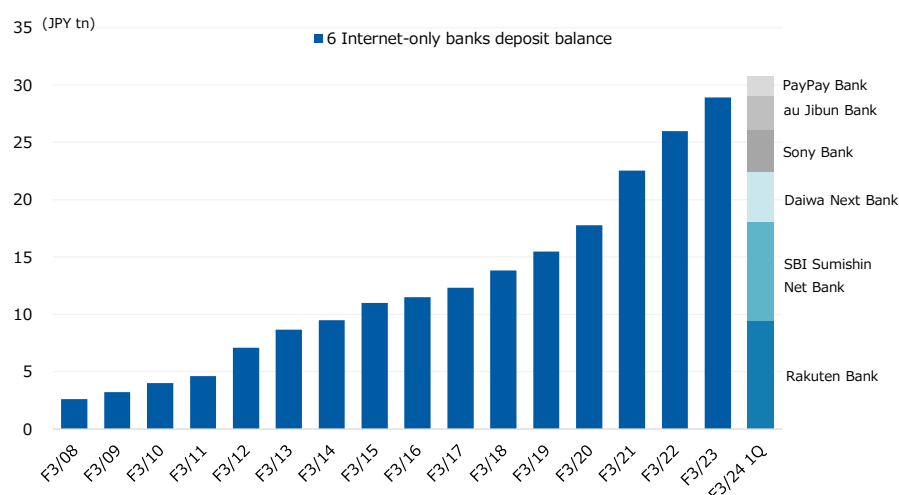
While BaaS business consists of a variety of monetization methods and models, as an indication of the growth potential via new alliance partners, SBI Sumishin Net Bank is looking at the B2C payments market (¥237trn), mortgage loan market (¥20trn), consumer loan market (¥1.3trn), credit card settlement costs (¥2.8trn), and the points program ecosphere (¥2.1trn). We expect earnings opportunities to arise in the mortgage loan and credit card settlement markets in particular. This indicates existence of a large earnings acquisition opportunity. One portion is fee income from home mortgage loan business (estimated at ¥300bn based on ¥20.8trn in annual provision value x 1-2%). There is also acquisition of fees related to cashless payments. SBI Sumishin Net Bank develops the main app for partner companies and conducts service that supplies the app and banking functions as a package. We think the retail industry, in particular, presents large latent demand. From a TAM perspective, we estimate that a portion of the latent market could be around ¥2.8trn, reaching by multiplying the value of transactions using credit cards by consumers (2022: ¥93.7trn) by participating merchant fees (3%).

Exhibit 45: Pace of customer acquisition: Japanese domestic fintech, digital banks/ neobanks and overseas challenger bank



Source: Company data, Morgan Stanley Research

Exhibit 46: Total deposits outstanding at main Japanese digital banks: Handful of digital banks driving share growth



Source: Company data, Morgan Stanley Research

Note: 6 Internet-only banks are SBI Sumishin Net Bank, Rakuten Bank, PayPay Bank, Sony Bank, au Jibun Bank (included from F3/09 data) and Daiwa Next Bank (included from F3/12 data).

Key nuances that matter

Potential for digital banks to extend their share in the retail arena

Digitization in the financial industry is advancing globally. In recent years, Japan's banking industry (especially in the retail area) has seen a rapid increase in the accounts and deposit balances at new special-ized online banks.

1. Number of accounts and deposits

By 2030, we think digital banks should reach 35-40mn accounts and their share of total deposit value should expand from about 2-3% at this point to 6-7% in the main case and about 13-14% in the bull case. Total account volume at newcomer digital banks has already attained a level on par with a single megabank in retail business with individuals (20-30mn accounts). We utilize megabank account volume (roughly 30mn at each bank) as the estimated account volume benchmark. Based on acquisition of 100,000-150,000 accounts per month at Rakuten Bank, which has the most accounts, we think total digital bank account volume could move within range of a level on par with or exceeding that of individual megabanks (about 35-40mn accounts) in 2030 if the pace continues. According to company disclosures as of end-June 2023, newcomer digital banks (SBI Sumishin Net Bank, Rakuten Bank, PayPay Bank, Minna Bank, and GMO Aozora Net Bank) had roughly 28mn accounts. The average deposit level per account was about ¥700,000 (this varies substantially among the banks with a range of ¥200,000 to ¥3mn because some banks have corporate accounts).

Exhibit 47: Digital bank TAM simulation: Share of deposits could rise from 2-3% currently to 6-7% in 2030 in our base case, or ~14% in our bull case

	Total # of accounts (mn)	# of Account 2022-2025 CAGR	Average Account size (JPY)	Estimated Deposit market share
Current (Estimate)	28		¥710,000	2.2%
Base	40	12%	¥1,500,000	6.6%
Bull	50	21%	¥2,500,000	13.7%
Bear	30	2%	¥800,000	2.6%

Source: Company data, Morgan Stanley Research, Morgan Stanley Research estimates.

Exhibit 48: Digital bank TAM (total addressable market) estimate sensitivity

Account size (JPY)	Total # of accounts (mn)											
	20	25	30	35	40	45	50	55	60	65	70	
	¥5,000,000	11%	14%	16%	19%	22%	25%	27%	30%	33%	36%	38%
	¥4,500,000	10%	12%	15%	17%	20%	22%	25%	27%	30%	32%	35%
	¥4,000,000	9%	11%	13%	15%	18%	20%	22%	24%	26%	29%	31%
	¥3,500,000	8%	10%	12%	13%	15%	17%	19%	21%	23%	25%	27%
	¥3,000,000	7%	8%	10%	12%	13%	15%	16%	18%	20%	21%	23%
	¥2,500,000	5%	7%	8%	10%	11%	12%	14%	15%	16%	18%	19%
	¥2,000,000	4%	5%	7%	8%	9%	10%	11%	12%	13%	14%	15%
	¥1,500,000	3%	4%	5%	6%	7%	7%	8%	9%	10%	11%	12%
	¥1,000,000	2%	3%	3%	4%	4%	5%	5%	6%	7%	7%	8%
	¥800,000	2%	2%	3%	3%	4%	4%	4%	5%	5%	6%	6%
	¥500,000	1%	1.4%	2%	2%	2%	2%	3%	3%	3%	4%	4%

Source: Company data, Morgan Stanley Research, Morgan Stanley Research estimates.

2. Home mortgage loan market

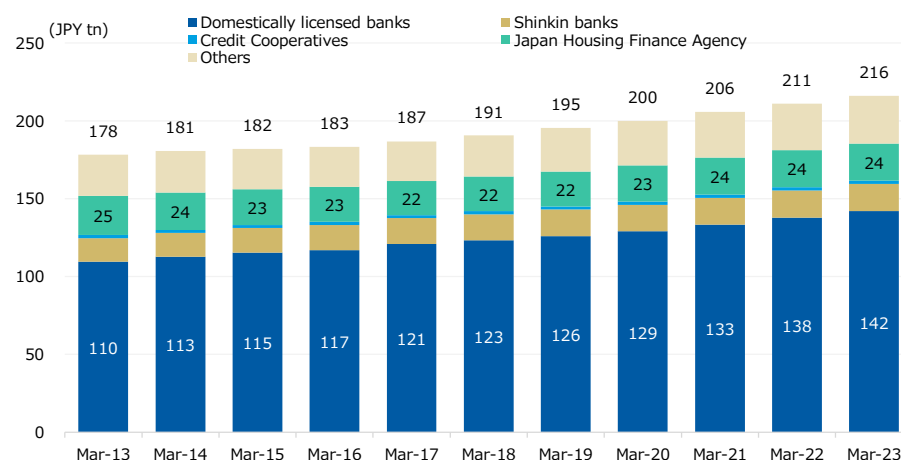
Japan's home mortgage loan market was worth roughly ¥216trn at the end of March 2023. Of this, the mortgage lending by Japan-licensed banks was approximately ¥141.9trn. Looking at market shares by individual banks among the Japan-licensed bank mortgage loan market, the three megabanks had a combined ¥33.4trn, or a 23.5% share. Major bank outstanding loan values and market shares were MUFG (not covered) at ¥14.6trn (10.3% share), SMFG at ¥11.0trn (7.8%), Mizuho FG at ¥7.8trn (5.4%), Resona Holdings at ¥16trn (11.8%), and SMTH at ¥10trn (7.4%).

In digital banks, SBI Sumishin Bank, Rakuten Bank, Sony Bank, AEON Bank, and au Jibun Bank (real estate loan) have a combined ¥12trn (8.4%). Their share of the overall market is still low. Meanwhile, digital banks have the advantage of low-cost operations because they do not run real-world branches, and this enables them to offer attractive deposit interest rates and keep mortgage loan rates and payment fees lower than at face-to-face banks.

In the mortgage loan business, while megabanks are unable to engage aggressively owing to weak profitability, digital banks have low cost as a strength and are expanding the customer base, mainly in the working-age generation that has a high smartphone usage ratio. The new loan origination market is ¥21trn a year, and we think digital banks are likely to expand their share over the medium term. SBI Sumishin Net Bank acquired Yuryo Jutaku Loan as a subsidiary in March 2021. This addition contributed to the roughly five-fold increase in the group's new mortgage loan origination value from ¥289.9bn in F3/12 to ¥1,485.2bn in F3/23 and more recently to a roughly 7% in F3/23. The bank's target for mortgage loan origination market share is 10% in F3/25, with the aim of reaching 20% in the future (assuming a ¥20trn market). Rakuten Bank is implementing a strategy of

increasing loan value with a suitable balance of various loans and purchased credits. While it has a certain amount of outstanding mortgage loan value, we think Rakuten Bank is intentionally curtailing this business.

Exhibit 49: Mortgage loan balance (end-of-period): About ¥216trn as of end-Mar 2023 and gradually expanding



Source: Japan Housing Finance Agency, Morgan Stanley Research

Potential in corporate business

The online banks have lending operations with a limited corporate customer base, and there are few examples of such banks being the primary lender for a business. However, we expect this number to increase, though remaining at a small ratio. All the online banks are working to increase their corporate user base. By being able to reduce bank processing costs, the online banks have made strides with small/mid-size businesses and single-person operations. And by digitalizing certain processes such as bookkeeping/accounting and documentation, there are also expected efficiency gains (in document retention & management). In digital banks, SBI Sumishin Bank, Rakuten Bank, Sony Bank, AEON Bank, and au Jibun Bank (real estate loans) have a combined ¥12trn (8.4%). Their share of the overall market is still low. Meanwhile, digital banks have the advantage of low-cost operations because they do not run real-world branches, and this enables them to offer attractive deposit interest rates and keep mortgage loan rates and payment fees lower than at face-to-face banks.

Global acceleration of fintech

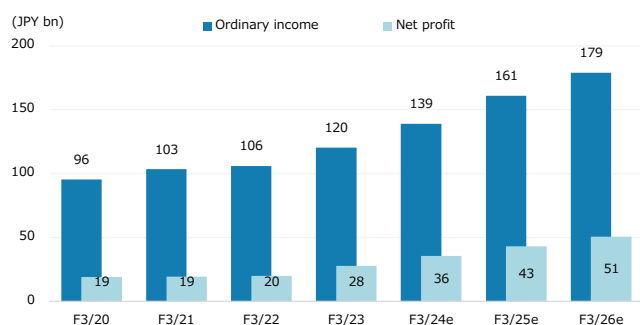
Online banks, also known as challenger banks, have been expanding globally, and in response to moves by overseas players in other industries to invest in or acquire Japanese fintech ventures, the expansion of the range of operations of Japan-origin fintech companies has accelerated, with digitization at incumbent financial institutions making advances.

How is Rakuten Bank positioned?

We think the listing of two digital banks in Japan in 2023 (SBI Sumishin in March, Rakuten Bank in April) sets the scene for autonomous mobilization of business strategies toward a growth stage. We think that a handful of players will drive growth in share and SBI Sumishin Net Bank is one of the main players. We may see stiffening competition between

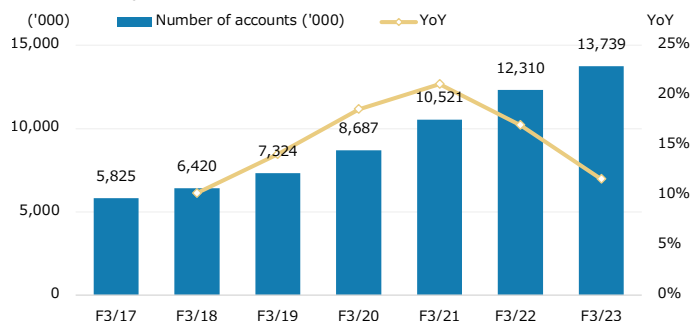
in-person and online in retail banking business.

Exhibit 50: Rakuten Bank earnings trend: We project average annual ordinary income growth of 12.9% over our forecast period (F3/24-26)



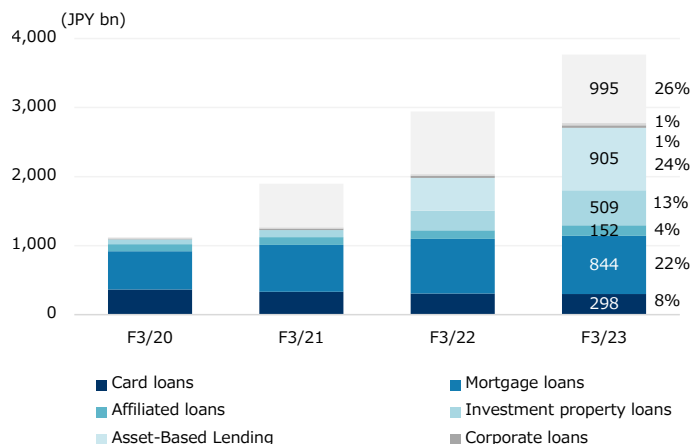
Source: Company data, Morgan Stanley Research. e=Morgan Stanley Research estimates.

Exhibit 52: Rakuten Bank: Number of accounts: With growth in the volume of accounts, the bank was able to increase revenue and profit by encouraging customers to use multiple services



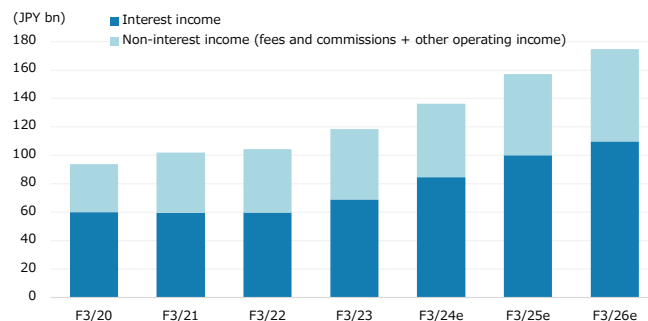
Source: Company data, Morgan Stanley Research

Exhibit 54: Rakuten Bank: Breakdown of loans (non-consolidated basis): Well balanced increase in lending across a diverse range of types



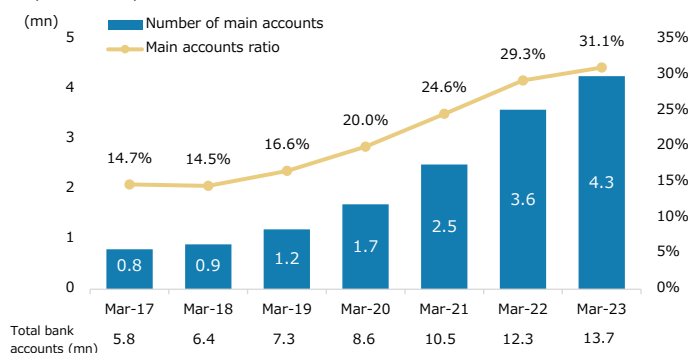
Source: Company data, Morgan Stanley Research

Exhibit 51: Rakuten Bank: NII about 60%, non-NII about 40%; we don't expect this structure to change



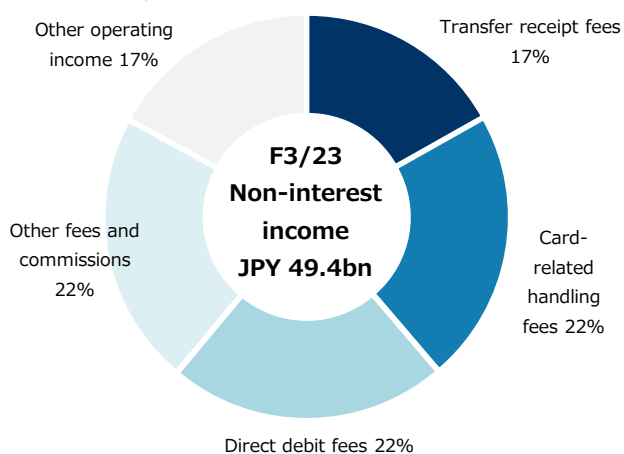
Source: Company data, Morgan Stanley Research. e=Morgan Stanley Research estimates.

Exhibit 53: Number of main accounts and main accounts ratio: Non-interest income also increased as growth in the number of accounts and progress in forming main accounts led to expanded uptake of services



Source: Company data, Morgan Stanley Research

Exhibit 55: Breakdown of non-interest income (non-consolidated)

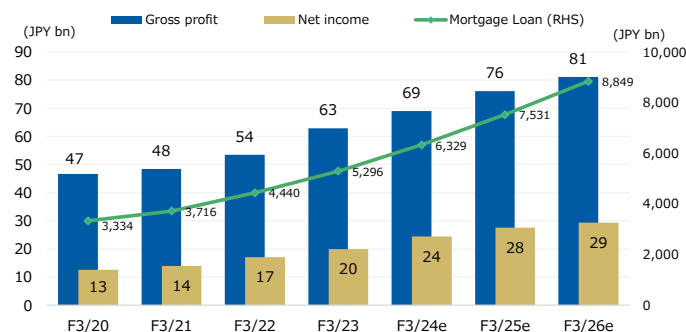


Source: Company data, Morgan Stanley Research.

How is SBI Sumishin Net Bank positioned?

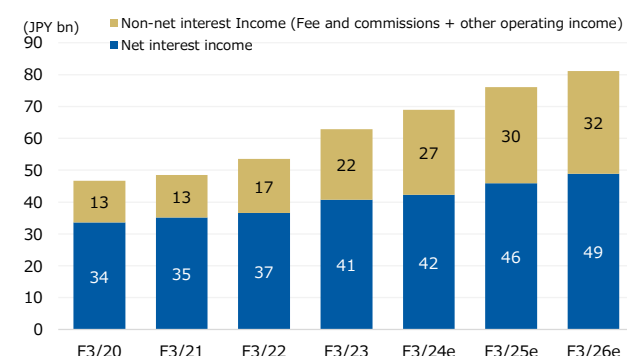
1. We expect average growth of +13.8% for consolidated NP over our forecasting period (F3/24-26). We believe the market already recognizes that operating leverage is higher than at conventional banks, and the appeal of the firm's ability to advance its business model.
2. However, we think further share price gains will hinge on valuation on the basis of SaaS-type indicators via rising profit exposure to BaaS that draws a line between SBI Sumishin and conventional banks.
3. Further, while the earnings structure is positively exposed to rising interest rates (more than 90% of the mortgage loans that make up much of the bank's lending are linked to the short-term prime rate), we also consider the relative magnitude of its interest rate sensitivity and risk-reward profile within our coverage.

Exhibit 56: SBI Sumishin Net Bank earnings: Has pursued growth to date by repeatedly pivoting into new growth areas



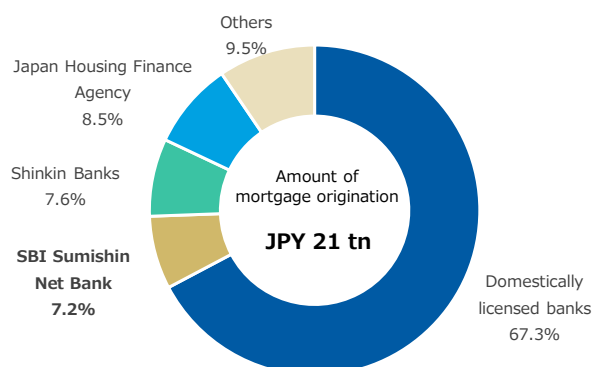
Source: Company data, Morgan Stanley Research. e=Morgan Stanley Research estimates

Exhibit 57: SBI Sumishin Net Bank: Net interest income (mainly mortgage loans) and non-net interest income (including BaaS business) are main sources of income



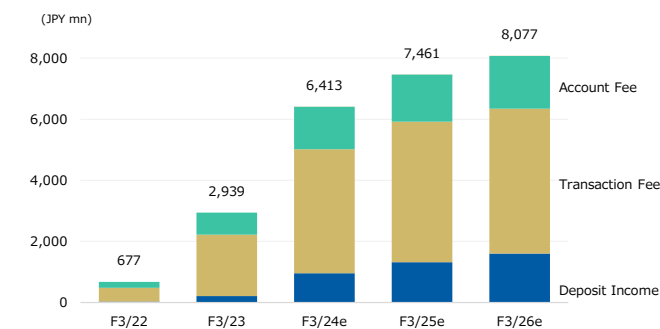
Source: Company data, Morgan Stanley Research. e=Morgan Stanley Research estimates

Exhibit 58: Mortgage loan origination market: SBI Sumishin Net Bank share is 7.2% (as of F3/23)



Source: Japan Housing Finance Agency, Morgan Stanley Research.

Exhibit 59: Total account earnings in BaaS alliance business: Increase of the profit exposure to BaaS is key

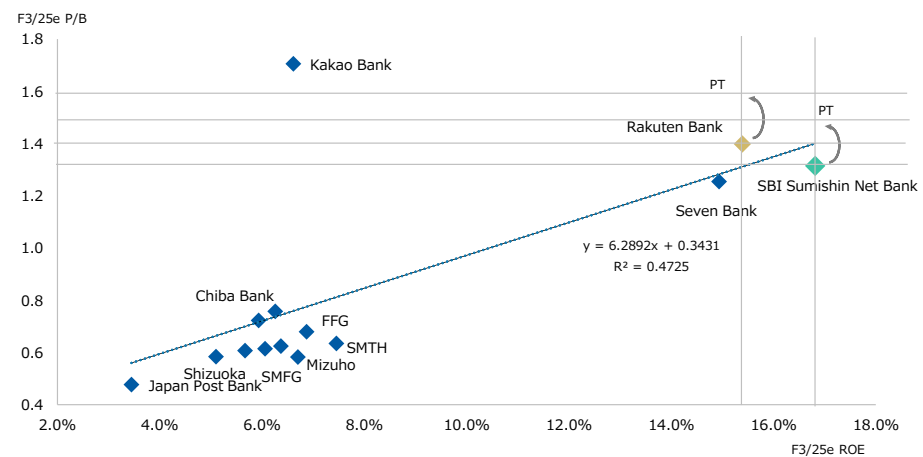


Source: Company data, Morgan Stanley Research, e=Morgan Stanley Research estimates.

Relevant research

SBI Sumishin Net Bank: Initiation: BaaS Holds Key to Earnings Growth (13 Nov 2023)

Exhibit 60: ROE-P/B regression analysis: We attach a multiple premium relative to traditional banks in our price targets for SBI Sumishin Net Bank and Rakuten Bank in light of strategic value and scarcity value



Source: Company data, Refinitiv, Morgan Stanley Research. e= Morgan Stanley Research estimates. Refinitiv consensus estimates for Seven Bank.

Exhibit 61: Main KPIs: Major digital banks & megabanks

	Digital Bank					Mega Bank		
	SBI Sumishin Net Bank	Rakuten Bank	PayPay Bank	GMO Aozora Net Bank	Minna Bank	SMBC	Mizuho Bank	MUFG Bank
Established	2007/9	2001/7	2000/10	2018/7	2021/5			
Parent companies/ Large shareholders	SMTB (50%) SBI Holdings (50%)	Rakuten Group, Inc. (100%)	Z Financial (46.57%) SMBC (46.57%) Fujitsu (5.26%)	Aozora Bank (91.24%) GMO Internet (4.38%) GMO Financial HD (4.38%)	Fukuoka Financial Group (FFG)(100%)	SMFG (100%)	Mizuho FG (100%)	MUFG (100%, Not covered)
Specialization	Deposits, loans (housing, consumer)	Smartphone apps, deposits, loans (housing, unsecured), joint services with Rakuten Card etc.	Online settlement, debit card, loans (personal unsecured loans, business loans)	Foreign currency, deposits, joint services with GMO Click Securities	Settlement, savings, consumer loans	Comprehensive finance	Comprehensive finance	Comprehensive finance
No. of accounts (thousands, Jun 23)	6,350	14,047	7,180	---	710			
No. of accounts (thousands, Mar 23)	6,146	13,739	6,960	---	600			
No. of accounts (thousands, Mar 22)	5,353	12,310	6,020	---	320	27,000.0	24,000.0	40,000.0
No. of accounts (thousands, Mar 21)	4,510	10,521	5,120	---	---			
YoY growth (Mar 22)	14.8%	11.6%	15.6%	---	87.5%			
Deposits (JPY bn, Jun 23)	8,587.0	9,488.0	1,688.8	---	24.4	161,003.6	114,967.3	205,737.9
Deposits (JPY bn, Mar 23)	7,977.7	9,129.8	1,666.9	413.5	22.8	149,948.9	119,182.8	203,348.6
Deposits (JPY bn, Mar 22)	7,115.8	7,765.3	1,461.8	293.5	5.9	141,015.2	112,531.4	194,249.3
Deposits (JPY bn, Mar 21)	6,293.9	5,765.5	1,216.9	174.8	---	134,685.6	111,043.3	193,113.1
YoY growth (Mar 22)	12.1%	17.6%	14.0%	---	289.2%	6.3%	5.9%	4.7%
Loans (JPY bn, Jun 23)	6,882.2	3,853.2	613.3	---	8.3	96,207.0	92,662.5	101,918.9
Loans (JPY bn, Mar 23)	6,606.5	3,780.5	624.5	191.1	7.0	94,307.4	90,346.1	99,399.9
Loans (JPY bn, Mar 22)	5,409.9	2,943.9	416.8	106.0	1.4	87,671.3	82,962.4	93,157.1
YoY growth (Mar 22)	22.1%	28.4%	49.8%	---	401.4%	7.6%	8.9%	6.7%
Loan to Deposit (Mar 23)	82.8%	41.4%	37.5%	46.2%	30.9%	62.9%	75.8%	48.9%
Loan to Deposit (Mar 22)	76.0%	37.9%	28.5%	36.1%	24.0%	62.2%	73.7%	48.0%
NP (JPY mn, F3/23)	19,932	27,692	4,482	-4,100	-4,336	805,842	555,527	1,116,496
NP (JPY mn, F3/22)	17,113	20,039	3,944	-3,977	-3,586	706,631	530,479	1,130,840
Earnings Structure (F3/23)								
Ratio of NII	68.6%	79.2%	81.2%	Mainly fee income	---	68.5%	62.2%	86.2%
Ratio of Fee Income	21.1%	10.0%	23.4%		---	26.8%	30.9%	29.0%
ROE (Mar 23)	13.4%	13.7%	4.6%	---	---	8.0%	6.6%	7.0%
OHR (Mar 23)	51.7%	50.1%	74.1%	---	---	61.5%	63.4%	64.6%
Client Attributes (Accounts, Mar 23)	Below 20 : 0% 20s : 11% 30s : 26% 40s : 28% 50s : 20% 60s : 10% 70s & above : 5% Male: 66%, Female: 34%	Below 20 : 3% 20s : 13% 30s : 24% 40s : 26% 50s : 21% 60s & above : 13% Male: 59%, Female: 41%	20s & below : 22% 30s : 22% 40s : 29% 50s : 19% 60s : 7% 70s & above : 2% Male: 70%, Female: 30%	---	Below 20 : 6% 20s : 40% 30s : 24% 40s : 16% 50s : 10% 60s & above : 4%	---	---	---
Regular deposit IR(Pre-tax)	0.001%	0.020%	0.001%	0.001%	0.001%	0.001%	0.001%	0.001%
Brokerage account linked preferential IR	0.010%	0.100%	---	0.110%	---	---	---	---
Other preferential IR	---	---	---	---	Saving account: 0.03%	---	---	---

Source: Company Data, Morgan Stanley Research

Note: For Megabanks, NP, ROE & OHR is consolidated basis, account numbers are our estimates. Deposits and Loans are parent, end of period data.

Korea

Joon Seok, Heewon Choi

Competitive landscape

Digital banks increasing their presence

There are currently three digital banks in operation in Korea, KakaoBank, KBank and TossBank. The first digital bank to come into operation was KBank, which launched in April 2017, and this was followed by KakaoBank's launch in June 2017 while TossBank was launched in October 2021. As of 3Q23, KKB is the largest digital bank with assets of W53tn followed by TossBank (W24.7tn) and KBank (W19.6tn).

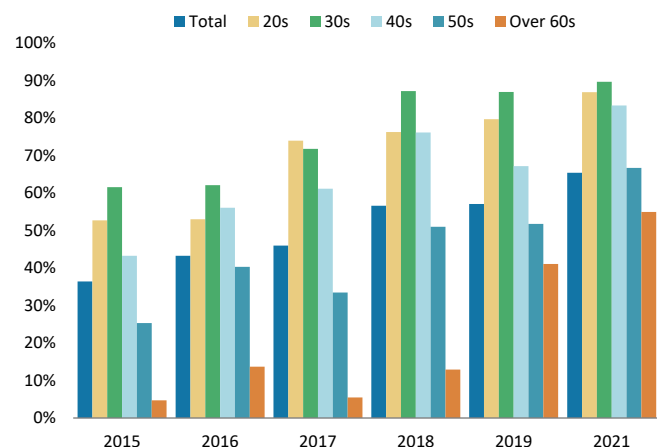
- All three digital banks kicked off banking businesses with unsecured loans, then proactively increased product offerings by launching secured loans such as mortgage loans and Jeonse loans to diversify loan portfolios and to manage asset quality.
- KakaoBank was the first to launch Jeonse loans in 2018, followed by KBank in 2021 and most recently Toss Bank followed suit by launching Jeonse loans in Sep 2023. Both KakaoBank and KBank offer mortgage loans, while TossBank hasn't yet launched mortgage products.

Exhibit 62: Korea: Digital Banks Comparison

	KakaoBank	Kbank	Toss Bank
Established	Jun. 2017	Apr. 2017	Apr. 2013
Major Shareholders	Kakao Corp. (27.2%) Korea Investment Securities (27.2%) + Others	BC Card (33.7%) Woori Bank (12.6%) + Others	Viva Republica (38.39%) E-LandWorld (10%) KBIZ (10%) Hanhwa Securities (9.1%) HFG (8.88%) Standard Chartered Korea (8.32%) + others
Specialization	Deposits, loans (unsecured consumer, jeonse), check cards	Deposits, loans (unsecured consumer), check cards	Deposits, loans (unsecured consumer, jeonse), credit cards
No. of accounts (deposits)	17.4mn	9mn	7mn
Deposits	W45.7tn	W17.2tn	W22.7tn
Loans	W37.1tn	W12.8tn	W11.2tn
ROE	6.37%	2.76%	-3.40%
Earnings Structure	NII: 82%	NII: 93%	NII: 111%
	Fees: 8%	Fees: 0.5%	Fees: -11%

Note: Data as of 3Q23. Source: Company Data, Morgan Stanley Research

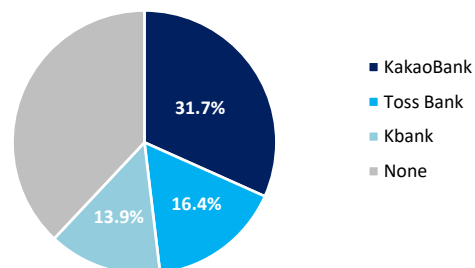
According to the BoK survey, overall penetration rate of mobile financial services (incorporating banking and payment services) has improved from 36% in 2015 to 65% in 2021 across all age groups, including those aged over 60 who have seen significant increase in mobile banking service usage through 2019 to 2021. Meanwhile, based on Realmeter survey in Nov 2022, KakaoBank was the most preferred among the three digital banks.

Exhibit 63: Korea: Mobile banking penetration rate by age group

Source: BoK, Morgan Stanley Research

Exhibit 64: Survey result on digital bank preference

RealMeter Poll: Digital Bank Preference (Nov 2022)



Source: Realmeter, Morgan Stanley Research

Incumbent banks vs. digital banks

KakaoBank's emergence has pushed banks to improve their mobile capabilities and to also move into mobile-based products in order to keep up with the digital banks. Commercial banks are now offering many loan products on mobile apps, and we expect this trend to continue as banks focus more on non-contact channels.

- Almost all the major banks are now selling retail loans through their mobile apps. In the case of Shinhan Bank, it has linked itself with Naver's real estate search platform, which is the largest in Korea (around 20% market share).

The top three banks have been pro-active in digitalization and also building up their mobile channels.

- All three major incumbent players have their own mobile banking apps that provide various financial services including banking, securities, payment and insurance. The apps are not much differentiated in terms of product offerings, and only have minor differences in credit limits or interest rate benefits available in mobile apps.
- In order to improve user experience, major commercial banks have been trying to aggregate separate apps launched by subsidiaries into a single 'super-app', encompassing multiple financial services - 'Star Banking' for KB Financial, 'SoL' for Shinhan Financial, and '1Q' for Hana Financial.

Exhibit 65: Korean commercial banks' mobile capability (as of Nov 2023)

Category	Description
KFG	Mobile Banking Apps Mobile Lending KB Star Banking / KB Pay (Payment) / KB Start Banking for Corporate Banking / Liv Next (for young generation) Unsecured retail loans: Various loans targeted towards salaried workers employed for more than one year, those who use KB for payroll accounts and etc. (Amount up to W350mn with 1~10 years of maturity) Mortgage Loans: Lending amount depends on the DSR and LTV. Maturities range between 10~50 years but only the borrowers aged under 34 are eligible for 40~ maturities. Group housing loans are also available through mobile banking apps. Jeonse: Up to 400mn with repayment at maturity (max. maturity at 25months) Autoloans: W40~60mn with amortized repayment of principal (1-10 years maturity), loans available for purchasing new or second-hand cars Loans using savings as collateral: Detailed loan amounts and procedures dependent on borrower
	Non-lending Services/Products Offered on Mobile - Daily banking activities (balance, transfer) - Time/Savings deposit products with various maturities and additional interest rate benefits available in mobile banking apps - Other types of financial products such as funds, insurance and FX deposits are also available in the apps - Provides customized services including Mydata which aggregates the information of accounts at Other financial institutions
	Mobile Banking Apps Mobile Lending Shinhan SOL (incorporates Shinhan SOL Bank / Pay / Securities / Life Insurance) Unsecured retail loans: Various loans targeted towards military personnel, employees and those with proof of income, small amount loan (pocket loan) of W5mn Housing Loans: Lending amount depends on the DSR and LTV. Maturities of up to 50 years. Jeonse Loans: Loans up to W500mn Autoloans: Up to W6mn with up to 10 years of maturity Loans using savings as collateral: Loan of up to 100% of the savings and applicable to less than W200mn
SFG	Non-lending Services/Products Offered on Mobile - Set to launch a unified banking app, Shinhan SOL, that includes all the financial services such as payment, securities and insurance - Daily banking activities (balance, transfer) - Time/Savings deposit products with various maturities and additional interest rate benefits available in mobile banking apps - Other types of financial products such as funds, insurance and FX deposits are also available in the apps - Provides customized services including Mydata which aggregates the information of accounts at Other financial institutions
	Fintech initiatives i) 'Shinhan Gold Transactions Certificate Service' launched in August, 2016 using block chain technology, ii) M folio - retail bank's first mobile roboadvisor for personal asset management, iii) Open banking interbank transfer allowed, improvement in UX (User Experience) and UI (User Interface), launched a new customized asset management app called 'My Asset' which provides customized recommendations based on personal data, iv) Shinhan Pay, a mobile payment solution, provides payment services and money transfers at several offline stores for Shinhan Card users. For Shinhan Bank users, it allows users to subscribe and activate mobile debit cards linked to bank accounts.
HFG	Mobile Banking Apps Mobile Lending Hana 1Q / Hana Pay Unsecured retail loans: Lending limits of up to W300mn Mortgage Loans: Lending amount depends on the DSR and LTV. Maturities of up to 50 years, with interest rates being fixed for the first 3 years Jeonse: Up to 400mn with repayment at maturity (max. maturity at 25months) Autoloans: Up to W500mn of loans provided for those who buy new / second-hand cars. Loans using saving as collateral: Maximum loan amount limited to W100mn SoHo Loans: Mobile services available for SoHo lending, guaranteed by public guarantee funds
	Non-lending Services/Products Offered on Mobile - Has a separate payment app (Hana Pay) from Hana Credit Card - Daily banking activities (balance, transfer) - Time/Savings deposit products with various maturities and additional interest rate benefits available in mobile banking apps - Other types of financial products such as funds, insurance and FX deposits are also available in the apps - Provides customized services including Mydata which aggregates the information of accounts at Other financial institutions
	Mobile Banking Apps Mobile Lending IBK Unsecured retail loans: Loans available for salaried workers. Loan limit up to W150mn Housing loans: Depending on different loanees, loan limit can be as high as W1bn for employed persons Loans using saving as collateral: Maximum loan amount limited to W100mn
IBK	Non-lending Services/Products Offered on Mobile - Daily banking activities (balance, transfer) - Time/Savings deposit products with various maturities and additional interest rate benefits available in mobile banking apps - Provides customized services including Mydata which aggregates the information of accounts at Other financial institutions

Source: Company Data, Morgan Stanley Research

Back in Apr 2021, Korean banking groups approached the financial authorities in hopes of launching their own digital banks amid the growing popularity of platforms such as KakaoBank (ETnews, Apr 1, 2021). While traditional banking groups have played a role in digital banks as consortium partners and investors, they have yet to establish their own online-only banking businesses.

- The Korea Federation of Banks, a body that represents commercial lenders, received a joint message from the banking groups, including the five major ones – KB, Shinhan, Hana, Woori, and NH – conveying their plans to enter the digital banking business if the financial authorities approve.

Based on various surveys, we find that digital banks generally are given higher ratings on user experience and also have higher usage rates compared to the incumbent banks.

- According to press (Safetimes, April 27), a survey carried out on satisfaction levels on financial apps shows that KKB and Toss are tied for first place, with the highest commercial bank app being Shinhan SOL.

Exhibit 66: Financial apps consumer satisfaction survey (1Q23)

App Name	Ranking	Satisfactory Score	vs. 2022
KakaoBank	Tied 1st	75.1	-0.9
Toss	Tied 1st	75.1	+0.2
Bank Salad	3	74.3	+0.2
Naver Pay	4	74.1	+2.1
Shinhan SOL	5	73.8	No change
NH Kok Bank	6	73.1	-0.9
Samsung Card	7	72.7	New entrant
Woori Won Card	8	72.3	New entrant
i-ONE Bank	9	72	-0.1
KB Bank	22	71.5	-0.3

Source: Media (Safetims, April 27, 2023). Based on results of survey published by Consumer Insight (Apr 27, 2023), Morgan Stanley Research

Overall regulatory and policy landscape

Regulatory framework and policy direction was initially favorable for digital banks...

The concept of developing digital banks in Korea emerged around 10 years ago, with the objective of digital banks being competitors to incumbent banks, taking advantage of changing customer attitudes toward e-commerce, a lower cost base (i.e., no branches), and also a more flexible business structure. While the Banking Act was the primary governing law for digital banks, there was a separate special Internet Bank Act that applied only to digital banks. Furthermore, digital banks, at launch, were given leeway in meeting capital and liquidity requirements. Hence, the market has largely understood that digital banks were being regulated under a different set of rules versus the incumbents.

- Digital banks in Korea are governed on the basis of the Banking Act and also the Act on Special Cases concerning Establishment and Operation of Internet-only Banks (Internet Bank Act).
- Note that KakaoBank is now conforming to the same liquidity requirements (i.e., liquidity coverage ratio, net stable funding ratio) of commercial banks, as the grace period ended as of 2019. For Basel3 capital requirements, a phase-by-phase full implementation was allowed, with full implementation to take place from 2023.

Exhibit 67: Key Differences on a Legal Standpoint Between Digital Banks and Commercial Banks

	Digital Banks	Commercial Banks
Ownership by non-financial entities	- Up to 34% - Previously, Banking Act stipulations were enforced but shareholding rules was amended in August 2018	- 4% Limit -10% in the case of including non-voting shares
Minimum capital requirement	- ₩25bn	- ₩100bn
Corporate lending business	- Digital banks cannot do corporate lending but under the SME Act they can provide credit to SMEs	- No restrictions

Source: Financial Services Commission (FSC), Financial Supervisory Service (FSS), Morgan Stanley Research

- Under the Banking Act, digital banks are able to provide the same services as the commercial banks. However, as they have only been in operation for a few years, and because the core banking system does not have all of the banking operation modules, digital banks will need to get the approval from the regulators prior to launching new products/services and for new business that they want to carry out.

Exhibit 68: Operations permitted under the Banking Act

	Digital Banks
Basic operations	- o Deposit-taking - o Loan origination - o FX transactions and services
Peripheral operations	- o Guarantees and underwriting of notes - o Payment services related e-Commerce - o Primary bank for local governments
Cross-selling related operations	- o Sale and intermediation of derivative products and derivative-linked products - o Underwriting and sale of government bonds - o Investment advisory - o Trust business - o Sale and intermediation of investment products - o Insurance agency business - o Credit card business

Source: FSC, FSS, Morgan Stanley Research

- While the two banking laws are the primary governing laws for digital banks, they have to adhere to other laws, including:
 - The Financial Investment Services and Capital Markets Act, as banks operate in financial investment businesses under the Banking Act.
 - Act on Corporate Governance of Financial Institutions, which governs the structure and management of financial companies.
 - Financial Consumer Protection Act, which is a set of rules to protect consumers from incurring losses or damages from financial products/services due to unlawful activities.

Policy environment turned less conducive for loan growth since 2H23

The regulators have been introducing new rules and also guiding lenders to curb retail lending since last year, but with results being mixed, the regulators are taking a stronger stance. What is different this time around is that the financial authorities are not just focusing on banks but on other second-tier lenders, as well, including card issuers, insurers, credit coops and savings banks.

We had expected KKB to be a beneficiary as the financial authorities had been pushing to change the competitive dynamics of the banking system, and we believed that digital banks will have a bigger role in relation to this policy path. Moreover, real estate lending rules have been eased in order to support the weakening residential property market, and retail lending demand seemed to be slightly better through 1H23, compared to last year.

However, recent regulatory conditions are not conducive for loan growth, as regulators have announced in August that they will closely monitor the household loan growth to manage credit risks with household credit-to-GDP ratio remaining well above 100%. In order to control the pace of household loan growth, regulators would monitor the digital banks' lending practices and asset quality management for their mortgage lending while clamping down on 50-year mortgages provided by commercial banks and insurers.

[S. Korea Financials: Clamping Down on Mortgage Lending and 50-year Mortgages \(24 Aug 2023\)](#)

For commercial banks, the key story that needs to play out is for a more favorable policy environment and for shareholder returns to show meaningful improvement, although this scenario looks uncertain at this time.

- Financial authorities are focused on boosting banks' absorption capacity. Hence, they plan to raise the minimum CET1 ratio and also have banks continue to make

pre-emptive provisions. We believe KKB will be less affected than peers because KKB has a very high CET1 ratio and growth is more important to KKB than boosting shareholder returns.

S. Korea Banks: Policy Storm or a Squall? (8 Mar 2023)

Banking service scope and ecosystem

Digital banks and commercial banks have differences in business scope with large corporate lending not allowed for digital banks. On the retail lending business, all commercial banks and digital banks are competing in the unsecured retail lending space while for mortgage lending KKB and KBank are relatively new players in the market.

Exhibit 69: Business scope comparison

																					
		BASIC ACTIVITIES			DEPOSITS		RETAIL LENDING			CORPORATE LENDING			PERIPHERAL/CROSS-SELLING SERVICES								
CATEGORY	Service Provider	View activity & balances	Fund transfer	FX remittance	Liberal Deposits	Time Deposits	Retail unsecured	Mortgage/Home equity	Jeonse	Mid/sub-prime	Large corporate	SME	SoHo	Payment services	Primary bank for local gov'ts	Sale/intermediation of loans	Sale/intermediation of derivative products	Investment advisory	Sale/intermediation of investment products	Insurance agency business	Credit card
 DIGITAL BANKS	KakaoBank	●	●	◐	●	●	●	●	●	●		○	●	●	○	●	○	○	○	◐	◐
	Kbank	●	●	◐	●	●		◐	◐	●		○	●	●	○	○	○	○	○	○	○
 COMMERCIAL BANKS	HFG (Bank)	●	●	◐	●	●	●	●	●	◐	●	●	●	●	○	●	●	◐	◐	◐	●
	KFG (Bank)	●	●	●	●	●	●	●	●	◐	●	●	●	●	◐	●	●	◐	◐	◐	●
	SFG (Bank)	●	●	●	●	●	●	●	●	◐	●	●	●	●	●	●	●	◐	◐	◐	●
 DIGITAL PLATFORMS	KakaoPay	●	●				◐	◐	◐	◐				●		◐			◐		
	NaverPay	●	●				◐	◐	◐	◐				●		◐					
	TOSS	●	●				◐	◐	◐	◐				●		◐					

Source: Company data, Morgan Stanley Research

KKB has relatively effective ecosystem owing to its linkage to KakaoTalk and also the wide partnership with many card issuers and brokerage firms. KKB's group account product cannot be easily emulated by other banks as it requires a connection to a talk platform such as KakaoTalk, which is the dominant talk app used by Korean consumers.

Key nuances that matter

Need to follow real estate market trends as this is where digital banks are growing

For the commercial banks, around 55-60% of the loans they originate are linked to the residential market including mortgages, jeonse loans, collective loans and real estate leasing business loans. The state of the residential property market is key to not only growth but also asset quality.

Initially, digital banks were focused on the unsecured retail lending business but since 2021 when KKB entered into the housing loan market with jeonse loans, digital banks have expanded their product offering into jeonse and mortgage loans. Until 1H23, digital banks have been increasing their exposures to mortgage loans to achieve higher loan growth and to diversify their loan portfolios in view of slight recovery in the housing market.

Exhibit 70: Digital bank market share

Loan Balance (Wtn)	2017	2018	2019	2020	2021	2022	2Q23
Bank System							
Total Loans	1,506.9	1,596.5	1,696.2	1,894.2	2,050.2	2,158.7	2,163.9
Corporate	817.0	857.3	906.3	1,020.3	1,113.5	1,221.6	1,240.1
Large Corp	161.6	161.1	159.6	184.1	191.3	228.9	237.0
SMEs	655.4	696.2	746.7	836.3	922.2	992.7	1,003.2
SoHo	288.9	313.8	338.5	386.1	423.0	442.7	444.2
Retail	659.5	712.1	766.8	849.2	909.5	902.2	890.1
Housing	463.7	493.6	533.2	583.2	628.8	644.3	642.2
Unsecured	195.8	218.5	233.6	266.0	280.7	257.8	247.8
Digital Banks							
Total Loans	5.5	10.3	16.3	23.3	33.5	47.3	50.5
Corporate						1.5	2.3
Large Corp							
SMEs						1.5	2.3
SoHo						1.5	2.3
Retail	5.5	10.3	16.3	23.3	33.5	45.8	48.2
Housing	-	0.8	2.4	4.7	10.3	15.6	16.7
Unsecured	5.5	9.5	13.9	18.6	23.2	30.2	31.5
Digital Bank MS (based on bank system, %)							
Total Loans	0.4%	0.6%	1.0%	1.2%	1.6%	2.2%	2.3%
Corporate						0.1%	0.2%
Large Corp							
SMEs						0.2%	0.2%
SoHo						0.3%	0.5%
Retail	0.8%	1.5%	2.1%	2.7%	3.7%	5.1%	5.4%
Housing	0.0%	0.2%	0.5%	0.8%	1.6%	2.4%	2.6%
<i>based on system loan incl. non-banks</i>	0.0%	0.1%	0.3%	0.5%	1.0%	1.5%	1.8%
Unsecured	2.8%	4.4%	5.9%	7.0%	8.3%	11.7%	12.7%
<i>based on system loan incl. non-banks</i>	0.9%	1.5%	2.1%	2.6%	3.0%	4.1%	6.7%

Source: FSS, Morgan Stanley Research

However, we believe that 1H23 recovery was short-lived, and expect the housing market to be on an extended down-cycle with an inflection coming at the earliest in 2H24. This implies less room for further system mortgage loan growth in the near-term. Therefore gaining more market share within the mortgage loan market amid weakening mortgage demand should be key to securing loan growth. Also, capturing refinancing demand for existing mortgages should bring further opportunities for loan growth.

Given this backdrop, KKB has been trying to gain market share in mortgage lending through (1) competitive pricing by leveraging its lower funding costs; and (2) mortgage refinancing: (i) KKB is able to keep its pricing competitiveness for lending given its lower funding costs supported by group deposits, while (ii) the company has been proactively penetrating customers with demand of refinancing its existing mortgages and jeonse loans taken out from other banks.

- As of 3Q23, around 57% of total deposits in KKB were low-cost deposits (vs. average 38% for other banks), leading to lower funding costs. KKB has been able to fund at lower costs owing to its group deposit products.
- More than half (51%) of KKB's mortgage loan balance came from refinancing as of 3Q23. KKB has recently launched refinancing service for jeonse loans as well.

However, as mentioned above, given the tighter regulatory monitoring on retail lending (particularly the mortgage lending by digital banks), not only the housing market trend

but the policy path remain to be seen for digital banks' loan growth. Also, we expect increasing competition among banks in the mortgage refinancing market as the current online refinancing loan platform is set to be expanded to include mortgage loans by the end of this year.

- The online one-stop refinancing platform was launched in May 2023 to allow borrowers to easily compare mortgage loan products and to refinance their loans faster. Currently the platform provides the service only for the unsecured loans, but will soon expand the offering into mortgage loans as early as December 2023 or early next year.

South Korea: Winter for Property Market; Credit Tightening (12 Nov 2023)

Have to meet mid-prime lending targets for unsecured retail book

In order to increase mid-prime lending, financial regulators introduced mid-prime lending targets for digital banks since May 2021. Digital banks are required to meet the year-end targets for the proportion of mid-prime loans – the loans provided to borrowers in the bottom 50% credit ratings – within their outstanding loan balance. Starting from 2021, digital banks have set their mid-prime lending targets for the next three years (through 2021 to 2023) so that the end-2023 level should be above 30% for all three digital banks.

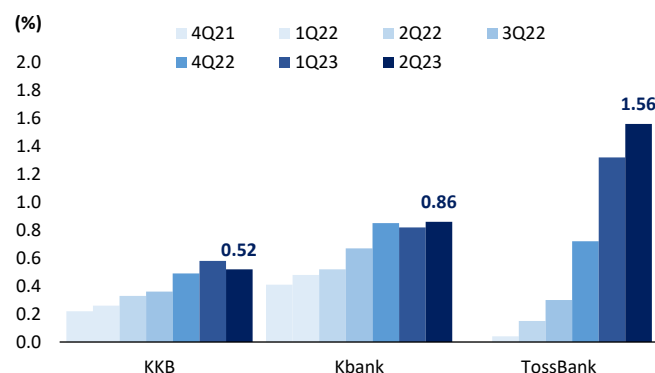
- Digital banks should make quarterly disclosures on their mid-prime loan proportions with detailed plans to achieve the annual target. Failure to meet the year-end target would not result in legal penalties but the regulators would consider the level of progress for mid-prime lending when approving new license or business for digital banks.

As of 3Q23, all three digital banks are running below the year-end target by 1.3~9.5%p, with TossBank presenting the largest gap at 9.5%p. However, it should be tough for digital banks to ramp up the mid-prime lending given the higher delinquencies, facing a dilemma between meeting the policy requirements and managing asset quality.

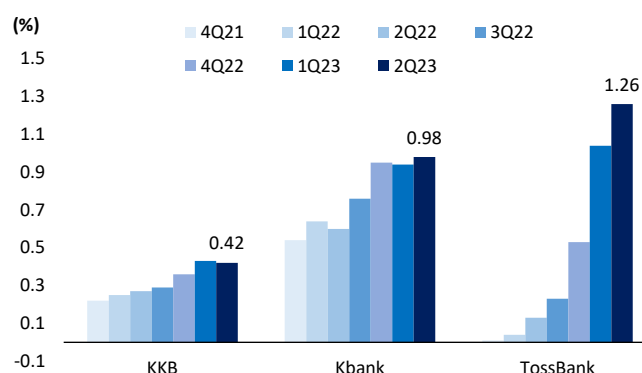
Exhibit 71: S.Korea digital banks: Mid-prime lending trend

(%)		4Q21	4Q22	1Q23	2Q23	3Q23	4Q23	Gap between 3Q23 and 23-end target
KKB	Actual	17.0	25.4	25.7	27.7	28.7		1.30
	Year-end Target	20.8	25.0				30.0	
Kbank	Actual	16.6	25.1	23.9	24.0	26.5		5.50
	Year-end Target	21.5	25.0				32.0	
TossBank	Actual	23.9	40.4	42.1	38.5	34.5		9.54
	Year-end Target	34.9	42.0				44.0	

Source: Company Data, Morgan Stanley Research

Exhibit 72: Delinquency ratio trend

Source: FISIS, Morgan Stanley Research

Exhibit 73: NPL ratio trend

Source: FISIS, Morgan Stanley Research

How is KakaoBank positioned?

KKB has the motor, mechanics and mettle to outperform peers

Compared to its local digital banking peers, KKB has better breadth of deposit and lending products with particular strength in low-cost deposit-taking via its group account product, which has yet to be emulated by another bank.

- KKB provides a near full breadth of retail loan products ranging from unsecured retail loans, jeonse loans, mortgage/home equity loans, and unsecured SoHo loans. Its competitor digital banks are missing some of the loan products KKB provides.
- KKB has strong low-cost deposit-taking capacity owing to its innovative deposit products – namely, its group account product.
- KKB's loan pricing tends to be more competitive than that of its peers owing to its lower funding cost (including free fund effect) and also its ability to cross-sell.

Furthermore, KKB has a stronger platform business with the scope of products and scale of operations being larger. The factors that set KKB apart from its peers are its "mini" platform targeting teens (14-year-olds and above), advertising, and also its soon-to-be-set-up online WM business.

On capital structure, KKB has a significantly higher CET1 ratio, meaning it has more than sufficient capital to continue to grow for many years from here.

Exhibit 74: Digital banks: Key financial metrics comparison

	(%)	2020	2021	2022	2Q23
ROA	Kakao Bank	0.47	0.65	0.67	0.81
	KBank	Loss	0.18	0.55	0.27
	Toss Bank	na	Loss	Loss	Loss
ROE	KKB	5.67	4.87	4.69	6.33
	KBank	Loss	2.05	4.74	2.74
	TossBank	na	Loss	Loss	Loss
CET-1	KKB	19.55	34.92	35.80	30.90
	KBank	17.27	17.31	12.74	12.35
	TossBank	na	na	na	na

Source: Company Data, FSS, Morgan Stanley Research

(1) In a higher rate environment, funding franchise and cost controls matter more:

Among the Korean banks, KKB has the lowest funding cost thanks to its strong deposit franchise, which has inherent advantages that cannot be emulated (i.e., group account). KKB continues to show stable growth in its group account balance and provide customers with innovative products, such as group account, 26-week installment account, and the new recording account.

KKB will have to continue to incur more SG&A expenses as it expands its business scope and scale over the next few years. However, the heavy spending appears to have been done, and incremental increases should be lower than we previously expected.

Exhibit 75: KKB vs. peers: funding costs

	2018	2019	2020	2021	2022	1H23
Funding Costs						
Incumbents						
KB Kookmin Bank	1.30%	1.36%	1.00%	0.70%	1.26%	2.46%
Shinhan Bank	1.41%	1.47%	1.09%	0.80%	1.44%	2.73%
KEB Hana Bank	1.49%	1.57%	1.16%	0.85%	1.57%	2.88%
Digital Banks						
KakaoBank	1.52%	1.61%	1.04%	0.79%	1.23%	2.36%
Kbank	1.90%	2.09%	1.26%	0.53%	1.24%	2.67%
TossBank				2.07%	2.22%	2.70%

Note: Based on average deposit yields. Source: FISIS, Company Data, Morgan Stanley Research

Exhibit 76: KKB vs. peers: Cost-to-income ratio

	2018	2019	2020	2021	2022	1H23
Cost-to-Income Ratio						
Incumbents						
KB Kookmin Bank	54.2%	53.1%	53.2%	51.0%	49.3%	45.6%
Shinhan Bank	47.0%	48.0%	48.4%	48.9%	44.4%	44.5%
KEB Hana Bank	50.3%	49.2%	47.7%	45.0%	40.9%	37.1%
Digital Banks						
KakaoBank	104.3%	74.7%	51.4%	45.1%	42.6%	39.8%
Kbank	424.3%	382.7%	326.1%	61.3%	37.5%	32.9%
TossBank	na	na	na	Loss	173.1%	59.9%

Source: FISIS, Morgan Stanley Research

(2) Better positioned to grow as KKB expands lending scope and has price

competitiveness: In 2022, the bank realized the lowest annual loan growth since its inception, reaching only 7.8% Y/Y, which is not that much higher than banking system growth of 5.0% Y/Y. KKB showed improved growth in 1Q23 (4.4% Q/Q), with prime unsecured lending being rekindled and mortgage loan growth picking up.

We believe KKB should be able to reach beyond its mid-teen loan growth target for 2023 and show strong growth from 2024, considering:

1. Mortgage loan TAM has been expanded – we forecast transaction volumes to be slightly higher in 2023 vs. 2022.
2. KKB's book has a lower amortizing proportion vs. commercial banks.
3. Jeonse demand is declining, but landlord collateralized lending demand to repay tenants is rising.
4. SoHo book should increase gradually, especially when secured SoHo loan product is launched.

Exhibit 77: KKB: Loan book portfolio

(Wbn)	2020	2021	2022	3Q23	Mse		
					2023e	2024e	2025e
Total	20,313	25,862	27,889	37,062	39,455	50,179	66,085
Housing + SoHo Loans	4,486	9,145	13,386	20,650	22,715	32,059	46,084
Unsecured Loans	15,827	16,717	14,503	16,412	16,740	18,120	20,001
Y/Y Growth (%)							
Total	36.5%	27.3%	7.8%	35.0%	41.5%	27.2%	31.7%
Housing + SoHo Loans	85.5%	103.9%	46.4%	63.6%	69.7%	41.1%	43.7%
Unsecured Loans	27.0%	5.6%	-13.2%	10.6%	15.4%	8.2%	10.4%
Mix (as % of total loans)							
Housing + SoHo Loans	22.1%	35.4%	48.0%	55.7%	57.6%	63.9%	69.7%
Unsecured Loans	77.9%	64.6%	52.0%	44.3%	42.4%	36.1%	30.3%

Source: Company Data, Morgan Stanley Research estimates (e)

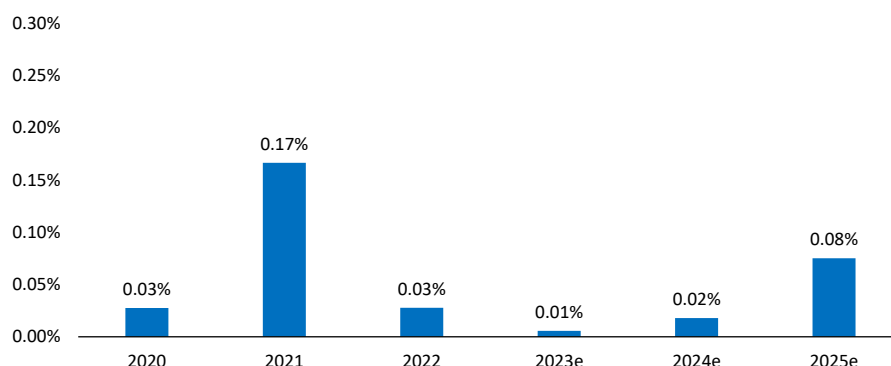
Exhibit 78: Korea Digital Banks and KKB Long-term Loan Market Share Forecast

(W in tn)	2021	2022	2023e	2024e	2025e	2026e	2027e	2028e
Unsecured retail								
System	311	307	297	312	331	349	368	388
Digital Banks Total	21	30	36	41	45	51	56	60
- Market share	6.9%	9.9%	12.0%	13.1%	13.7%	14.6%	15.2%	15.5%
KakaoBank	15	15	17	21	23	26	29	31
- Market share	4.9%	4.7%	5.8%	6.6%	6.9%	7.4%	7.8%	8.0%
Housing retail								
System	735	755	767	794	825	859	895	932
Digital Banks Total	10	16	26	34	43	51	59	65
- Market share	0.9%	1.4%	2.3%	2.9%	3.5%	4.0%	4.4%	4.7%
KakaoBank	9	13	21	27	34	38	43	47
- Market share	1.2%	1.8%	2.8%	3.4%	4.1%	4.5%	4.8%	5.1%
SoHo								
System	909	1,020	1,057	1,105	1,165	1,223	1,284	1,348
Digital Banks Total		1	4	9	16	22	28	33
- Market share		0.1%	0.4%	0.8%	1.4%	1.8%	2.2%	2.5%
KakaoBank		0	1	3	9	13	17	20
- Market share		0.0%	0.1%	0.3%	0.8%	1.1%	1.3%	1.5%
Total								
System	1,955	2,081	2,120	2,211	2,321	2,431	2,547	2,668
Digital Banks Total	32	47	66	84	104	124	143	159
- Market share	1.6%	2.3%	3.1%	3.8%	4.5%	5.1%	5.6%	5.9%
KakaoBank	25	28	40	51	66	77	88	98
- Market share	1.3%	1.3%	1.9%	2.3%	2.8%	3.2%	3.5%	3.7%

Source: FISIS, Morgan Stanley estimates

(3) Proactive push into new businesses accentuates KKB's standing as a platform: KKB

is a bank, but it has better motor and mechanics to diversify its platform business. This is being illustrated by its growing loan-referral business, the successful penetration of "mini" (its sub-brand for teens) in the teen customer segment, and a newly launched advertising business. Moreover, KKB will also be adding an investment product sales function when it receives its full license – expected sometime in 2023 – and we believe KKB will become another major player in the online wealth management market.

Exhibit 79: KKB: Net fee income trend and forecasts

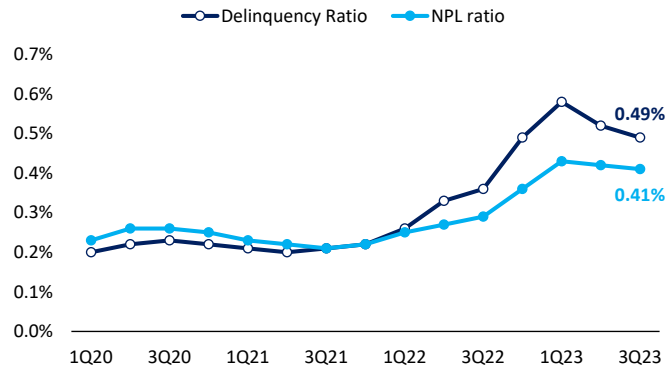
Source: Company Data, Morgan Stanley Research estimates (e)

Credit risks become more severe due to macro slowdown but manageable

Credit costs have been rising as the mid-prime book has been growing and the tougher macro conditions for retail borrowers are leading to rising delinquencies. Management

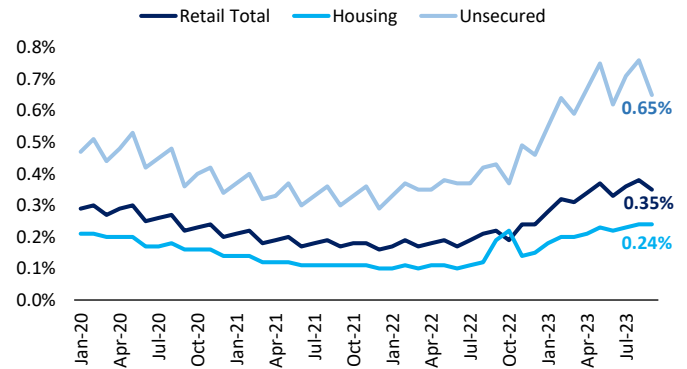
expects delinquencies to be on an upward trajectory throughout 2023 and for credit costs to be higher.

Exhibit 80: KKB: Delinquency and NPL ratio trend



Source: Company Data, Morgan Stanley Research

Exhibit 81: Korean banks: System retail delinquency trend



Source: FSS, Morgan Stanley Research

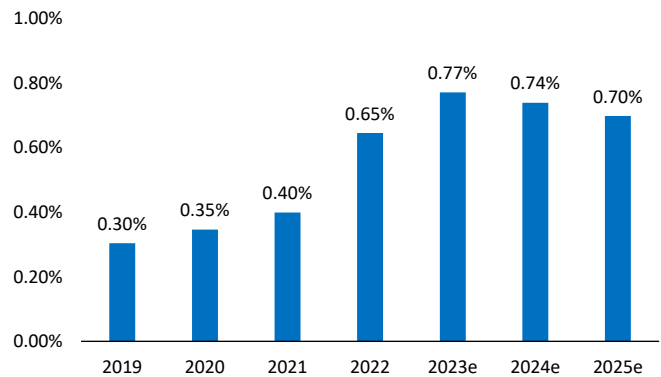
If macro conditions become tougher and the real estate market becomes weaker than we currently expect, credit costs could come in higher.

Exhibit 82: KKB: Quarterly credit cost ratio trend



Source: Company Data, Morgan Stanley Research

Exhibit 83: KKB: Credit cost trend and forecasts



Source: Company Data, Morgan Stanley Research estimates (e)

Relevant research

[Kakao Bank Corp: Proving Its Mettle \(15 May 2023\)](#)

[Kakao Bank Corp: Mechanics, Motor & Mettle to Lead Digital Banking \(11 Oct 2021\): Initiation Report](#)

Latin America (Brazil)

Jorge Kuri, Jorge Echevarria, Andrew Geraghty, & Victoria Kang

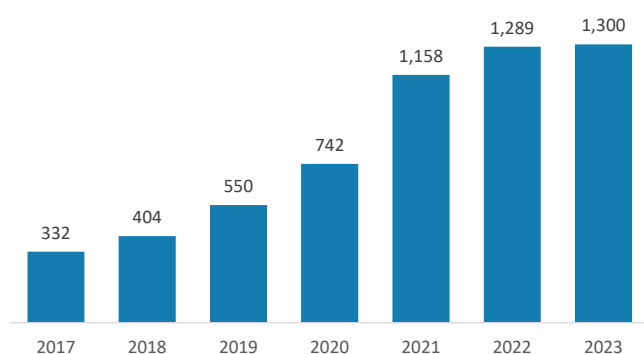
Competitive landscape

Digital banks in operation

The number of fintech lenders that operate in Brazil is quite robust today, having grown notably in the last five years. Of the 1,300 fintech companies operating today, 231 are considered digital lenders, according to Distrito. And while we do not have the full details of the exact product offerings for each one, we know that good deal provide a broad range of products and services — including checking/savings accounts, debit cards, credit cards, personal/payroll loans, and more — evidently putting them in the digital bank category. Payment companies also account for a good deal of the fintechs in the country, at 171. Back office, technology, and digital services, are the other key sub-sectors where fintechs are operating.

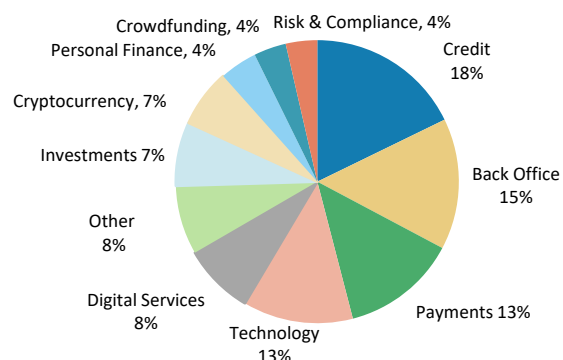
But the elevated interest rate environment has seemingly stalled the stellar growth of fintech in Brazil, with the segment accelerating rapidly during the Covid Pandemic. However, with the Brazilian Central Bank currently cutting rates (as of Aug-23), we would not be surprised to see a re-acceleration of fintech growth over coming years. Note: back in 2021, there were already nearly 1,200 fintech companies, of which 157 were considered digital lenders and 174 payments companies (we suspect many of the payment names have expanded into lending).

Exhibit 84: Total number of fintech companies in Brazil



Source: Distrito, Morgan Stanley Research

Exhibit 85: Breakdown of fintech companies in Brazil



Source: Distrito, Morgan Stanley Research

Of the various digital banks operating in Brazil, two pure-play names are public: Nubank and Inter. Many digital players have been able to amass large client bases, with a number of them boasting >20 million users. This includes digital native players (like Nubank and C6), offerings launched by incumbents (like iti and Next), along with traditional players that have pivoted into a fully digital approach (like Inter). However, many Brazilians have 2-3, or more, bank accounts. Hence, the challenge for fintechs has been winning "principality" of the banking relationship and monetizing these large client bases. Nubank stands out in this regard, as it is well underway in building a sizable and profitable operation in Brazil, reaching more than 90 million clients and a profit run rate

close to US\$1 billion. But (in our view) as a result of said challenges, many have yet to go public given the market's current focus on profitability over growth. Currently, we cover the two publicly traded pure-play digital banks in Brazil: Nubank (NU.N) and Inter (INTR.O).

Below please find a summary list of some of the largest and most relevant digital banking players in Brazil.

Exhibit 86: Key digital banking players – Summary & value proposition

Company	Summary / Value Proposition	Total clients	Relevant Ownership
	- Tech-forward company offering financial products and services such as personal loans, credit cards, and insurance, among others. It's main focus is payroll loans. - The company complements a full digital offering with close to 900 small, cost-efficient "hubs" across Brazil.	> 6 mm	-
	- In 2019, the company focused on disbursing payroll and auto loans through in-person, third-party channels. - Today, it has a broad product mix (including credit cards and personal loans among others) that is completely available online.	>25 mm	BTG Pactual holds a 74% stake of Banco Pan
	- Leverages tech expertise to offer complex products at a low cost. - Offers a wide range of products digitally, such as intl. debit and credit cards, investments, SME accounts, and more.	~25 mm	JP Morgan holds a 46% stake of C6 Bank
	- Bradesco's fully digital banking platform that offers a simplified, bureaucracy-free, and innovative user experience. - Offers digital account, credit card, personal loans, PIX, cashback, and others. Integrated the digital wallet Bitz.	~7 mm	Fully owned by Bradesco as of 2021
	- Completely free and full-service digital account made possible by branchless distribution. "Super App" with banking and non-financial services enables higher client engagement (includes a marketplace, insurance distribution, and more).	30 mm	-
	- Itaú's free digital account for clients which offers products and services in a simplified manner. - Translate lower cost to serve into lower prices for digital banking products to attract younger consumers.	~24 mm	Launched by Itaú
	- Including underserved, low to middle income individuals into the financial system with a free, simple, and fully-digital account offering. - Offers personal loans, credit cards, investment products and a range of other digital banking solutions.	~26 mm	-
	- Provide younger, hyper-connected consumers with a superior client experience. - Offer tailored product and financial management recommendations as consumers move along in their lifecycles. - Is becoming a segment of Bradesco, using synergy of the back-office and broadening product scope.	~15 mm	Launched by Bradesco
	Providing access to a "global financial life" with an international debit card, transfers, and foreign exchange transactions.	1-2 mm	-
	- Self-reinforcing business model built on four core principles: advanced proprietary technology, customer centric culture, extraordinary customer experience, and utilization / optimization of data. - Credit cards, account/debit card, personal loans, payroll loans, BNPL, insurance, investments, and more.	90 mm	-
	- Disrupt and democratize financial services in Brazil by providing an end-to-end digital banking ecosystem that is safe, affordable, simple and mobile-first for both merchants and consumers. - Ideally, complements the merchant acquiring services for SME customers.	30 mm	Launched by PagSeguro
	- Originally gaining popularity as a digital wallet, it is now a two sided ecosystem offering a wide range of products and services to both consumers and sellers. - Will be integrating with digital bank Banco Original which focuses on commercial and agribusiness lending.	>70 mm (total) >30 mm (DAU)	-
	- Meet the financial needs of under-banked individuals with a range of basic products and simple user experience. - Offers products for individuals and corporate entities such as a fully digital account, credit cards, personal loans, and more.	2 mm	Launched by Santander

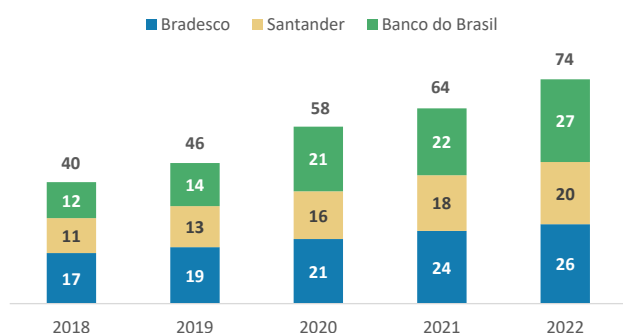
Source: Company Data, Morgan Stanley Research. Note: Nubank clients are as of 3Q23 — the company had 90 mm total clients including Mexico and Colombia. Also, a handful of the total client figures are somewhat dated (like Next as of 4Q22) and may under/over-represent the actual number.

Incumbent banks vs. digital banks

Tougher competition has pushed incumbent banks to improve user experience, lower prices, and adopt digital at a fast pace. Faced with the rapid proliferation of fintech companies (as shown above), the large banks have been pushed to invest millions to

develop better digital capabilities and become more flexible, agile, and client-centric. Indeed, incumbents have been tapping their deep pockets to close the gap vs. challengers, co-opting and/or acquiring successful parts of the fintech offering. In our view, most banks are doing a good job of closing the gap with fintech players in digital, customer service, and user experience. And cost savings via increasing digitalization are gradually allowing them to compete on price. We think Itaú understood the need to evolve earlier more than most and has made the most impressive strides thus far — for an overview of its business transformation please see our dispatch from last year's investor day: [Key Takeaways from Itaú Day](#).

Exhibit 87: Digital active clients by incumbent bank (MM)



Source: Company Data, Morgan Stanley Research. Note: This excludes clients of digital bank offerings like Next at Bradesco. Itaú stopped disclosing the figure.

The digital push by incumbent banks and shift in consumer behavior has, in turn, resulted in a rapid ramp-up of digital clients.

The total number of digital active clients at the incumbent banks has grown quite rapidly in recent years. As of 2021, Itaú led the pack with 28 million digital active clients; however, the company no longer releases this figure. In addition, as shown above Bradesco and Itaú's digital ventures — Next and iti — have already amassed significant scale, at about 15 and 24 million clients respectively (many of which were not account holders at these banks before; 78% at Next).

Nevertheless, incumbents have been somewhat slow to monetize new, digital-only customers. Disclosure for the digital ventures of incumbents is quite limited, but commentary from the banks indicates that most have been hesitant to quickly push monetization through

lending. Itaú, for example, has said it wants to better understand these relatively new customer cohorts, and thoroughly test its underwriting models before lending at scale. Bradesco has also slowed down credit growth for low-income consumers after idiosyncratic delinquency issues arose this past year, and we suspect this includes lending for many of the clients at Next and Digio. We know, for example, that as of Dec-22, Digio's total loan portfolio only amounted to R\$3 billion (i.e., less than US\$1 billion). It certainly seems as though incumbents still have work to do when it comes to successfully underwriting risk with younger, lower income, digital-first customers — something that Nubank has a clear track-record of executing effectively.

Overall regulatory and policy landscape

In general, the regulatory environment has been supportive of digital/challenger banks, as they increase competition and broaden financial inclusion within the system. Indeed, the regulator has implemented several initiatives over the past handful of years to eliminate barriers to entry for challengers and make the banking landscape more competitive, some of which include:

- **Lock-In on Payrolls.** In 2018, the regulator eliminated the long-standing lock-in on payrolls, which had historically allowed the banks to buy thousands of individual clients by negotiating payroll agreements with their employers, rather than by offering attractive products and services that were appealing to consumers. This change allowed millions of Brazilian workers to start freely choosing which bank or fintech company (even those without full banking licenses) receives their payroll

deposits.

- **Positive Credit Registry.** In early 2020, Brazilian banks started being required to report positive data to the credit bureaus every 10 days at no cost, creating the first large-scale positive credit registry in the country. This shift in regulation has helped dismantle one of the key pillars supporting the traditional banking oligopoly in Brazil — the lack of a proper credit bureau. Greater transfer of information within the system allows challengers to better price for risk, which in turn results in lower interest rates on loans, higher approval rates, lower defaults, more sophisticated products, and ultimately, higher credit penetration over time.
- **Open Finance.** The central bank is currently in the process of implementing open finance through four incremental phases. Phase 1, which consists of sharing data with participating institutions (i.e., characteristics and prices of retail banking products), was implemented in Feb-21. Phase 2, which consists of sharing consumer data, was implemented between Aug and Oct-21. Phase 3, which consists of payment initiation transactions, including wire transfers, PIX (which in and of itself has accelerated digital banking), boletos, and more, kicked off in Oct-21. And Phase 4, which consists of sharing data on insurance, investments, pension, FX, acquiring, and other products, was launched in Dec-21 and reached the consumer data sharing stage in Sept-23. The idea with open finance is to increase competition and give consumers ownership of their own financial data, allowing better/cheaper products and services. However, consumers need to give explicit permission in order to release their data to a third party, i.e., an opt-in model, potentially limiting the amount of data available to third-party providers through open finance vis-à-vis positive credit bureaus.

However, counter to popular belief, Nubank and most large fintechs operate under similar regulatory guidelines to incumbent players. Some market participants argue that Nubank, and most fintech companies in Brazil, benefit greatly from regulatory asymmetry versus the large incumbent banks; and, that if forced to transition to similar regulation, profitability, growth, and capitalization would be impaired. We disagree. Our thinking is twofold:

- 1) As of today, Nubank's lending activities in Brazil are subject to the same minimum capital requirements applicable to most financial institutions in Brazil. Nubank and Inter (both operating under S3 conglomerate designations) have a minimum capital adequacy ratio (CAR) of 10.5%, vs 11.5% for S1 conglomerate banks (i.e., the incumbents). And importantly, Nubank previously had an agreement with the Brazilian Central Bank to maintain a CAR of 14.0% — well above that applicable to incumbents — which was revoked in July 2022. Hence, the company has proven that it can operate efficiently at those levels. We would also highlight that the current capital position of both Nubank and Inter is much higher than the minimum required of them.
- 2) There is nothing in the Brazil product road-map for Nubank that requires a conversion from the current Financeira to a Conglomerate bank license.

Nonetheless, regulatory risks remain, namely a potential cap on revolving credit card interest rates. In early October, the Brazilian Congress approved a proposal to cap revolving credit card rates at 100% p.a. — much lower than the extraordinarily high levels of >400% p.a. today. The high prices, however, are intrinsically linked to the interest-free

installments available with credit card purchases, the so called *parcelados*, which account for ~50% of total credit card balances and create subpar returns on credit cards for most issuers. Note that the industry has until the first week of January to present a counter-proposal, which we think is likely to happen as such a cap could have profound ramifications for the availability of credit. We believe that the final outcome will ultimately result in a minimal impact for Nubank and incumbent banks alike. For more details, please see [The Beginning Of The End For Parcelados?](#).

Banking service scope and ecosystem

The incumbent banks in Brazil certainly offer more products than most fintechs / challenger banks... As outlined in our in-dept [Product Comparison](#) from [The Rise of Digital Banks](#) insight report back in early 2022, large incumbents like Itaú and Bradesco have the most complete product portfolios. These players are very large conglomerates with operations touching all key product verticals in the banking space — including Payments, Credit, Insurance, Investments, Non-Financial Products, and Customer Service.

... But the largest fintechs have been gaining ground. The largest digital banks (i.e., Nubank and Inter) have been continuing to develop their "super-apps" that allow retail banking customers to access most key payments and credit products in a quick and simple fashion, which has translated into high NPS scores (around 90+ at Nubank in both Brazil and Mexico and 84 at Inter). Nubank for example, just rolled-out payroll loans earlier this year, which is the largest source of consumer credit in Brazil (please see [The Rise of Digital Payroll Loans](#)). And some of the fintech players also engage in the sale/intermediation of key products that they do not want to directly underwrite. For example, Nubank offers home equity loans via a partnership with a partner (as it prefers to avoid these "asset-heavy" lending products) and insurance via a partnership with Chubb. We would highlight that some of the fintechs are even going beyond financial services, like Inter with Inter Shop, its in-app e-commerce marketplace that provides access to products from hundreds of popular merchants.

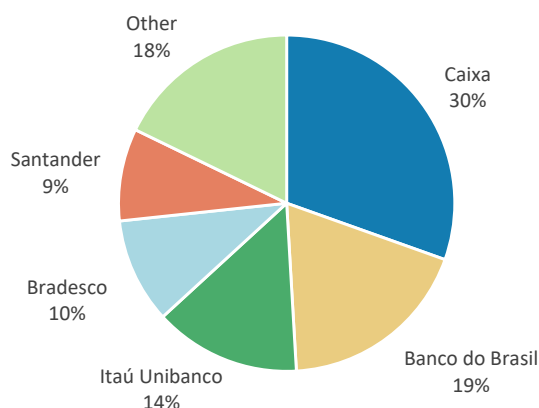
Key nuances that matter

Banking in Brazil has traditionally not been as competitive and efficient as in most places around the world, hence the opportunity for fintech disruptors was/is outsized. For much of recent history, Brazil's highly concentrated banking market has resulted in limited competition, very high prices — arguably near the very top of global banking — and generic and non-client centric products and services. This favorable setup for the incumbents allowed them to sit back, charge high, and make attractive returns on capital without necessarily having to be leaders in efficiency, innovation, customer service, or IT. And this was largely the case until ~5 years ago, when fintech was truly starting to emerge and when regulators began to take a more pro-competition stance (as mentioned above).

Looking back to five years ago, the top five banks controlled ~80% of loans and deposits — Banco do Brasil, Itaú, Bradesco, Caixa Economica Federal, and Santander Brasil. And these players also had (and largely still have) dominant shares in other

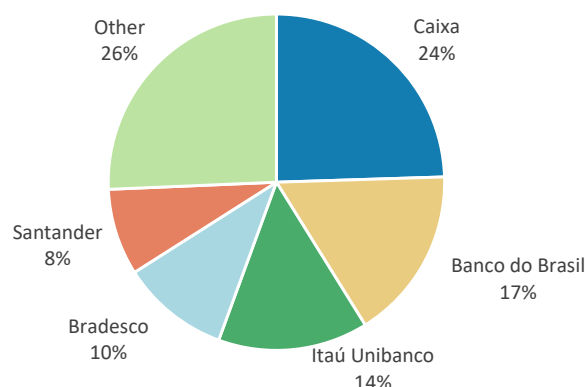
products, such as asset management, insurance, brokerage, and merchant acquiring. However, the emergence of fintech challengers has slowly begun to chip away at this market share, while also growing the size of the market (particularly in unsecured retail lending).

Exhibit 88: 2017: Consumer lending market share (% total)



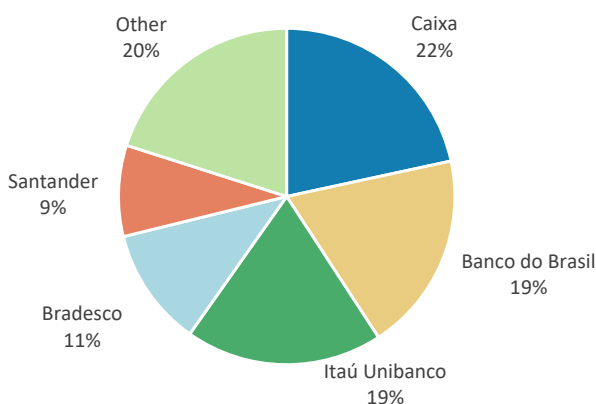
Source: BCB, Morgan Stanley Research

Exhibit 89: 2Q23: Consumer lending market share (% total)



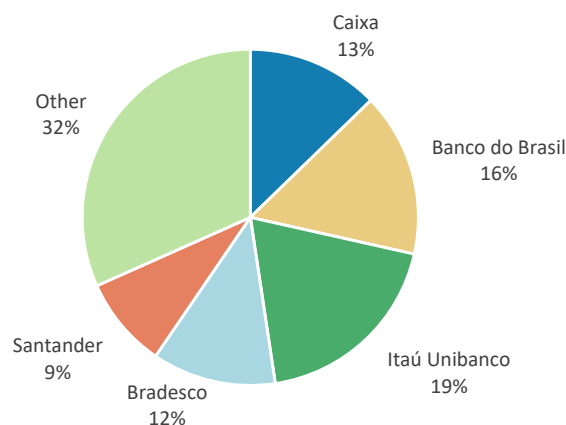
Source: BCB, Morgan Stanley Research

Exhibit 90: 2017: Total deposits market share (% total)



Source: BCB, Morgan Stanley Research.

Exhibit 91: 2Q23: Total deposits market share (% total)



Source: BCB, Morgan Stanley Research.

Three key pillars helped bolster the banking oligopoly in Brazil: i) the lack of a proper credit bureau, ii) broad physical presence, and iii) a lock-in on payrolls (two of which we address above in the Regulation section).

Pillar #1: Credit Bureau: The lack of a proper credit bureau, and the efforts of the large incumbent banks to keep the status quo, impeded the transfer of information in the system for the longest time, restricting the ability of challengers to properly price for risk. This was a meaningful driver of elevated prices and generic products in Brazil. As recently as five years ago, the Brazilian banks were earning average consumer lending spreads (ex-mortgages) of close to 30%.

Pillar #2: Physical Presence: Over the past few decades, Brazilian banks had built massive and far reaching distribution networks, including branches, mini-branches, agents, and non-shared ATM networks. The top five banks had 85% of the all branches in Brazil and ~80% of the ATMs. These vast networks have historically represented a significant barrier to

entry in the marketplace. Of course, today the banks have been reducing their footprint in order to compete with digital players on prices while maintaining cost-efficiency. Please see [Branch Rationalization](#).

Pillar #3: Lock-in on Payrolls. As discussed earlier, the long-standing lock-ins on payrolls (eliminated in 2018) allowed the banks to buy thousands of individual clients by negotiating a payroll agreement with their employers, rather than by offering attractive products and services. This, in combination with the lack of a proper credit bureau in Brazil, limited the consumer's ability to freely choose banking relationships.

Nonetheless, digitalization and help from regulators has opened the door for competition and fintech disruption. Nearly ubiquitous smartphone usage, a young/digitalized population, pro-competition policies by regulators, and an acceleration towards digitalization post-Covid, has driven impressive fintech growth in Brazil as outlined earlier. And while incumbents are catching up with digital peers, many players (like Nubank) have capitalized on the under-served nature of Brazilian consumers. Given the highly concentrated banking market, and continued growth of the lending industry, the revenue pool for fintech players to tap into in Brazil is massive. To see some of our latest retail lending TAM estimates, please see [here](#).

More on Nubank (NU.N)

Nubank is one of the world's largest and fastest growing digital banking platforms.

Founded in 2013 by CEO David Veléz, a Colombian financial engineer, the company is already one of the largest financial institutions in Brazil and one of the top new credit card issuers in Mexico and Colombia. At the core of the product offering is a distinctive no-frills, no-fee credit card with a fully-digital app experience. However, the company is now further monetizing its 90 million customer base through the cross-sell of other products, including personal loans, payroll loans, Buy-Now Pay-Later, investments, insurance, and more. For more details about the business and its key competitive advantages, please see our initiation report from January 2022, [A Purple-Sky Growth Opportunity](#).

We think the company could reach US\$100 billion valuation by 2026. Nu is well underway in building a sizable and profitable operation in Brazil, reaching more than 90 million clients and a profit run of over US\$1 billion this year. However, as we outline in our recent Insight report — [The Path To \\$100 Billion](#) — we think it can become an even larger business, growing revenues threefold in the next five years by cross-selling products to its very large and loyal client base and successfully replicating its business model in Mexico and Colombia. We get to these figures with our new product-by-product and country-by-country model, that leverages proprietary unit economics builds in Mexico and Brazil that provide greater visibility on six key products, along with proprietary TAM builds.

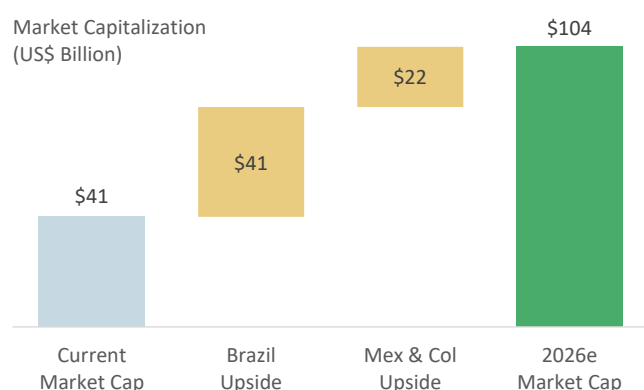
Key Findings From Our Latest Insight Report (from Oct-23) Are Summarized Below:

1) Higher product penetration in Brazil and success in Mexico and Colombia could unlock US\$60 billion in market cap. Our new product-by-product and country-by-country model suggests higher product penetration in Brazil — mainly personal, payroll, and BNPL loans — and successful customer growth and product penetration in Mexico

and Colombia could add an additional US\$60 billion in market cap.

Methodology: We use our discount rate to push forward our residual income-based 2024 PT of US\$16 to year-end 2026 — implying Nubank's fair value will reach US\$104 billion, or US\$20.9 per share, three years from now, implying a P/E multiple of 22.1x.

Exhibit 92: Increasing penetration in Brazil and success in Mexico and Colombia...



Source: Morgan Stanley Research estimates (e). Price as of December 11, 2023

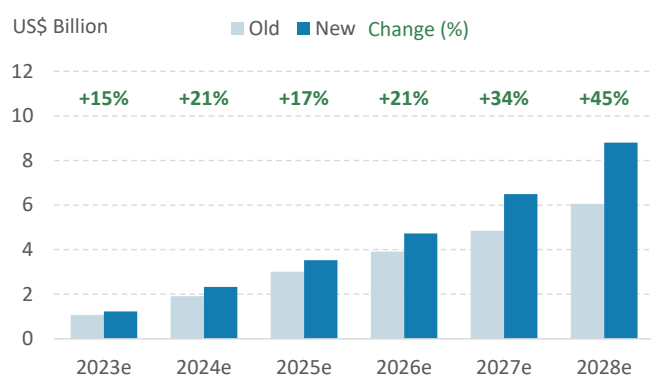
Exhibit 93: ... can make Nubank a US\$100 billion market cap stock by 2026.

Discount Rate	14.2%				
US\$	2024e	2025e	2026e	2027e	2028e
Fair Value / Share	16.0	18.3	20.9	23.9	27.3
EPS	0.47	0.71	0.95	1.30	1.76
P/E	34.3x	25.9x	22.1x	18.4x	15.5x
Market Cap (\$Bn)	80	91	104	119	136

Source: Morgan Stanley Research estimates (e)

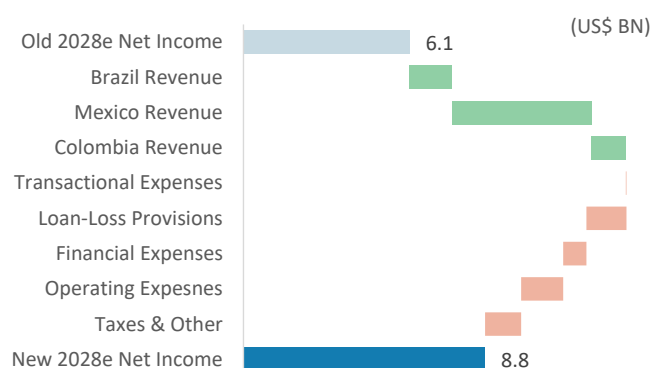
2) We now see a sustainable ROE of >35% between 2025 and 2028. The outputs of our detailed models suggested that our prior consolidated numbers for Nubank looked low. Hence, we recently increased our net income estimates from 2023 to 2028 ([link](#)). We now see a sustainable ROE of 35-36% between 2025 and 2028, versus 30% before. Note, while our prior model tried to capture some upside from Mexico and Colombia, it was not based on the detailed bottom-up product-by-product and country-by-country revamp we undertook this time around.

Exhibit 94: We have recently increased our 2028 NI estimate by 46%...



Source: Morgan Stanley Research estimates (e)

Exhibit 95: ... as we include Mexico and Colombia in our model

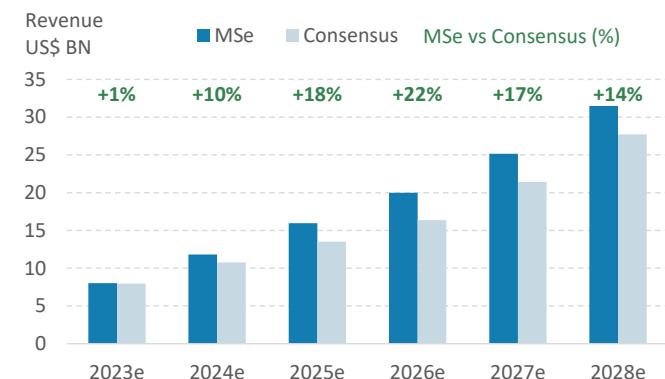


Source: Morgan Stanley Research estimates (e)

3) Not only is the market underestimating the opportunity to increase product penetration in Brazil, but we also think consensus is not pricing in much related to Mexico and Colombia. Based on our granular product-by-product and country-by-country model, our 2024 through 2028 consolidated net income numbers are 15 to 25% above consensus. In addition, our detailed work on unit economics gave us greater conviction to forecast attractive gross profit margin expansion going forward. And this, combined with

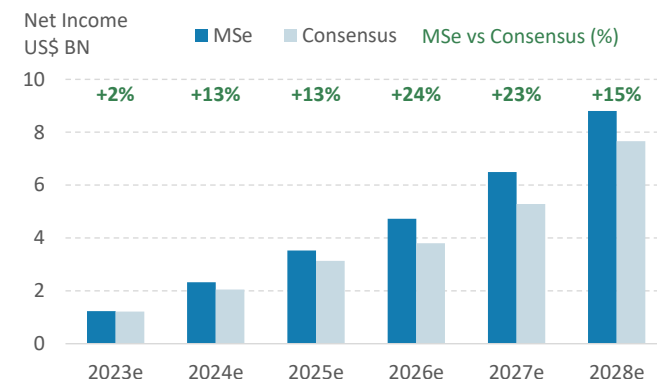
operating leverage from Nubank's highly scalable business model, explains why our new net income estimates are well above consensus.

Exhibit 96: Our 2024 to 2028 revenue estimates for Nubank are 10-25% above consensus...



Source: Visible Alpha, Morgan Stanley Research estimates (e). Consensus as of December 5, 2023

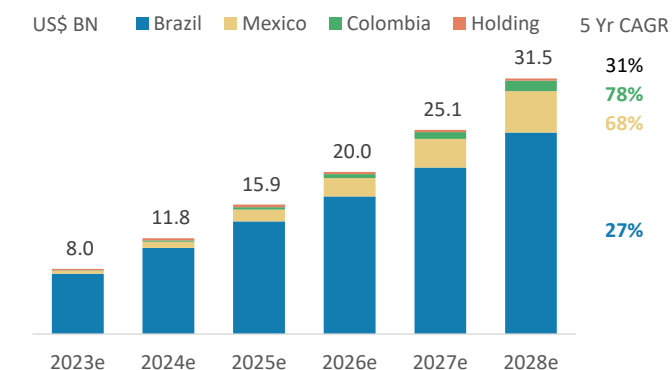
Exhibit 97: ... and our net income numbers are 1-30% higher, as we see attractive margin expansion



Source: Visible Alpha, Morgan Stanley Research estimates (e). Consensus as of December 5, 2023

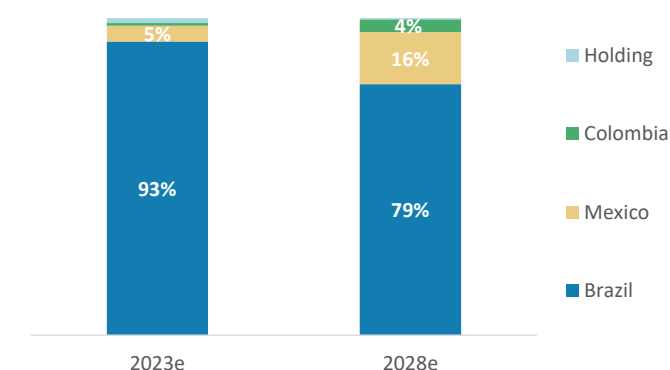
4) We see Mexico and Colombia making up 20% of total revenue in 2028, up from 6% currently. This means Mexico and Colombia revenue should grow at five-year CAGRs of 68% and 78%, respectively. Hence, we now see revenues reaching US\$31.5 billion in 2028. Our revenue forecasts are based on detailed unit economics models (see [Mexico & Brazil Unit Economics](#)) and reasonable customer growth and product penetration assumptions in each country. More on this below.

Exhibit 98: We increased our 2028 revenue forecast by 30%, reaching US\$31 billion in 2028



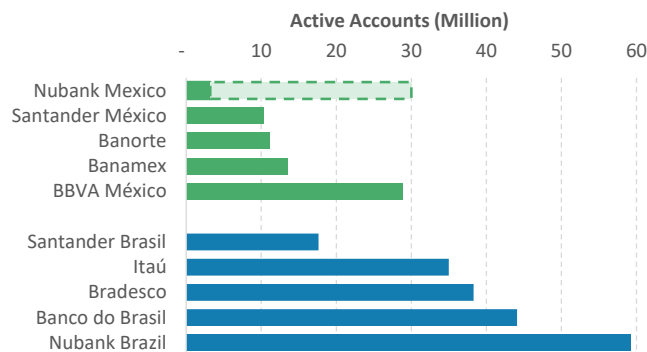
Source: Morgan Stanley Research estimates (e)

Exhibit 99: As Mexico and Colombia ramp-up, we see them reaching 20% of revenue in 2028

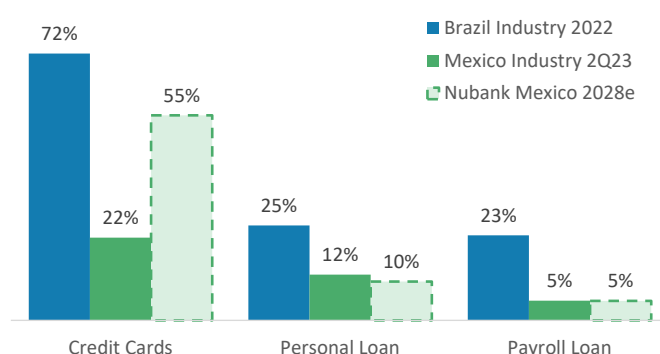


Source: Morgan Stanley Research estimates (e)

5) Rightsizing the opportunity in Mexico, we calculate Nubank can capture ~30 million active clients and increase penetration of products significantly over the next five years. We came to these conclusions by considering the following: the number of clients owned by looking at Nubank's performance relative to the incumbent banks in Mexico and Brazil, including the number of active clients and product penetration. We also took into account demographics and smartphone penetration in each country. All told, we are optimistic on the growth prospects for Nubank in Mexico given the powerful drivers of financial inclusion and electrification of financial services. We estimate ~11.5 million individuals in Mexico already applied to Nubank, supporting our view that its 3.6 million client base could grow much faster.

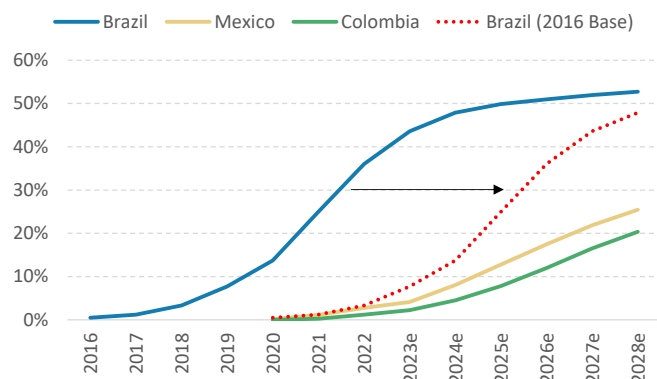
Exhibit 100: Nubank can capture ~30 million active customers in Mexico by 2028...

Source: Company Data, Morgan Stanley Research estimates (e)

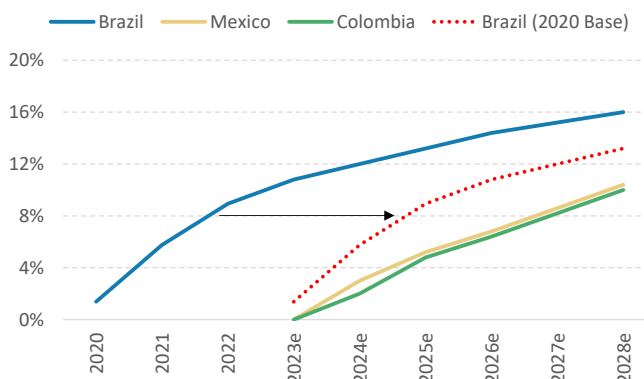
Exhibit 101: ... increasing penetration of personal and payroll loans from 0% to 5-10%

Source: BCB, CNBV, Morgan Stanley Research estimates (e)

And we think our estimates for client growth and product penetration outside of Brazil are quite reasonable, if not conservative. Indeed, our current projections for client growth in Mexico and Colombia imply a much more gradual ramp-up versus what occurred in Brazil over the past handful of years. We are also modeling to a more gradual increase in penetration of key lending products. Looking at personal loans, which were launched back in 2019 in Brazil, we estimate that reaching 8% penetration will take almost two years longer outside of Brazil, a conservative view.

Exhibit 102: Active clients by country (% of adult population)

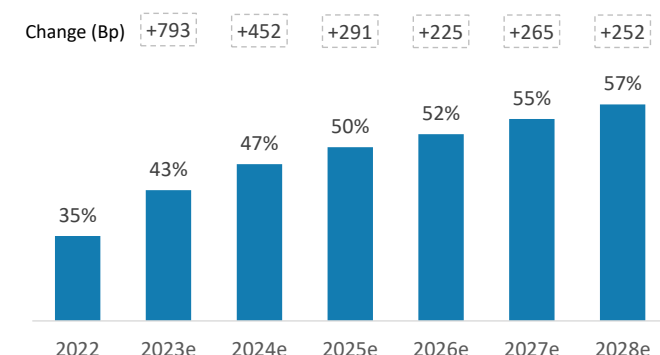
Source: Company data, Morgan Stanley Research estimates (e)

Exhibit 103: Penetration of personal loans (% of active clients)

Source: Company data, Morgan Stanley Research estimates (e)

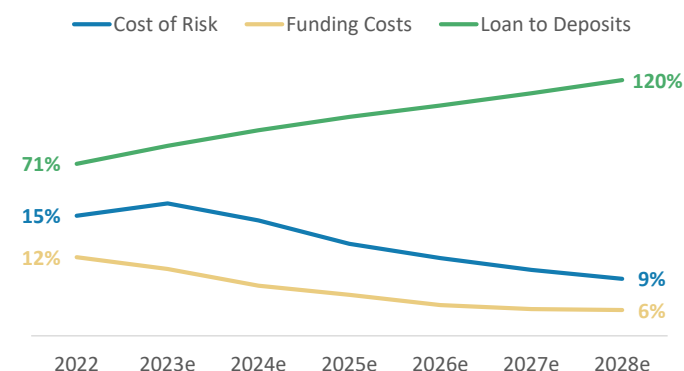
6) Our new estimates build in attractive gross profit margin expansion. Our model yields a gross profit margin of 57% in 2028, up from 43% this year. The expansion is mostly due to revenue growth outpacing that of loan-loss provisions and financial expenses, as a result of the following assumptions: lower cost of risk, lower cost of funding, and higher leverage. First, our long-term cost of risk forecasts in all countries trend lower toward, but slightly above, the historical industry average of each product. Mix shift is also an important driver of lower cost of risk. We forecast lower funding costs in response to falling interest rates and stable remuneration on a growing deposit base. And we assume leverage will increase as loan growth continues to outpace deposits.

Exhibit 104: We forecast attractive gross profit margin expansion over the next five years...



Source: Morgan Stanley Research estimates (e)

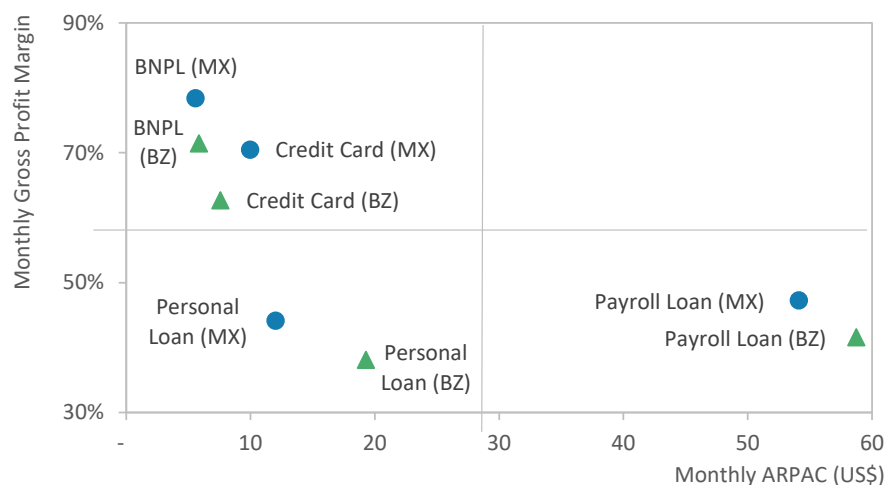
Exhibit 105: ...fueled by lower cost of risk, lower cost of funding, and higher leverage



Source: Company Data, Morgan Stanley Research estimates (e)

7) Our unit economics work shows that lending is more profitable in Mexico relative to Brazil, with higher gross profit margins on a product-by-product basis. Higher share of interest earning-balances in cards, along with lower funding cost and lower cost of risk in personal and payroll loans, are a few examples of the drivers of higher gross profit margins in Mexico. Note that the assumptions in our models are tailored to Nubank's offering and clients base, aiming to replicate its unit economics in Mexico and Brazil. Our estimates are supported by data from the central bank, channel checks, and scraping of Nubank's disclosures. Importantly, the company does not provide unit economics by product or country.

Exhibit 106: Nubank's credit offering offers greater profitability in Mexico vis-à-vis Brazil.



Source: Morgan Stanley Research estimates (e)

8) Credit cards can generate exceptional returns in Mexico, thanks to higher share of interest-earning balances. Our unit economics work show that credit cards in Mexico have 50% greater gross profit contribution compared to Brazil, with gross margins of 70% versus 63%. The main reason behind this gap is that Mexican issuers have a higher share of interest-earning balances and collect interest income for more days, which more than offsets higher financial expense due to greater working capital needs relative to Brazil. For more details on the particularities of the Brazilian credit card market (such as interest free

installments, "parcelados") please see [The Power of Cards](#).

To illustrate this, we ran a sensitivity analysis of Nubank's credit card unit economics in Mexico. Evidence from our analysis shows Nubank's gross profit would be 5% higher for every 5% increase in share of revolving loans and 10% higher for every 5% increase in share of interest with installment. For instance, the gross profit be ~27% higher if we assume interest-earning balances represent ~75% of card expenditures, compared to 55% in our base case.

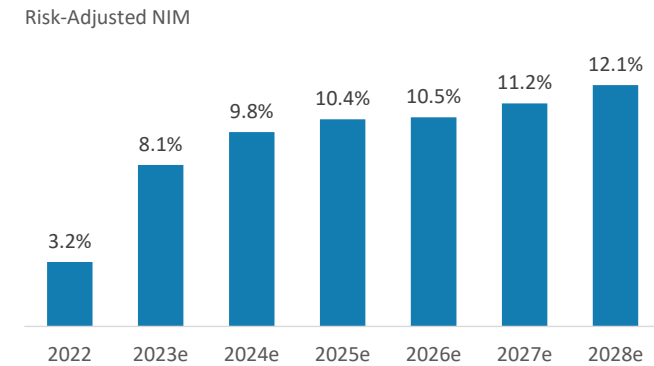
Exhibit 107: Mexico – Nubank's monthly Gross Profit sensitivity to interest-earning balances

Gross Profit		Revolving (% of Total)				Change in Gross Profit vs Base Case			
		40%	45%	50%	55%				
Installments with interest (% of Total)	5%	6.1	6.5	6.9	7.2	-13%	-8%	-2%	3%
	10%	6.6	7.0	7.4	7.8	-5%	0%	5%	11%
	15%	7.2	7.6	8.0	8.3	2%	8%	13%	19%
	20%	7.8	8.1	8.5	8.9	10%	16%	21%	27%

Source: Morgan Stanley Research estimates (e)

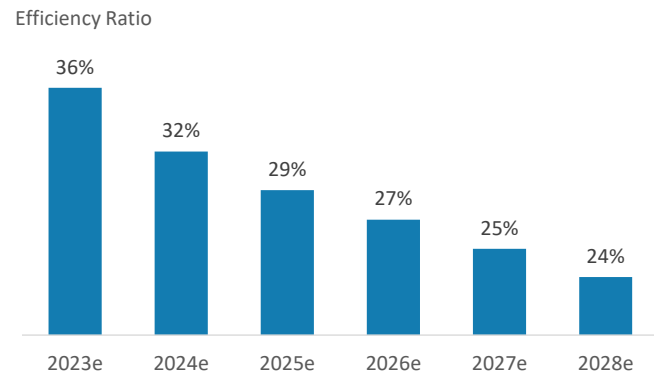
9) Nubank's scalable business model with a high percentage of fixed costs offers further operating leverage as it expands beyond Brazil. Indeed, we see the efficiency ratio improving from 36% in 2023 to 24% in 2028. And based on our unit economics framework, we forecast that the monthly ARPAC should jump from US\$10.0 to 22.8 over the next five years, while the monthly cost to serve per client should increase only from US\$0.9 to 1.6 over the same period.

Exhibit 108: We expects gradual NIM expansion going forward, after strong performance this year.



Source: Company Data, Morgan Stanley Research estimates (e)

Exhibit 109: We also forecast operating leverage will boost earnings growth and profitability

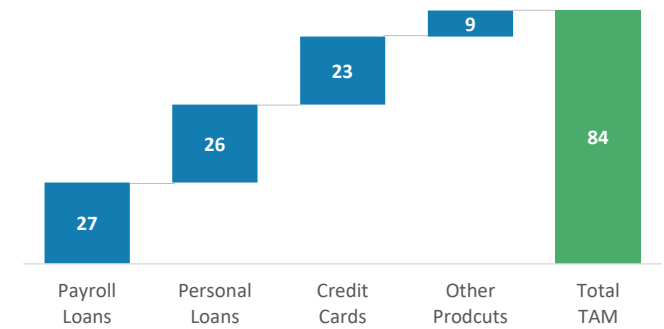


Source: Morgan Stanley Research estimates (e)

10) We calculate the current revenue TAM is US\$84 billion in Brazil and US\$17 billion in Mexico and we see Nubank's market share doubling to 16% by 2028. Our top-down TAM models focus on credit cards, personal loans, payroll loans, Buy-Now Pay-Later (BNPL), retail investments, and insurance. Given limited data on these addressable markets, our proprietary analysis leverages multiple sources along with our country-level expertise of banking and lending products in Brazil. Overall, we get to a combined revenue

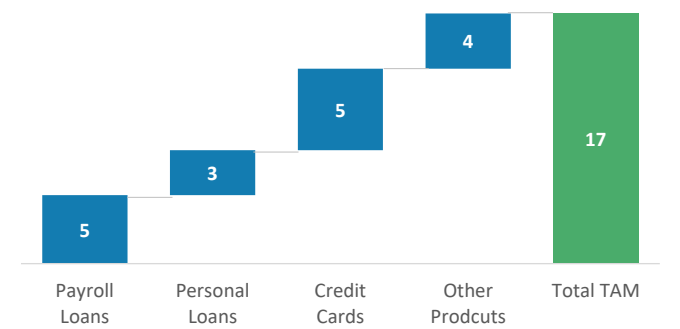
TAM in Brazil and Mexico of over US\$100 billion, which represents a meaningful white space opportunity for Nubank looking at our combined revenue estimates in these countries — US\$8.0 billion in 2023, or 8% market share, growing to US\$31.5 billion in 2028, or ~16% market share.

Exhibit 110: We estimate Brazil revenue TAM is currently US\$84 Billion...



Source: Morgan Stanley Research estimates (e). Note: Others include insurance, investments and BNPL.

Exhibit 111: ... four times greater than Mexico's US\$17 Billion revenue TAM



Source: Morgan Stanley Research estimates (e). Note: Others include insurance, investments and BNPL.

For more details, please see the full note which was published on Oct. 18th, 2023: [The Path To \\$100 Billion](#). Note that some of the charts above have been updated since our last report, given that we updated estimates reflecting 3Q23 results and new macro forecasts.

Relevant research

Digital Banks: [The Rise of Digital Banks](#), [The Power of Cards](#), [The Rise of Digital Payroll Loans](#), [The Rise of Latam BNPL](#), and [PIX FAQs](#).

Nubank: [The Path To \\$100 Billion](#) (video [here](#)) and [A Purple-Sky Growth Opportunity](#) (Initiation).

Inter & Co: [2023 Investor Day](#) and [Execution Risk and Valuation Make This a Show-Me Story](#) (Initiation).

ASEAN (ex Singapore)

Nick Lord, Selvie Jusman, Yvonne To, Aitong Li

Competitive landscape

Digital banks in operation

The countries where there are significant digital banking operations are Indonesia and the Philippines. In Thailand the central bank, Bank of Thailand, is still in the process of finalising selection criteria and regulation for virtual banks. BNM in Malaysia awarded five digital banking license in April 2022, with the first bank, GXBank, beginning operation in December 2023.

There are a few digital banks in operation in Indonesia including Bank Jago, SEA Bank, Neo Commerce, Bank Raya and Superbank. Some of the banks are part of the wider super app ecosystem such as Bank Jago, SEA Bank and Neo Commerce whilst some of the banks are subsidiaries of incumbent banks such as Bank Raya by BRI and Hibank by BNI.

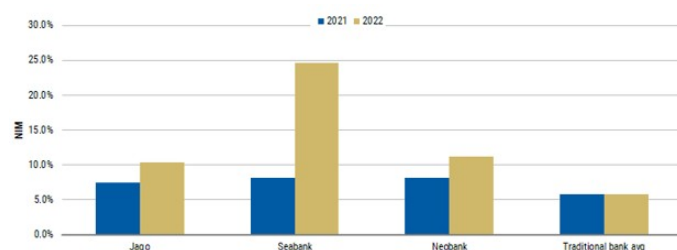
In Philippines, Banko Sentral Philippines issued six digital bank licenses in 2022. The digital banks include Union Digital Bank, Tonik Digital Bank, Maya Bank, Overseas Filipino Bank, and UNObank. BSP has imposed a three year moratorium on the digital banking licenses to monitor and the performance and the impact of the digital banks on the Industry

Incumbent banks vs. digital banks

We view incumbent and digital banks are not in direct competition, with the exception of their funding franchise. In the two underbanked and underserved markets, Indonesia and the Philippines, the presence of digital banks will help to serve SME and retail markets which are currently underserved by incumbent banks. Historically, these segments have been served by the informal market. The growth of digital banks would help to increase accessibility of funding to this group.

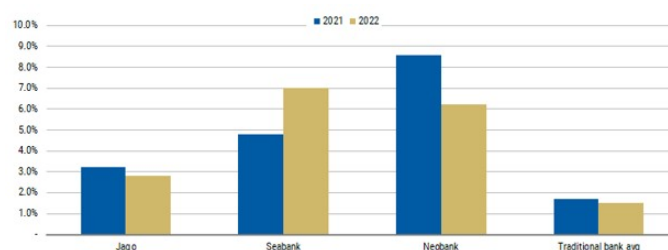
Incumbent banks have been building and continually investing in their digital presence. They view digital presence as a complement to their physical presence and part of a multi channel approach. We view in Indonesia, it will not be an easy feat for digital banks to build a superior digital platform and interconnected ecosystem to rival existing banks given the level of investments from the incumbent banks. This is also evident in the level of funding cost of the digital banks that are materially higher than the incumbent banks. However, we see room for the banks in the Philippines to bridge the gap or offer a good digital banking platform as we view banks in the Philippines still have to catch up in their technology investment compared to their ASEAN peers.

Exhibit 112: Different risk profile is reflected in the NIM between digital banks vs. traditional bank



Source: Company reports, Morgan Stanley Research

Exhibit 113: Digital banks tend to offer higher deposit rates, reflected in higher funding costs



Source: Company reports, Morgan Stanley Research

Overall regulatory and policy landscape

In Indonesia, Bank Indonesia does not differentiate between digital banks and traditional banks and digital banks are regulated just like a traditional bank. (See [ASEAN Financials: OJK's new regulations on banks/ digital banks and digital offerings \(20 Aug 2021\)](#)) There is no particular advantage from regulatory or policy landscape viewpoint to be a digital bank in Indonesia, however, the financial services authority has earlier cautioned against loose labelling of digital and require banks to have limited physical presence to be qualified as a digital bank.

In the Philippines, the launch of digital banks is aligned with BSP's aims to digitalise 50% of transactions and encourage at least 70% of Filipino to open financial accounts by 2023. BSP has been supportive towards the creation of digital banks whilst being prudent in its implementation - it has capped the number of licensed digital banks that can operate in the country to six with a three year moratorium on the grant of digital banking licenses even though it is reported that BSP is considering of removing the moratorium ([see news report](#))

Thailand (BoT) and Malaysia (BNM) have been more careful in the launch of digital banks. Proposal and plans are still under consideration and regulators seem to require players to have clear value propositions and they are not allowed to engage in value destructive competition to gain market share.

Exhibit 114: Summary of ASEAN digital banks regulation

	Indonesia	Philippines	Thailand	Malaysia
How many licenses to be provided and/or approved?	Not applicable	1 digital banking license	License applications are expected to open in 2024 and business operations to begin in 2025.	3 licenses to conduct traditional, or Islamic finance
Who can apply?	Indonesia does not have a license specific to digital banks. The license falls under the commercial banking licensing regime. With OJKR 12/2021, a business can now establish a new bank with a digital banking business model (or transform an existing conventional bank into a digital bank).	Both local and foreign companies, provided that the bank maintains a principal or headquarters in the Philippines	Under discussion	Preference given to an application where the controlling interest is held by Malaysians
Other Requirements	(i) having one physical head office, (ii) having a business model with innovative and secure technology, (iii) having the ability to manage a prudent and sustainable digital banking business model, (iv) having adequate risk management, and (v) applying personal data protection.	assessment of the transparency of ownership and control structure; suitability of shareholders, including the ultimate beneficial owners (UBOs); fitness and propriety of directors and senior management; and adequacy of capital, including sources of initial capital and ability of shareholders to provide additional financial support, where needed. The assessment shall likewise cover the strategic and operating plan, including an appropriate system of corporate governance, risk management and internal controls, projected financial condition, consumer assistance mechanism, and compliance with anti-money laundering (AML) and combating the financing of terrorism (CFT) regulations.	(1) robust corporate governance (2) use of agile, secure, resilient and high availability technology and IT system (3) the ability to access, manage and utilize diverse types of data (4) applicant's ability to deliver the Green Line in an effective and sustainable manner. Green Line refers to the ability to leverage technology, data and digital service expertise to offer financial services that will bring about new value proposition to serve the needs of customers. (5) prohibits the change of shareholders that are deemed critical to the success of the virtual bank during the first 3-5 years of its operations (restricted phase)	Offer banking services in underserved and unserved segments, including MSMEs as well as retail. Offer meaningful access to affordable and affordable financial solutions to financial customers. Track record of operating in a regulated environment is preferred alongside ability to apply transformative technology, access robust customer analytics, financial strength and for an Islamic license requestor Shariah expertise. A holder of more than 50% of the shares may need to create a separate holding company. Need a sustainable business model and path to profitability and an exit plan
Minimum paid-up capital	Bank BNI (Indonesian Legal Entity) and KCBN (Branch Office of a Bank domiciled abroad): IDR 10tn	PHP 10tn	minimum registered capital is expected to be 5 billion baht	RM200m (USD25m) in foundation phase (the initial 3-5 years, during which BNM can observe performance and risks); RM300m (USD37m) thereafter
Deposit/loan size restrictions	Follows the law governing commercial banks	No specific restriction for digital banks	Under discussion	Maximum assets of RM20n (USD200n) in initial 3-5 years.
Capital and liquidity Rules	Follows the law governing commercial banks	Same as local banks	Under discussion	During the foundation phase licensed digital banks will be subject to more simplified requirements for capital adequacy (min TCR of 8%, no capital buffer requirements), liquidity and public disclosure. Exempt from stress testing.
Business restrictions	Follows the law governing commercial banks	No specific restriction for digital banks. Digital banks are allowed to perform any of the following services: a. grant loans, whether secured or unsecured; b. accept savings and time deposits, including basic deposit accounts as defined under Sec. 213, G. except foreign currency deposits, as defined under R.A. No. 6426, as amended; d. invest in readily marketable bonds and other debt securities; commercial papers and accounts receivable, drafts, bills of exchange, acceptances or notes arising out of commercial transactions; e. act as correspondent for other financial institutions; f. act as collection agent for non-government entities; g. issue electronic money products subject to the guidelines provided under Sec. 202, h. issue credit cards; i. buy and sell foreign exchange; and j. present, market, sell and service microinsurance products subject to the guidelines provided under Sec.113-6.	Under discussion	Digital Banks will not be allowed any physical branches (outside their sole place of business), but they will have access to the shared ATM network and cash out services offered by PayPal. They can use an agency network. Digital Banks will need to comply with requirements under PSA 2013 and PSA 2013 on standards of prudential and business conduct and consumer protection as well as AML and anti-terrorism financing rules

Source: Company reports, Morgan Stanley Research

Banking service scope and ecosystem

In Indonesia, products and services offerings also range from free or low fee payment system, higher deposit products, and unsecured lending such as BNPL. Some of the digital banks in Indonesia also provide channelling to other multi finance companies as they could leverage on their balance sheet to finance companies that do not have the balance sheet. Ecosystem is an important factor in Indonesia as through these ecosystem, the digital banks typically first acquire their customers.

Exhibit 115: Indonesia digital bank ecosystem

Digital Bank	Ecosystem
Jago	GoTo
HiBank	BNI, Mayora, SEA Ltd
Sea Bank	SEA Ltd
Superbank	Grab
Neo Commerce	Aku Laku

Source: Morgan Stanley Research

Product and services offerings from the digital banks offered by the digital banks mainly target retail and SME. In the Philippines, digital banks offer retail products such as free or low fee payment and transfer services, credit/ debit card, deposit products and both secured and unsecured loans. Some of the digital banks also offer products for businesses such as payment systems, business loans and deposits. Some of the digital banks have partnerships with incumbent banks/ telcos such as Union Bank and Maya Bank but we have yet to see a big ecosystem tie-up in the Philippines of the level seen in Indonesia, due to the different market structure.

How is Bank Jago positioned?

Bank Jago is Indonesia's digital bank pioneer. Its first-mover advantage, seasoned management team and leverage to its shareholder, GoTo's online ecosystems put it in a strong position to lead the digital bank path in Indonesia. Bank Jago focuses on millennials and the younger generation, who are more tech-savvy than the older generation for its funding franchise and to date it has successfully established a respectable funding franchise for a digital bank with CASA ratio of 73% as of September 2023. On the lending side, Bank Jago has adopted partnership lending which allows it to manage credit quality prudently. Its primary lending target market is the SME market which is underbanked in Indonesia.

Relevant research

[Bank Jago Tbk PT: Upgrading ARTO to OW \(10 May 2023\)](#)

HK and Singapore

Nick Lord, Selvie Jusman, Yvonne To, Aitong Li

Competitive landscape

Digital banks in operation

Hong Kong was early in granting digital, or virtual banking licenses and issued eight licenses, with all of the holders starting operations in 2020. Two of the virtual banks are backed by local banks (Mox - Standard Chartered and Livi - BoCHK), with the rest backed by mainland technology (Tencent, Xiaomi, WeLab), or financial groups (PingAn/One Connect, Ant, ZhongAn). The other local banks have chosen to focus on their inhouse digital offerings.

In Singapore, there are five digital banks, two have digital full bank licenses, GXS Bank, backed by Grab and Singtel, and MariBank which is owned by Sea Ltd. In addition, there are two banks with digital wholesale banking licenses - ANEXT (part of Ant Group) and Green Link Digital Bank. Finally, Trust Bank (backed by Standard Chartered and NTUC Enterprise) has a full banking licence, which is outside of the digital banking license programme. The other large local banks, Maybank, HSBC and Citi all have digital offerings, but none have followed Standard Chartered by creating a stand alone business.

Note that Ant Financial and Standard Chartered have licenses in both markets.

Incumbent banks vs. digital banks

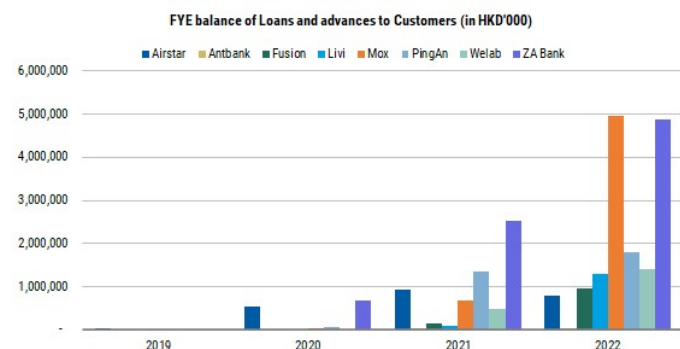
Both HK and Singapore already have very deeply penetrated banking markets, with only pockets of underserved customers. Singapore is also a competitive market and HK is becoming more so. In addition, the incumbent banks are large, with the scale and propensity to invest heavily in technology. The larger banks in both markets have also focussed on creating the correct softer skills to enable them to behave more like technology companies. For example DBS talks about how [it has changed the way it works to adapt to new technologies](#), whilst one of the key strategic objectives of HSBC over the last few years has been to 'energise for growth: to be 'fit for the future'. All of this creates a difficult environment for digital banks to thrive, despite high mobile phone ownership, excellent 4G/5G mobile networks and digitally savvy populations.

Exhibit 116: ASEAN and HK Digital Environment

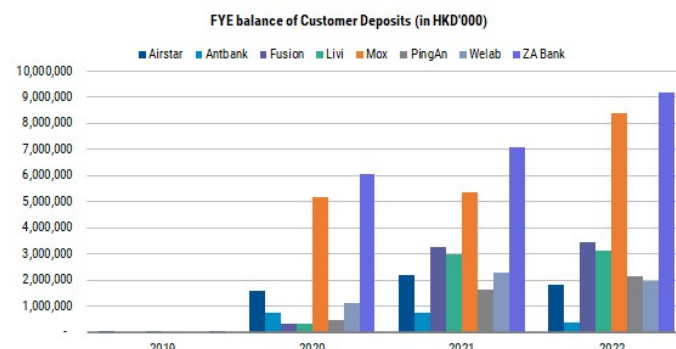
2022	Unit	Hong Kong	Singapore	Indonesia	Malaysia	Thailand	Philippines
Mobile penetration	%	305%	166%	128%	128%	145%	146%
Average data usage per month	GB/month	6.8	11.6	9.8	24.1	31.5	10.6
Network Strength: Average download speed	Mbps	36.9	61.8	17.3	32.2	18.4	23.6
Network Coverage: Average availability	% of time	99.4%	99.3%	96.4%	97.3%	99.3%	96.4%

Source: Open Signal, Company Data, Morgan Stanley Research

The HK digital banks have been operating for 2-3 years. By end-2022 they collectively had just 0.15% of system loans and 0.2% of system deposits. Mox and ZA Bank have much larger loan and deposit balances than peers, and both appear to be sustaining good growth momentum – based on current data, they look most likely to be successful.

Exhibit 117: HK virtual banks - loan balances

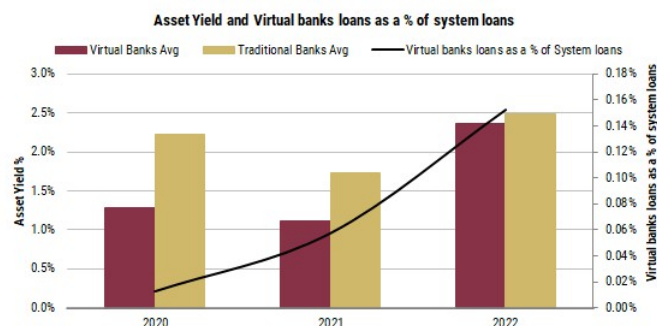
Source: HKMA, Company Reports, Morgan Stanley Research

Exhibit 118: HK virtual banks - deposit balances

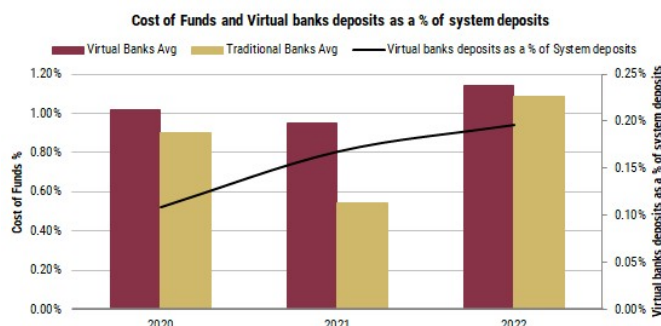
Source: HKMA, Company Reports, Morgan Stanley Research

Whilst market share remains modest, yields and cost of funding have improved as the banks have levered their balance sheets. NIMs across the sector have risen to 1.5% (vs. 1.4% for traditional banks). Ping An/One Connect has the highest NIM, followed by Airstar and ZA Bank. Mox's NIM has recently inflected to ZA's level. Loan to deposit ratios across the sector now sit at 49% (vs. 68% for the sector), whilst assets to equity have improved to 4.5x in 2022 from 3.5x a year earlier. The virtual bank's CET1 ratio has fallen to 67%!

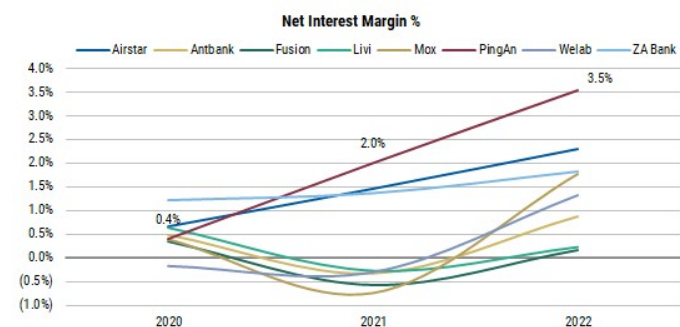
ZA, Mox and Livi, also have healthy fees at 25-33% of total revenues. All the banks still have cost:income ratios over 100% in 2022, whilst ZA, Mox, Ping An and Airstar have the least negative PPOP as a % of average assets and the least negative RoAA.

Exhibit 119: Loan market share and asset yields relative to incumbent banks

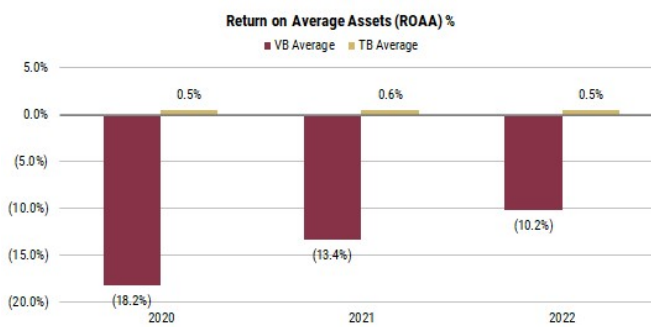
Source: HKMA, Company Reports, Morgan Stanley Research

Exhibit 120: Deposit market share and funding costs compared to local banks

Source: HKMA, Company Reports, Morgan Stanley Research

Exhibit 121: HK Virtual Bank NIMs

Source: HKMA, Morgan Stanley Research

Exhibit 122: VB RoAA compared to listed banks

Source: HKMA, Company Reports, Morgan Stanley Research

In Singapore, it is still too early to properly judge performance. GXS and Maribank have experience strong initial demand leading MAS to allow them to move out of the initial phase limits on deposit balances. However, Trust looks to be the bank most likely to succeed as it utilises Standard Chartered's digital banking knowledge gleaned from the launch of Mox in HK and NTUC's customer base (2.4m Link rewards programme members, 1.7m insurance policy holders) . It aims to be Singapore's fourth-largest retail bank by customers by 2024 and to break even by 2025. It had 700k customers in May 2023.

Overall regulatory and policy landscape

The HKMA was relatively early in introducing a specific virtual bank regulatory environment. In doing this, HKMA was looking to introduce more competition to the banking sector, with the aim of promoting innovation in fintech, enhancing user experience and fostering financial inclusion (including for SMEs). At the same time, HKMA was keen to ensure the system's safety. HKMA insists that the controlling shareholder is to be a well regulated bank, or financial institution, or if not, ownership needs to be via a locally incorporated holding company.

The introduction of digital banks in Singapore was part of MAS efforts to encourage the incumbent Singapore banks to adapt to the changing tech environment and has been driven by a desire to allow new players to fill gaps in the existing market, whilst not undermining the local banks. The full digital licenses are only available to companies headquartered in Singapore and controlled by Singaporeans, although foreign companies can be involved in a JV. The digital wholesale licenses are available to local and foreign banks.

In both HK and Singapore, the regulators have encouraged the launch of digital banks, whilst making sure that the stability of the local banking system is not undermined.

Exhibit 123: Summary of HK and Singapore Digital banking Regulations

	Singapore	Hong Kong
How many licenses to be provided and/or approved?	2 full digital banking licenses, 2 digital wholesale licenses, and 1 full banking license (Trust Bank)	8 virtual banks
Who can apply?	- Full digital bank: Companies headquartered in Singapore and controlled by Singaporeans. Foreign companies are eligible if they form a j.v. with a local company and the jv meets the HQ and control requirements. Will need to be locally incorporated. - Digital wholesale bank: Singapore or foreign companies, but it must be locally incorporated	Financial and non-financial companies, must operate as locally incorporated banks. Generally, HKMA wants and entity controlling >50% of a virtual bank to be a well regulated bank, or financial institution. If it is not a bank, the ownership will need to be through a holding company incorporated in HK. The supervisory conditions with respect to capital, risk management, and financial disclosures will be applied to this holding company.
Other Requirements	A track record in operating an existing business. Clear value proposition on how the bank can serve existing unmet, or underserved needs. Sustainable business model including business plans and financial projections (MAS does not allow value-destructive competition to gain market share). Exit plan	Sufficient resources, credible business plan, ability to develop requisite IT infrastructure and ability to commence business as soon as the licence is granted. Exit plan
Minimum paid-up capital	- Full digital bank: S\$15m (US\$11.1m) to progressively increase to S\$1.5bn (US\$1.1bn) when full licence issued. - Digital Wholesale Bank: S\$100m (US\$74m)	HK\$300m (US\$38.5m)
Deposit/loan size restrictions	- Full digital bank: Initial cap set at S\$50m, can only accept deposits from a limited set of depositors. Individual deposit limit S\$75k. This will increase over time subject to MAS criteria, once S\$1.5bn is committed, deposit cap is removed. Deposits covered by deposit insurance scheme (upto S\$75k, in-line with scheme rules). - Digital Wholesale Bank: Not allowed to take deposits from individuals, except fixed deposits of at least S\$250k. Bank is free to accept and maintain business deposit accounts from SMEs and corporates.	No deposit cap (Under deposit protection scheme, deposits upto HK\$500k are protected)
Capital and liquidity Rules	Capital same as local banks. For full digital license initially the banks must keep 15% minimum liquid assets	Capital: Same as local banks
Business restrictions	- All banks one physical place of business only. - Digital Full licence: initially can offer simple credit and investment products and operate in not more than 2 overseas markets. These restrictions will be lifted over time. No minimum balance, or fall below fees. Compliance with unsecured credit rules, can offer cash back services through EFTPOS terminals at retail merchants, but no access to ATM or CDM network. - Digital Wholesale Bank: Only conduct activities within proposed business scope	All retail services including SME banking. No minimum balance and fall below fees
Timeline of launches	Applications submitted by 31st Dec 2019. 2 full digital banking licenses, 2 digital wholesale licenses, and 1 full banking license (Trust Bank) has been issued. All the digital banks launched in 2022, except for Maribank in March 2023.	All banks launched in 2020

Source: MAS, HKMA, Morgan Stanley Research

Banking service scope and ecosystem

Service Scope

With the exception of the two Singapore banks with Wholesale licenses, all of HK and Singapore's digital banks serve retail customers. In HK, Ant Bank, Livi and ZA Bank offer services to SMEs as well, whilst Ping An One Connect and Maribank in Singapore gather retail deposits, but only make SME loans.

Amongst the HK banks all the banks (except Ping An One Connect) offer savings and time deposits alongside personal loans. Ant Bank, Livi and Mox have added BNPL to their loan product. Most also offer FX services. Outside of this Mox is the only bank with a credit card, whilst Livi and ZA Bank are the only ones with an insurance brokerage. Fusion, WeLab and ZA have investment offerings. ZA Bank has the most comprehensive product offering. In Hong Kong, virtual banks, have been credited with a number of product innovations, including virtual account opening, numberless cards, savings offset accounts for personal loans and short term salary advances to aid liquidity management.

In Singapore, the retail digital banks offer savings accounts and GXS and Trust offer personal loans as well. Trust has the most developed product offer also offering credit cards and insurance brokerage.

Exhibit 124: HK and Singapore Digital Bank Service Summary

	Launch Date	Retail	SME	HK License			SG License	Shareholder	Value Proposition
				HKMA	SFC	IA	License Type		
Hong Kong									
Airstar	11-Jun-20	Yes	No	Yes	No	No	n.a.	Xiaomi, AMTD	Basic retail banking products (savings a/c and personal loans)
Fusion Bank	21-Dec-20	Yes	No	Yes	Yes	No	n.a.	Tencent	Basic retail banking products + remittance, fund investments
Mox Bank	22-Sep-20	Yes	No	Yes	Yes	No	n.a.	Standard Chartered	Basic retail banking products + the only VB that issues credit card
Welab Bank	30-Jul-20	Yes	No	Yes	Yes	No	n.a.	Welab	Basic retail banking products + fund investments
Ant Bank	28-Sep-20	Yes	Yes	Yes	No	No	n.a.	Ant Group	Basic retail and corporate banking products (savings a/c and loans)
Livi Bank	12-Aug-20	Yes	Yes	Yes	Yes	Yes	n.a.	BOCHK, JD, Jardine	Basic retail + remittance + insurance and credit card brokerage. Basic corporate banking products.
Ping An VB	29-Sep-20	Yes	Yes	Yes	No	No	n.a.	OneConnect	Basic retail (no retail loans). Big focus on SME banking products. Has the most number type of SME loans displayed on website.
ZA Bank	24-Mar-20	Yes	Yes	Yes	Yes	Yes	n.a.	ZhongAn Online	Basic retail banking products + remittance, fund investments, insurance brokerage. Basic corporate banking products.
Singapore									
GXS Bank	31-Aug-22	Yes	No		n.a.		Digital Full Bank	Grab, Singtel	Basic retail
Maribank	14-Mar-23	Yes	Yes		n.a.		Digital Full Bank	Sea Ltd	Basic retail (no retail loans) + investments. Basic corporate banking products.
Trust Bank	1-Sep-22	Yes	No		n.a.		Full Bank	Standard Chartered	Basic retail + the only digital bank with credit card, FX and insurance brokerage.
Annex Bank	2-Jun-22	No	Yes		n.a.		Digital Wholesale Bank	Ant Group	Basic corporate banking products.
Green Link	3-Jun-22	No	Yes		n.a.		Digital Wholesale Bank	Greenland, Linklogis	Basic corporate banking products.

Source: Company Reports, MAS, HKMA, Morgan Stanley Research

IT background

Singapore has introduced a number of programmes to digitise the financial services infrastructure. For example it was early in introducing real time payments (FAST) and the apps that allow these to be used digitally both to pay merchants and to transfer cash seamlessly between individuals (Paynow). SGQR is a standardised QR code that enables equal access to payments for all banking apps and e-wallets enhancing interoperability. This prevents the segmentation of the financial system, whereby you can only use one wallet at a particular merchant, and benefits all, whilst preventing large banks or technology players from creating an outsized advantage. Finally, the creation of Singpass and myinfo, makes it easier to share credentials between accredited players - for example, Trust Bank can onboard a customer in ~3 minutes (compared to 5 minutes for HK digital banks) and over 90% of customers are on-boarded without digital intervention. This lowers the cost of digital bank account opening, whilst reducing KYC risks, and will eventually help drive open banking in Singapore. Singapore is also increasingly becoming a cashless society - for example, free cheque usage ended in Singapore on 1st November 2023 and corporate cheques will be phased out by 2025.

In Hong Kong, the HKMA has announced a number of initiatives to support Hong Kong's move to the 'new era of Smart Banking'. These include full connectivity of digital retail payments through the Faster Payment System (FPS), similar to Paynow in Singapore and developing an open API framework. HKMA has also pushed the 'Banking made easy initiative', which is focussed on adapting regulation to meet the needs of a more digital banking system. For example, HKMA has facilitated remote customer on-boarding and allows banks to use alternative data to assess credit risk, rather than having to rely on traditional measures such as proof of income.

Key nuances that matter

Both HK and Singapore are financial hubs, this means that local banks have quick access to leading international talent and are rapidly exposed to global trends. HSBC, Citi and Standard Chartered are relevant competitors in both markets and the dominant banks in both HK and Singapore are global, or regional players. In addition both regulators have been early in pushing for tech adoption within the financial services industry.

Relevant research

[Mapping the New Normal: Rise of the ASEAN Super Apps \(20 May 2020\)](#)

Disruption Decoded: Who will win in e-payments – banks, independents, or telcos? (24 Jul 2018)

Europe

Giulia Aurora Miotto, Alastair Nolan, Kerry Shaw

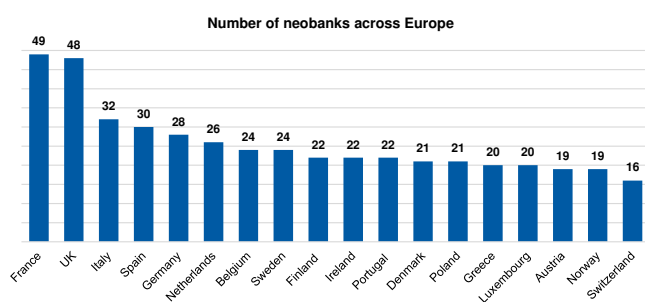
Competitive landscape

Digital banks in operation

In Europe there are more than 10 'neobanks' per country, with more than 30 in the UK, although their business models are varied in terms of regional/client focus (such as SME or teenagers, etc.) and revenue mix (see Exhibit X). For most, the majority of revenues are derived from fees, with lending still only a small contributor, while for others interest income is the main source of revenue (e.g Atom Bank). However, what most typically have in common is a more user-friendly interface, a slicker app and the claim that they are modernizing the existing incumbent offering, usually through more transparent, cheaper and inclusive services.

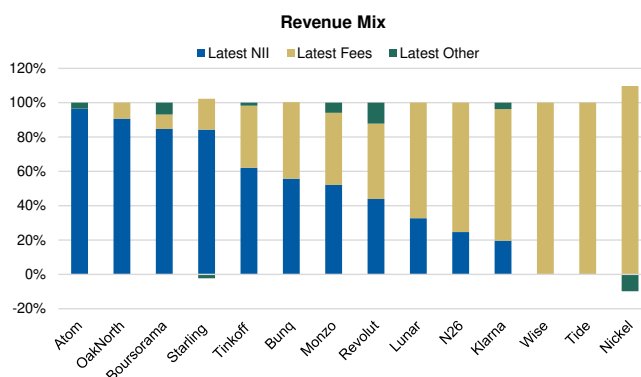
We think the set-up is tougher in Europe for neobanks than in other parts of the world due to: 1) an already oversupplied banking market (as the ECB often points out, making the case for banks' consolidation); 2) lack of scale, vs for example the US, which has about the same population as the entire Eurozone (329mn vs 342mn), but only one language and a more homogenous legal system; and 3) lower interchange fees, which are capped at 0.2% for debit and 0.3% for credit. We expect only a few players to emerge, those that are able to get to scale first, and better monetize customers, via cross-selling of different products to complement a compelling current account offering.

Exhibit 125: Number of neobanks present in each market



Source: Neobanks.app, Morgan Stanley Research, Nov-23

Exhibit 126: Neobanks at a glance: revenue mix varies materially across the spectrum



Source: Company data as of latest available, Morgan Stanley Research. Note that some details have been intentionally removed from consideration in this report because, under applicable law and/or Morgan Stanley policy, Morgan Stanley may be precluded from issuing such information with respect to the relevant issuer at this time

Incumbent banks vs. digital banks

We think there are a few key differences between neobanks and incumbent banks in Europe:

1. Neobanks typically have slicker apps and higher net promoter score, while traditional banks tend to have less user friendly apps, with some exceptions (we would call out KBC

for example, with its AI powered digital assistant Kate);

2. Yet, neobanks usually have a narrow product focus, with traditional banks offering a broader set of products, which typically results in stronger, stickier relationships with customers (as an example, at the time of our survey, in 2021, neobanks typically had ½ product per customer, vs incumbents having a higher number of customers with 2, 3, or more products with the same bank.

3. Traditional banks have a "deposit franchise", i.e. they are the last ones to increase interest rates following a central bank interest rate hike, while neobanks typically offer higher rates in their savings / term products.

4. Neobanks are not as well versed as traditional banks when it comes to regulation, which has resulted in a few issues - for example Revolut is yet to obtain a banking licence in the UK, Monzo has faced FCA probes over potential money laundering breaches, and Bafin has imposed customers growth caps on N26 until certain KYC / onboarding / transaction reporting systems were fixed.

5. Not having any branches, and being built on a more modern, cloud-native, modular infrastructure, neobanks typically argue that they have a material cost advantage vs incumbents when at scale. While most neobanks are for now still loss making, we would point to Starling (ROTE 23% in FY2022) as an example here.

Path to Disruption – our AlphaWise Digital Banking Consumer Survey

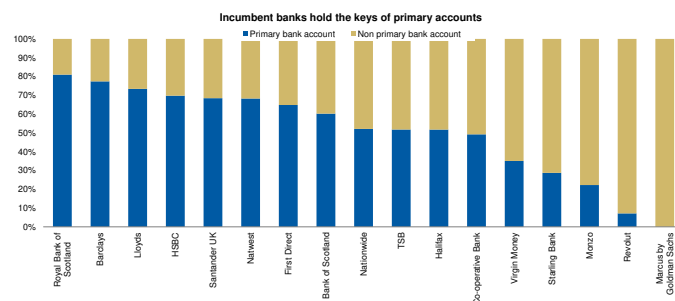
In Feb-22 we carried out a proprietary Alpha Wise survey across over 4k consumers in France and the UK to explore the disruptive potential of neobanks. We don't expect that the story has changed materially since then, and continue to believe that disruption to legacy players is still in an early stage. This is, in our view, due to three key factors highlighted by our findings: 1) incumbent banks still manage the majority of primary account relationships; 2) neobanks offer a more narrow and less profitable product range vs incumbent banks; 3) incumbents better tailor their offering to users' different banking priorities. Conversely, neobanks do screen better when it comes to appealing branding, ease of use, lower fees and cash reward incentives, while they screen lower on other attributes such as trust, range of products on offer and attractiveness of pricing on credit products. All-in, this suggests to us that despite some neobanks leading on consumer experience, they still lag behind incumbents when it comes to product breadth, scale and financials.

Incumbents hold the key to the primary account relationship. Despite the majority of the banked population owning multiple accounts (in the UK/France only 35%/42% of the population has just one account), most still hold their primary account at incumbent banks. Some neobanks do not yet offer accounts covered by the deposit insurance scheme (e.g. Revolut), but even for those with banking licences like Monzo/Starling, only 22%/29% of users are primary account holders. This compares to 77% of Barclays customers (see [Exhibit 127](#)). Fintechs would argue that although users may not receive their salary into their neobank account, they can still extract higher fees without having to manage excess liquidity. However, this is much less beneficial in a higher rate environment where excess liquidity can be reinvested in higher yielding assets. Further still, our analysis showed that

only a small number of users would say their most used account is different from the one into which they receive their salary.

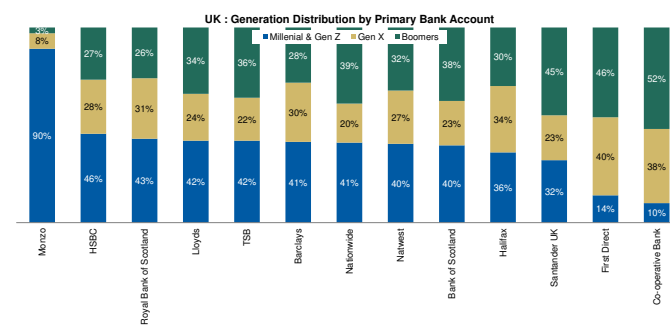
Demographics debate. In the UK, Monzo's primary account users are skewed to Millennials and Gen Z, who represent c.90% of primary relationships, while The Co-operative Bank has the highest skew to the older population (see [Exhibit 128](#)). The trend is less clear in France, as Boursorama (the leading digital bank) actually has a demographic spilt similar to that of an average incumbent bank and is more skewed to the most affluent part of the population, likely driven by other dynamics such as the fact it was founded in 1998 and has a strong trading/investment offering. Across overall users, customers of all generations and income are keen to use neobanks as secondary accounts.

Exhibit 127: Primary vs non-primary bank account: Incumbents have 60%+ of primary bank account penetration, vs around 20-30% for Starling / Monzo



Source: AlphaWise, Morgan Stanley Research

Exhibit 128: Monzo is skewed to a younger cohort

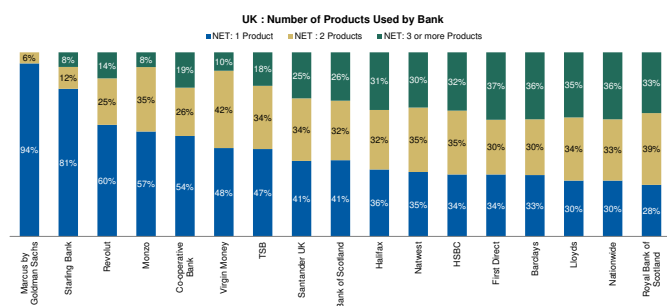


Source: AlphaWise, Morgan Stanley Research

Satisfied with the level of service. In the UK, 59% of non-primary account holders at Monzo hold their primary accounts elsewhere as they are happy with the level of service they receive there. This is similar to what Boursorama customer's said (56% of customers are happy with the service they already receive from their primary bank). Other reasons customers gave for not switching included inconvenience (26% of Monzo customers, 17% of Boursorama customers) and lack of branches (16% of Monzo customers and 23% of Boursorama customers).

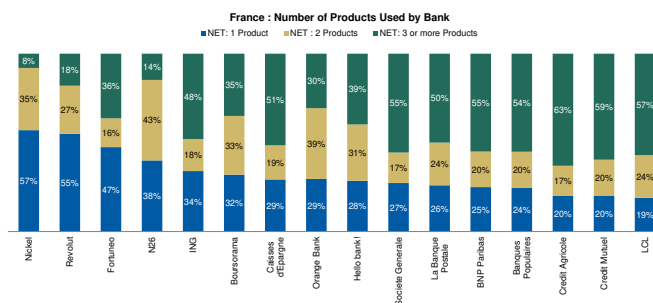
Capturing cross-selling opportunity. Challengers have fewer products per client vs incumbents, with most still in "build out" mode. Product penetration is also lower for neobanks (1.42 vs 2.15 at incumbents in the UK on average). In particular, neobanks are progressing very slowly in terms of lending, with very few currently offering mortgage products. We think it is virtually impossible for neobanks to build a profitable business model based on transaction banking alone, (unless it is FX-related, like in the case of Wise) given most products are zero-fee/low margin, and we see the ability to move into more profitable banking verticals as mission critical.

Exhibit 129: UK: Number of products used by each bank: neobanks tend to have fewer products per customer



Source: AlphaWise, Morgan Stanley Research

Exhibit 130: France: Number of products used by each bank: this shows a similar result to the UK



Source: AlphaWise, Morgan Stanley Research

Overall regulatory and policy landscape

While Europe has been more open to granting banking licences to startups than the US for instance, we think ultimately these players will be held to the same standards, and we think this will hold true for fintechs that market themselves as a neobank, although currently they operate with an e-money licence. Overall, therefore, we would expect more scrutiny in the coming years. If prior experience of incumbent banks is any guide, we would expect larger problems to come from neobanks that have grown over different countries quickly, because that could make it difficult to incorporate different legal and compliance requirements.

Banking service scope and ecosystem

While banks in Europe (especially continental Europe ex UK) have a holistic relationship with their customers, and offer most financial products (from current accounts to investments, insurance, leasing, etc), with the best run banks owning an asset manager, an insurance company and having as much as eight or more products per customers on average (particularly in markets like France, or Italy), neobanks still seem to us in 'build out' mode, with most offering a limited product suite. Moreover, products often span non financial services, for example Revolut offering a service similar to Airbnb for booking holidays.

Key nuances that matter

2022-23 saw a material shift in focus within neobanks, from growth at all cost, to profitability, with for example Monzo, Starling, Klarna, Atom, N26 and more all highlighting how profitability is key to their strategy very upfront in their shareholders' letter with 2022 annual reports. There are currently more than 10 neobanks in every developed market in Europe and as many as 30+ in France and in the UK – we think this is more than the market can support, considering that Europe is already overbanked. We expect more scarce venture capital funding to lead to consolidation, with few neobank leaders emerging among those that can reach profitability fast. We also think a drop in valuations may provide incumbents with the opportunity to acquire the much-needed better customer interface, rather than develop it in-house.

Relevant research

[Neobanks: how does the path to profitability look?](#)

[Neobanks: Exploring disruptive potential](#)

[Video: How Disruptive Are Neobanks?](#)

MENA

Nida Iqbal

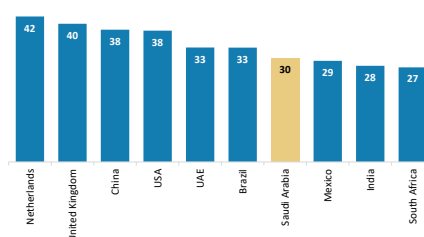
Competitive landscape (Saudi Arabia)

Digital banks in operation

Digital banks is still a nascent theme in Saudi Arabia, with only three digital banks launched. Compared to its peers, Saudi Arabia is still in its early days of digital banks adoption. We understand that Saudi only has three digital banks authorized to operate in Saudi, out of which two have launched operations and one is in pre-ops stage. That said, we see Saudi Arabia as a fertile ground for the emergence of digital banks, as there is a young, tech savvy population who will be an important driver of banking and payment innovation adoption, we think. Saudi Arabia has a young population with a median age of 30 years, and near 70% of the population below 40 years of age. We think the age profile of the Saudi population is one of the main factors behind the country's elevated adoption of smartphones, use of Internet and social media.

Exhibit 131: We think Saudi Arabia's population will drive the digital banking and innovation adoption...

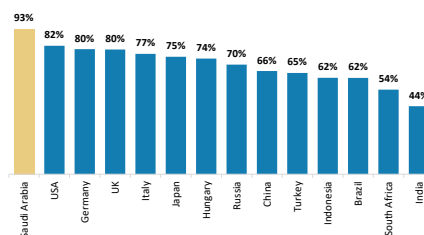
Population Median Age (2021)



Source: United Nations, Morgan Stanley Research

Exhibit 133: Smart phone penetration in Saudi Arabia is very high relative to other countries

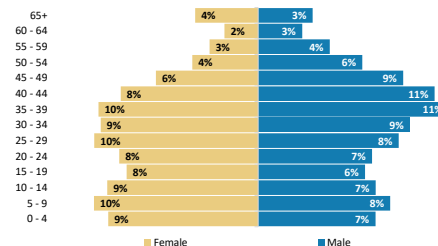
Smartphone Ownership (% of population, 2021)



Source: NewZoo, Statista, Morgan Stanley Research

Exhibit 132: ... with 54% and 43% of women and men, respectively, being under the age of 30

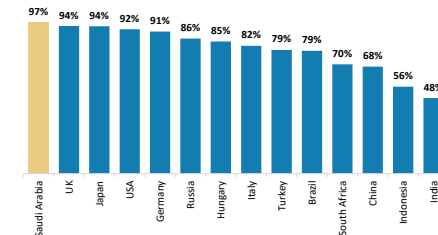
Saudi Arabia Population Breakdown by Age and Sex



Source: GASTAT, Morgan Stanley Research

Exhibit 134: With a 97% Internet penetration offering a large addressable market for open banking and digital payments

% of population using the internet (2021)



Source: Euromonitor, Morgan Stanley Research

Moreover, the Saudi Government has been active in supporting innovation in the sector. Saudi Arabia launched the Financial Sector Development Program (FSDP) as part of its Vision 2030 programme, which includes initiatives such as: 1) allowing non-banks to provide payment services (such as payment initiation and digital wallets) through the issuance of Payment Services Provider regulations, 2) the creation of Fintech Saudi, an

institution which aims to act as a catalyst for the development of the fintech industry in Saudi Arabia by supporting the development of infrastructure, building capabilities, and the companies; 3) the introduction of a QR payment system. As of 2022, Saudi Arabia had 147 fintechs operating in the Kingdom, an increase of 79% from last year. Of these fintechs, 30% are in the payments development segment, the leading segment, in front of lending and finance at 19%. Vision 2030 targets the Kingdom to reach 525 fintechs by 2030, with the sector contributing SAR13bn to the country's GDP.

Exhibit 135: Since the launch of Fintech Saudi, the number of fintechs has increased by ~15x...

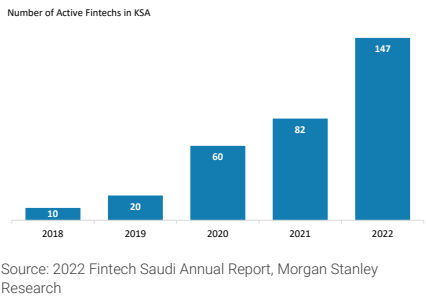


Exhibit 136: ...and SAR3.95bn has been invested into the fintech industry

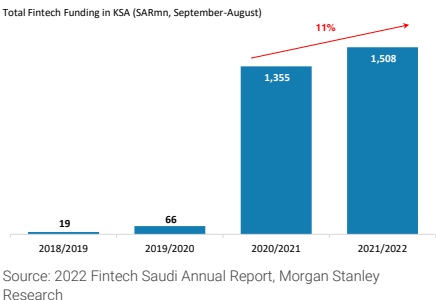


Exhibit 137: 37% of the total fintechs in the kingdom are in the idea stage...

Breakdown of Fintechs in KSA by Stage

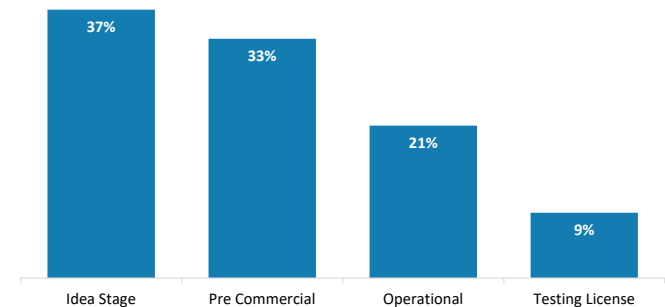
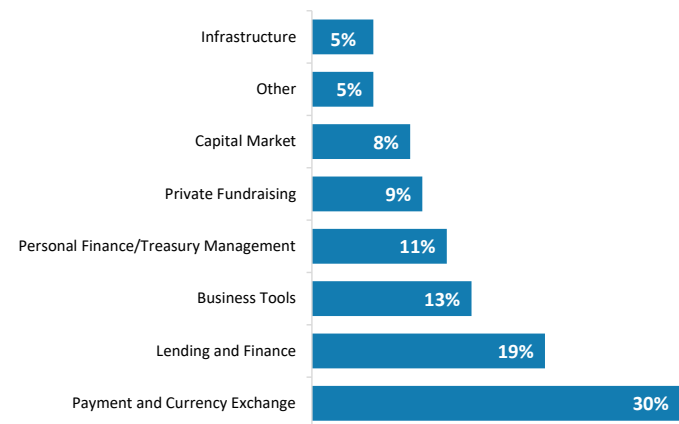


Exhibit 138: ...with 30% of active fintechs in Saudi Arabia being in the payment and currency exchange space...

Breakdown of Fintechs in KSA by Industry (as of August 2022)



Incumbent banks vs. digital banks

We see digital functionality and banking apps as the next battlefield vs branches in the past. With that, to face the fintech threat, investment in digitisation is key for the Saudi banks. We think the emergence of challengers will have the effect of pressuring incumbent banks to improve their digital offering further in order to compete with the emerging digital banks. As Saudi consumers get more acquainted with a digital product offering, traditional banks will need to review their branch strategies, in our view, as there will be reduced demand for services offered in person.

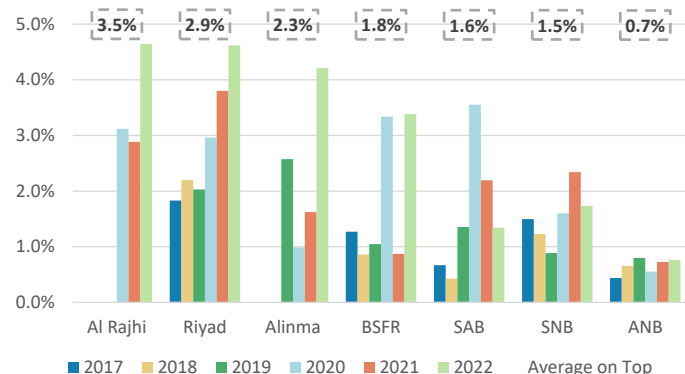
Our analysis of investment in technology for the Saudi banks over the last five years.

We have analysed the gross and net additions of technology related assets for the banks under our coverage over the last five years (when available). Net technology related

assets have grown at an average 24% CAGR over the last two years, and banks have spent between 0.7-3.5% of revenues on technology capabilities over the last five years. **Al Rajhi** and **Riyad** bank stack up as having made the highest levels of investment, whilst Al Rajhi and SNB have seen the highest increase in net technology related assets in the last two years, and **Alinma** ranks fourth in terms of IT investment growth. In addition, we note that Rajhi and Alinma have launched their own fintechs, Neoleap and Alinma Pay, positioning them well for potential upside as the shift towards digitisation takes place in Saudi Arabia.

Exhibit 139: Al Rajhi has led the way in investing in new technology assets, followed by Riyadh and Alinma

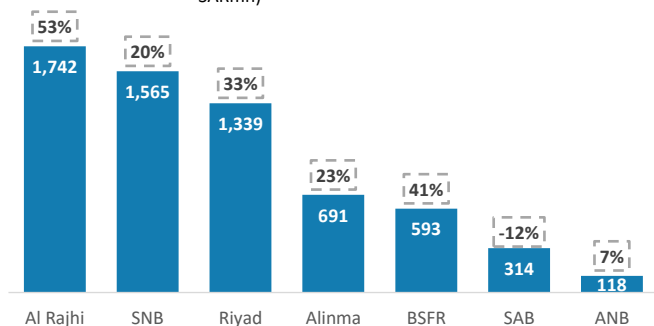
Gross additions to technology assets / operating income



Source: Company data, Morgan Stanley Research

Exhibit 141: The net value of technology related assets for banks under our coverage have grown at an average ~24% CAGR in the last two years

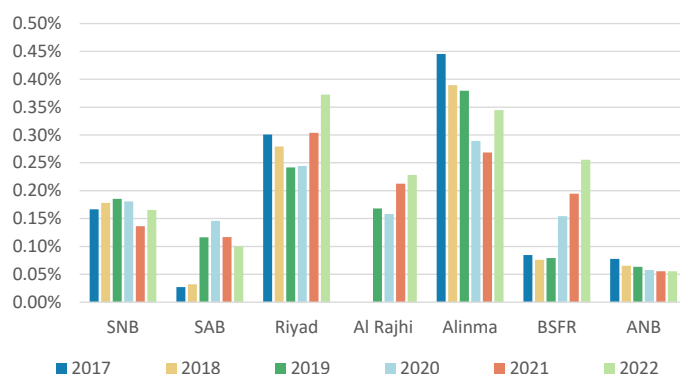
Net Value of Technology Related Assets (2020-22 CAGR on top, SARmn)



Source: Company data, Morgan Stanley Research

Exhibit 140: Reflected in a growing percentage of net technology related assets as a % of total assets

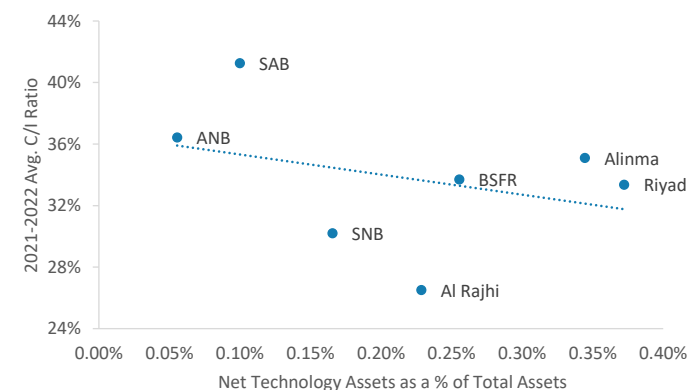
Net Technology related assets as % of total assets



Source: Company data, Morgan Stanley Research

Exhibit 142: Correlation between cost-to-income ratio and technology related assets is negative but is still low

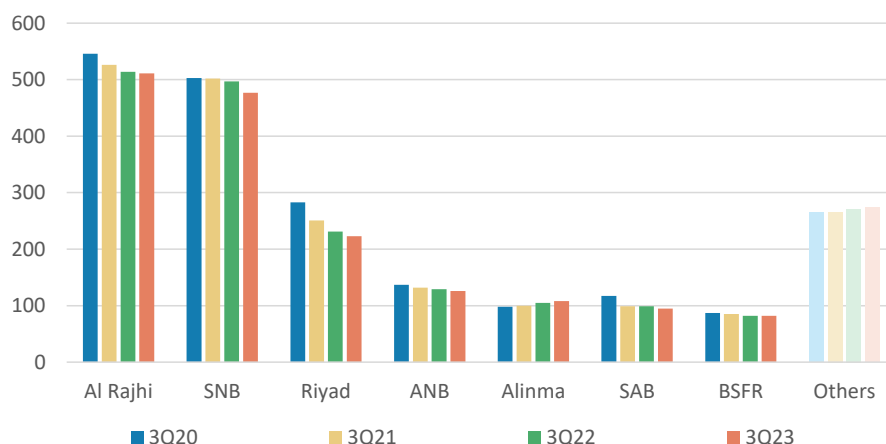
Net Value of Technology Related Assets (SARmn)



Source: Company data, Morgan Stanley Research

Exhibit 143:

With growing digital operations, we would expect the inverse correlation between technology related assets and cost to income ratio to increase over the coming years. An example of potential cost reductions is through the reduction of number of branches

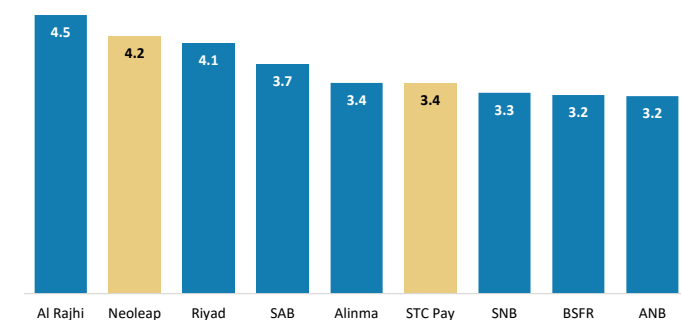
Number of Bank Branches in Saudi Arabia

Note: SNB is pro forma for 3Q20. Source: Company data, Morgan Stanley Research

Banks that have invested more in technology in recent years also appear to have more customer engagement. Our analysis of app downloads, app usage and ratings shows that the banks that have invested more heavily (proportionally) in technology related assets in the last several years are also the ones that appear to have: 1) a higher app rating (see [Exhibit 144](#)), more accumulated downloads (see [Exhibit 146](#)), and lastly (and more importantly) higher consumer engagement (see [Exhibit 150](#)). Al Rajhi, Riyadh Bank and Alinma stand out. We believe that banks with apps offering a wide range of products as well as a smooth and intuitive UI will be able to attract more business to their franchise (be it in the form of fees, loan originations or deposits).

Exhibit 144: Al Rajhi and Riyadh stand out as the banks with the best rated apps, and the only ones with +4 stars

App Ratings (Google Play and App Store Weighted Average)

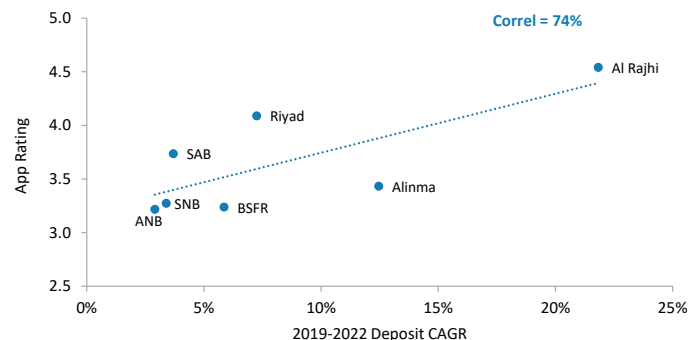


Source: Apple App Store, Google Play Store, Morgan Stanley Research

Exhibit 145: Banking apps the next battleground for deposits:

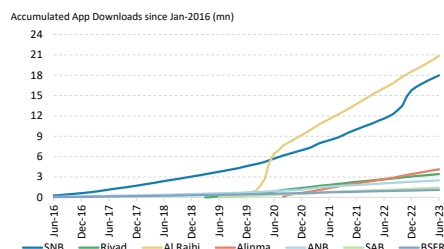
Correlation between deposit growth and app ratings is elevated

App Ratings x 2019-2022 Deposit CAGR



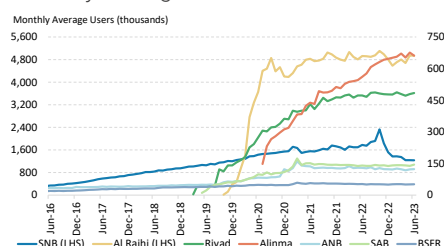
Note: SNB is pro forma. Source: Company data, Apple App Store, Google Play Store, Morgan Stanley Research

Exhibit 146: Al Rajhi leads the way with the most app downloads, having accumulated 21mn downloads since 2016



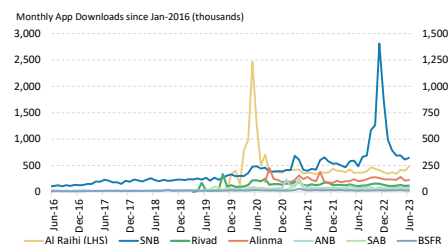
Source: SensorTower, Morgan Stanley Research

Exhibit 148: Not surprisingly, SNB and Al Rajhi are also the apps with the most monthly average users...



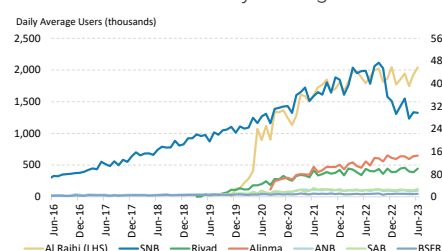
Source: SensorTower, Morgan Stanley Research

Exhibit 147: SNB is the second most downloaded app, with its number of monthly downloads increasing over time



Source: SensorTower, Morgan Stanley Research

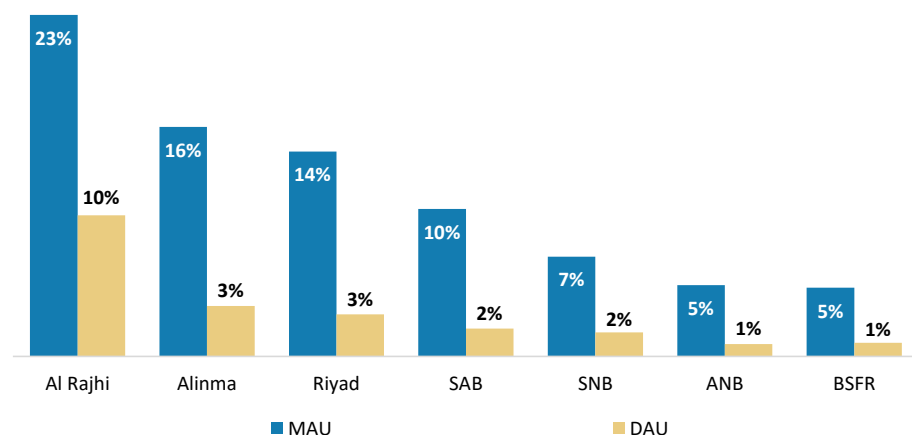
Exhibit 149: ... and daily average users



Source: SensorTower, Morgan Stanley Research

Exhibit 150:

Al Rajhi, Alinma and Riyadh have the most active customer base in its app
MAU and DAU divided by accumulated downloads



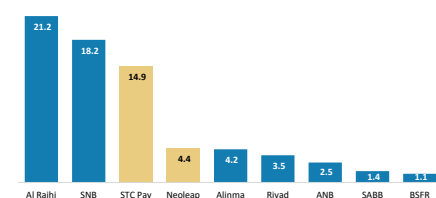
Source: SensorTower, Morgan Stanley Research

Though mostly on the payments front, we have started to see digital solutions emerge for payments; we flag STC Pay and Neoleap as the most advanced in customer traction: In our view, STC Pay, a company backed by Saudi's largest Telco operator, Saudi Telecom, is the most advanced out of its peers, having accumulated 15mn downloads since it was launched in 2018, which compares to Al Rajhi's accumulated downloads of 21mn and SNB's 18mn, according to SensorTower. Al Rajhi's Neoleap payment app "Urpay" also stands out, with 4.4mn accumulated downloads, and an elevated ratio of MAU to

accumulated downloads (33%).

Exhibit 151: STC Pay and Al Rajhi's Neoleap have accumulated more downloads than most of the incumbent apps

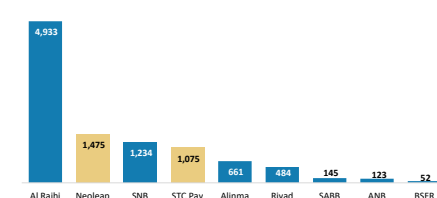
Accumulated Downloads per App (mn, as of Jun 2023)



Note: data on Alinma Pay not available. Source: SensorTower, Morgan Stanley Research

Exhibit 152: With Neoleap's Urpay app having an elevated MAU to accumulated downloads ratio (33%), indicating high customer engagement

Monthly Average Users (thousands, as of Jun 2023)



Note: data on Alinma Pay not available. Source: SensorTower, Morgan Stanley Research

Exhibit 153:

We view Al Rajhi's Neoleap value proposition as very compelling and see the subsidiary as a testament that incumbents will not be complacent with the ongoing sectorial changes

Feature	Description
Neoleap (Al Rajhi)	
Digital Wallet	Neoleap has an app called Urpay, a digital wallet that enables consumers to make payments through the app, including purchases, local and international transfers. The user is also able to issue digital cards for cash withdrawals and online purchases.
Payment Gateway	Called Neogate, the service enables merchants to accept credit card and mada card payments through the payment page on their website or via a sharing link
Stores Services	The service Neostore provides merchants with an integrated solution to build an online store within moments, linking it to multiple payment methods, different shipping and delivery companies, and more additional services such as e-marketing and inventory management
Cashier services	Neoeqr provides shop owners (such as restaurants and markets) the ability to receive customer requests, pay directly and issue invoices to them
Point of Sale	Neopos enables merchants to accept customer payments on mobile POS devices or their mobile phones by installing a payments application on their devices.
Issuing and Processing	Enable financial institutions and banks by managing their payment card issuing and processing, while they focus on sales and revenue growth

Source: Company data, Morgan Stanley Research

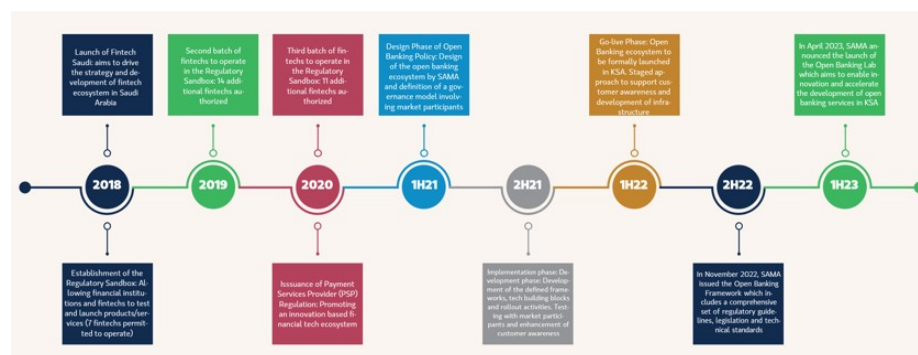
Overall regulatory and policy landscape

In Feb-2020, SAMA (Saudi Arabia Monetary Authority) laid out a set of guidelines/requirements for digital only banks. In doing this, SAMA was looking to promote the establishment of an innovation-based financial technology ecosystem and at the same time ensure the system's safety. Under SAMA's guidelines, digital-only banks must be set up in the Kingdom of Saudi Arabia (KSA) as joint stock companies. Its promoters must have technology and financial industry related experience and knowledge, and the financial capacity to support setting up the digital-only bank. Furthermore, digital banks are required to have a physical presence in the country as their principal place of business, but are not expected to establish physical branches and also must follow the same principles of corporate governance as for other conventional banks. On the capital requirement front, while the guidelines do not specify any minimum requirements, digital banks are required to submit an internal capital adequacy assessment plan (ICAAP) and an internal liquidity adequacy assessment plan (ILAAP), based on which SAMA will assess the adequacy.

We think open banking regulation creates a more level playing field for challengers to compete with incumbents. In January 2021, SAMA issued its Open Banking policy, setting out Open Banking regulatory guidelines as well as its go-live stage plan. In the first half of 2021 we understand that SAMA worked with market participants to design the open banking ecosystem as well as the definition of a governance model. In the second phase of its three-step plan, the implementation phase, SAMA developed the defined frameworks, tech building blocks and the rollout of activities. In 1H22, the Open Banking ecosystem was formally launched by SAMA, with Malaa Technologies as the first fintech company to obtain a permit from the Saudi Central Bank to engage in open banking activities.

Following its formal launch in 1H22, the open banking ecosystem regulations were developed in 2H22. In November 2022, SAMA issued the Open Banking Framework which includes a comprehensive set of regulatory guidelines, legislation and technical standards to enable banks and fintechs to provide open banking services in Saudi Arabia. The key elements of the framework include: 1) Use Cases: an overview of the prioritisation and details of use cases, 2) Business Rules: details on the requirements that must be followed by banks and fintechs if they want to provide open banking services, and 3) KSA Standards: these include customer experience guidelines, API specifications, implementation requirements and operational guidelines. In April 2023, SAMA announced the launch of the Open Banking Lab which aims to enable innovation and accelerate the development of Open Banking services in KSA. The Open Banking Lab will also be responsible for ensuring conformance with the Open Banking Framework.

Exhibit 154: Regulatory framework time line of the Saudi banks open banking system



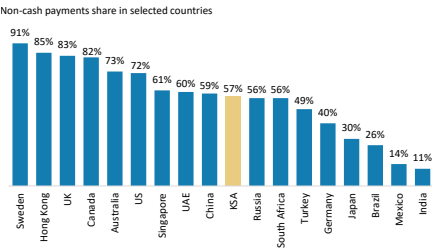
Source: SAMA, Morgan Stanley Research

Banking service scope and ecosystem

A key pillar of Vision 2030 is to develop Saudi Arabia into a global Fintech hub to foster competition and innovation in financial products. Saudi Arabia's young demographics, ongoing cultural shifts, and tech savvy population set the stage for innovation adoption and traditional banking disruption. The share of cashless transactions for individuals is targeted to grow to 70% by 2025 and 80% in 2030 (see [here](#)). FSDP also aims to develop and promote growth of the payments sector. Payments are not just about payment processing revenues - they are about getting access to deposits, customer data, and translating this into financial products - putting incumbent banks' revenues at risk. Penetration of new payment options given the shift in Saudi Arabia, such as digital wallets and account to account transfers, is set to grow with Open Banking. According to the [FIS Global Payments Report 2023](#), digital wallets already represent 16% of point of sale

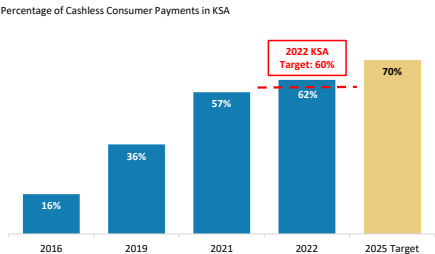
payment methods and 22% of e-commerce transaction values, whilst account-to-account transfers are 16%, and FIS expects this shift to continue. In fact, FSDP targets for 80% of total transactions to be cashless by 2030, increasing 18pp vs. the 2022 level of 62% (2pp above the targeted 60% level for the year). We think that key governmental initiatives such as SARIE instant payments alongside the emergence of new fintechs in the payment sector will play a key role in helping the Kingdom achieve its target.

Exhibit 155: KSA has room to grow its share of non-cash payments with only 57% (as of 2021) of payments being digital...



Note: as of 2021. Source: SAMA National Payments Usage Study, Morgan Stanley Research

Exhibit 156: ... but is on the right track with KSA's non-cash transactions in 2022 surpassing the country's target by 2 pp



Source: SAMA, Saudi Payments Company, Morgan Stanley Research

Relevant research

[Saudi Banks: Open Banking Decoded: Payments: What's at stake?](#)

South Africa

James Starke

Competitive landscape

Digital banks in operation

There are a few pure play digital banks in South Africa, two of which are Bank Zero and Discovery bank, both unlisted. These banks all launched operations within the last five years.

South Africa		
	Bank Zero	Discovery Bank
License granted	2019	2017
Launched	Soft: 2020, Full: 2021	Soft: 2019, Full: 2021
Largest Shareholder	Not disclosed	Discovery
Customer Base	Retail: Mass market, price sensitive consumers. Business: cost conscious SMEs	Mid-market to affluent customers
Key Services	Both Retail and Business transactional accounts, debit cards, savings and cash investment accounts, does not do credit	Retail transactional accounts, debit cards, savings and cash investment accounts. Mortgages added in 2023.
Key differentiation	Patented debit card with enhanced security features, allowing debit cards to have three different card numbers, which enhances security / reduces fraud. Also tech stack internally developed to reduce costs/reliance on 3rd party IT vendors	Banking products integrate with the Vitality rewards program so beneficial client behaviors are rewarded with improved pricing
No. of Customers	Not disclosed	0.7m

Bank Zero was started by Michael Jordaan (former CEO of FNB (FirstRand's retail and business banking franchise) and Yatin Narsai (Head of FNB Retail banking) along with 5 other co-founders and holds a Mutual banking licence (in contrast to peers which hold commercial banking licenses).

Discovery Bank is 100% owned by Discovery Limited, a South African insurer, and describes itself as a shared-value or behavioral bank. Discovery aim to combines technology and digital capabilities with it's expertise in behavior change (via the Vitality rewards program) to incentivize clients to behave in ways that are beneficial to the bank and in turn Discovery shares some of the value back to the client via rewards and improved product pricing. The value proposition is designed to integrate with Discovery's existing financial products including life insurance, medical aid, car & home insurance and investments.

Exhibit 157: Financial performance summary

Item	Bank Zero				Discovery Bank				System			
	Jun 21	Jun 22	y/y	3yr	Jun 21	Jun 22	y/y	3yr	Jun 21	Jun 22	y/y	3yr
NIR					214	318	+49%	+122%	177 381	152 273	-8%	+5%
NIR					882	633	(28%)	+4%	126 981	135 645	+7%	+2%
Revenue					1 096	951	(13%)	+17%	304 362	327 917	+8%	+3%
Opex					(1 673)	(1 597)	(5%)	+25%	(179 059)	(190 160)	+6%	+3%
PP&OP					(577)	(646)	(12%)	+41%	(24 709)	(37 757)	+50%	+4%
Credit losses					(272)	(87)	(78%)	+99%	(46 532)	(31 548)	(32%)	+1%
Headline earnings					(766)	(772)	(2%)	+62%	55 029	74 813	+36%	+4%
Gross Loans (Rbn)	0.00	0.00	n/a		3.8	4.3	+14%	+589%	4 581	5 005	+9%	+5%
Deposits (Rbn)	0.02	0.09	+473%		9.0	10.9	+21%	+693%	5 144	5 660	+10%	+7%
Loan/Deposit Ratio					43%	40%			89%	88%		
NIM					2.8%	3.3%			3.2%	3.3%		
NIR/Deposits					3.3%	3.2%			3.5%	3.6%		
NIR/Deposits					13.6%	6.4%			2.5%	2.5%		
CLR					175%	197%			59%	58%		
CLR					7.1%	1.6%			1.1%	0.7%		
NPL Ratio					n/a	n/a			5.1%	4.5%		
Provision / Gross loans					12.2%	9.3%			n/a	n/a		
ROA					-5.7%	-4.8%			86.0%	114.0%		
ROE					-16.9%	-15.7%			11.3%	14.4%		

Source: Company data, SARB, RMB Morgan Stanley Research. Note: Limited disclosures available on Bank Zero.

Profitability rather than growth, remains a challenge for digital banks, centered around generating enough revenue to cover operating costs. While improving balance sheet scale may help, we note that fee structures (ie NIR), relative to deposits are already favourable compared to the South African banking system, leaving these banks vulnerable to competitive pressures and an ever increasing balance sheet targets to scale to.

Incumbent banks vs. digital banks

The sector remains concentrated in the larger established banks, with the 6 largest domestic banks (in a system with 39 banks) controlling c92% of system assets off c85% of system equity. The digital banks remain relatively immaterial to the system with 0.5% of system assets and 1.1% of system equity.

Exhibit 158: Balance sheet market share per filings with the Prudential Authority

Market share (% of system)	Top 6							Chas	
	Total	ABG	FSR	NED	SBK	CPI	INL	DISY	ZI
August 2023									
Core loans	100.0%	22.7%	21.5%	18.2%	23.5%	2.2%	8.1%	0.1%	0.0
Total retail advances	100.0%	22.7%	21.6%	17.3%	27.8%	4.2%	4.3%	0.3%	0.0
Mortgages	100.0%	23.8%	20.3%	14.5%	34.4%	0.2%	6.6%	0.0%	0.0
Installment finance	100.0%	23.9%	24.2%	36.1%	14.7%	0.0%	1.4%	0.0%	0.0
Unsecured retail	100.0%	19.1%	22.8%	10.7%	21.5%	17.3%	0.8%	1.1%	0.0
Credit Card	100.0%	26.2%	25.5%	10.6%	23.3%	5.7%	1.2%	3.5%	0.0
Overdrafts	100.0%	28.4%	21.2%	12.3%	37.1%	0.5%	0.0%	0.1%	0.0
Other unsecured	100.0%	14.2%	21.6%	10.6%	18.3%	25.7%	0.7%	0.0%	0.0
Non-retail advances	100.0%	22.7%	21.6%	19.0%	19.8%	0.6%	11.1%	0.0%	0.0
Commercial property	100.0%	17.4%	8.3%	35.1%	19.3%	1.0%	19.1%	0.0%	0.0
Non-retail installment finance	100.0%	20.9%	27.6%	13.7%	27.8%	1.9%	6.7%	0.0%	0.0
Overdrafts	100.0%	21.9%	36.7%	10.0%	18.9%	1.2%	0.0%	0.0%	0.0
General Commercial	100.0%	25.2%	24.3%	15.1%	19.2%	0.2%	9.9%	0.0%	0.0
Provisions	100.0%	19.5%	18.7%	14.9%	25.8%	11.3%	1.9%	0.3%	0.0
Private sector debt instruments	100.0%	6.0%	40.5%	15.9%	25.1%	0.0%	11.9%	0.0%	0.0
Repos and interbanks assets	100.0%	17.8%	14.1%	9.3%	30.5%	2.4%	9.3%	0.1%	0.0
Liquid assets	100.0%	17.2%	22.5%	15.9%	19.1%	5.6%	7.1%	0.5%	0.0
Total Assets	100.0%	20.4%	21.3%	15.8%	24.1%	2.8%	7.9%	0.3%	0.0
Total deposits	100.0%	21.4%	21.8%	17.6%	22.5%	2.8%	8.0%	0.3%	0.0
Retail deposits	100.0%	21.0%	22.6%	16.5%	20.0%	8.2%	8.2%	0.8%	0.0
Commercial deposits	100.0%	18.8%	29.6%	15.8%	22.3%	0.4%	6.5%	0.0%	0.0
Wholesale funding	100.0%	23.0%	22.1%	24.5%	35.0%	0.2%	11.1%	0.0%	0.0
Interbank funding	100.0%	64.9%	9.1%	10.8%	16.3%	0.0%	2.4%	0.0%	0.0
Foreign Currency Funding	100.0%	17.3%	19.1%	14.0%	22.0%	0.3%	8.5%	0.1%	0.0
Repo & Collat borrowing	100.0%	40.3%	17.4%	10.0%	18.4%	0.0%	11.5%	0.0%	0.0
Total Equity	100.0%	18.1%	19.0%	13.5%	20.6%	6.5%	6.7%	1.0%	0.0

Note: Wholesale funding includes deposits and other funding (loans, debt securities etc) from the financial sector (eg pension funds, insurers, unit trusts, PIC etc). Interbank funding includes ZAR denominated deposits and other funding from banks.

Source: SARB, RMB Morgan Stanley Research

Exhibit 159: Balance sheet growth per filings with the Prudential Authority

YoY - August 2023	System	ABG	FSR	NED	SBK	CPI	INL	DSY	ZER
Core loans	5.4%	4.1%	9.4%	6.2%	2.2%	9.9%	7.4%	18.0%	n/a
Total retail advances	6.0%	6.7%	7.4%	6.5%	4.3%	5.8%	5.8%	18.0%	n/a
Mortgages	5.4%	4.7%	6.8%	7.8%	4.4%	-24.1%	5.5%	n/a	n/a
Installment finance	7.4%	11.6%	9.7%	6.2%	-0.5%	-81.6%	18.8%	n/a	n/a
Unsecured retail	6.4%	8.5%	7.0%	2.9%	6.6%	7.5%	-2.3%	18.0%	n/a
Credit Card	9.1%	9.1%	12.2%	2.4%	2.0%	19.4%	7.4%	20.9%	n/a
Overdrafts	0.6%	1.1%	-8.9%	22.7%	1.0%	-32.2%	n/a	-67.2%	n/a
Other unsecured	5.9%	10.4%	6.6%	0.5%	11.6%	6.5%	-9.4%	n/a	n/a
Non-retail advances	4.9%	2.0%	10.9%	5.9%	-0.1%	45.5%	7.9%	n/a	n/a
Commercial property	4.1%	8.5%	9.7%	1.8%	0.2%	59.2%	4.0%	n/a	n/a
Non-retail installment finance	15.7%	9.3%	30.1%	17.0%	7.0%	137.7%	12.6%	n/a	n/a
Overdrafts	0.6%	-18.0%	16.0%	5.1%	1.6%	45.2%	-100.0%	n/a	n/a
General Commercial	4.6%	2.5%	7.8%	8.8%	-1.6%	9.2%	14.2%	n/a	n/a
Provisions	12.7%	14.0%	6.7%	15.9%	14.9%	14.7%	20.6%	18.9%	n/a
Private sector debt instruments	-6.6%	7.4%	2.6%	15.6%	-18.5%	n/a	-32.1%	n/a	n/a
Repos and interbanks assets	18.3%	-18.1%	35.6%	8.9%	56.2%	-15.3%	74.2%	27.3%	99.4%
Liquid assets	4.2%	16.1%	-0.5%	7.4%	-0.1%	16.3%	-2.7%	15.3%	0.1%
Total Assets	5.9%	1.9%	8.1%	5.1%	7.1%	9.0%	11.1%	21.0%	71.7%
Total deposits	8.2%	7.7%	12.0%	6.4%	6.5%	7.8%	7.7%	28.4%	120.4%
Retail deposits	7.9%	5.3%	5.0%	10.0%	9.9%	7.5%	14.1%	31.2%	145.5%
Commercial deposits	5.4%	0.4%	10.0%	4.7%	3.0%	8.5%	14.2%	n/a	81.0%
Wholesale funding	11.4%	15.7%	18.4%	2.4%	9.9%	49.2%	-0.4%	-72.8%	n/a
Interbank funding	8.2%	1.1%	37.8%	52.8%	-5.7%	-93.2%	114.0%	n/a	n/a
Foreign Currency Funding	8.3%	29.4%	6.1%	24.2%	-9.5%	35.1%	30.7%	136.7%	n/a
Repo & Collat borrowing	6.9%	-18.0%	50.5%	1.2%	190.5%	n/a	76.3%	n/a	n/a
Total Equity	2.8%	3.7%	-1.7%	-2.0%	10.0%	11.3%	-6.1%	5.5%	21.8%

Source: SARB, RMB Morgan Stanley Research

While deposit growth has shown good momentum for all digital banks, their share of system deposits remains limited and profitability elusive, suggesting the narrow business models, largely focused on consumers without a physical point of presence, may have to be adapted. With current offerings may not hold broad enough appeal to achieve the scale required to cover operating costs, we expect these digital banks will have to broaden offerings in the pursuit of scale. This is already underway with Discovery Bank now offering mortgages. With this, will come increased scale and complexity, which in turn drives operating costs further.

Overall regulatory and policy landscape

South Africa does not have specific regulations or policies that pertain specifically to digital banks, although the South African Reserve Bank permission required to operate exclusively online. There are three types of banking license in South Africa, Commercial bank, Mutual bank or Co-Operative Bank. The SARB then further approves activity/products. Discovery Bank operates on locally controlled (commercial) banking licenses while Bank Zero opted for the mutual bank license. Mutual Bank has lower minimum prescribed capital requirements (c25x lower per Bank Zero), which Bank Zero reasoned that mutual bank structure allows for efficient use of capital and that can be passed onto customers. More generally South Africa has 39 banks which includes 14 locally controlled banks, 3 mutual banks, 6 co-operative Banks, 4 foreign controlled banks and 12 foreign branches. General requirements have been summarized below.

Exhibit 160: Summary of South African banking regulations

South Africa	
How many licenses to be provided and/or approved?	No specific "digital bank" license, normal banking license requirements apply.
Who can apply ?	There are no restrictions on natural or juristic persons nor local versus foreign persons applying for banking licences, however applicants will need to demonstrate the required capital, skills, experience and capacity to operate a bank
Other Requirements	Prescribed fees (less than R100k) to be paid.
Minimum paid-up capital	Commercial bank: R250m Mutual bank: R10m Co-operative bank: R5m
Size limits	Minimum capital requirements as above, else no size limits.
Capital and liquidity Rules	11.5% minimum total capital, 4% minimum leverage ratio and other minimum capital adequacy ratios as prescribed by the Prudential Authority. Newly established banks may be subject to a higher minimum CAR in their formative years, depending on the risk profile of the proposed operations. Foreign-bank applicants are expected to meet comparable capital adequacy standards.
Business restrictions	Based on the category of license (Commercial, Mutual, Co-operative)
Timeline	6-9 months

Source: SARB, Morgan Stanley Research

Banking service scope and ecosystem

Exhibit 161: Business Scope - Comparison

		Bank Activities				Deposits			Retail Lending				Corporate Lending			Peripheral/Cross-selling Services																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
Category	Service Provider	View activity & balance	Fixed transfer (International)	FX services	FX remittance	Fixed Deposits	Time Deposits	Student loan	Retail unsecured	Mortgage/home equity	Collection	Multiple prime	Legal corporate	SME	SOE	Guarantee	Payment services (Debit card)	Primary bank for local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local 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govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local govt	Local 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govt	Local govt	Local govt	Local

Source: Company data, Morgan Stanley Research

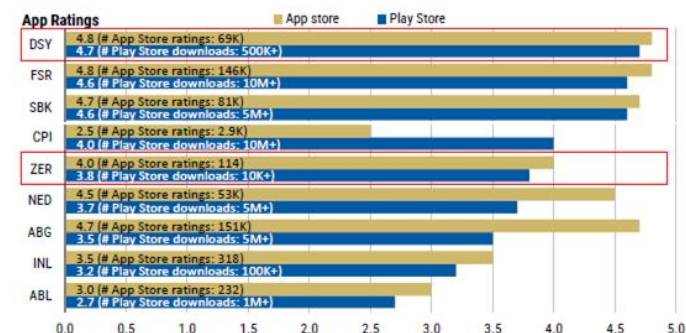
Digital banks tend to focus on consumer transactional activities and deposit taking rather than corporate banking.

While Bank Zero has business banking products (targeting small and medium size enterprises) the scale and breath of product is limited. Retail lending tends to be unsecured, when it is done. Interestingly Discovery Bank recently launched a mortgage product, which we sense was necessary to improve relevance/client primacy in its mid-to-affluent market segment. We see the choice of retail lending these digital banks do as being a direct function of their ability to derive a data advantage to feed into their credit underwriting strategies. For example, Discovery Bank can leverage its size Vitality program data insights into their credit card business.

Cross-selling tends to be limited among digital banks in South Africa. Given that digital banks are relatively new, their operating models are relatively simple. In time, we do see

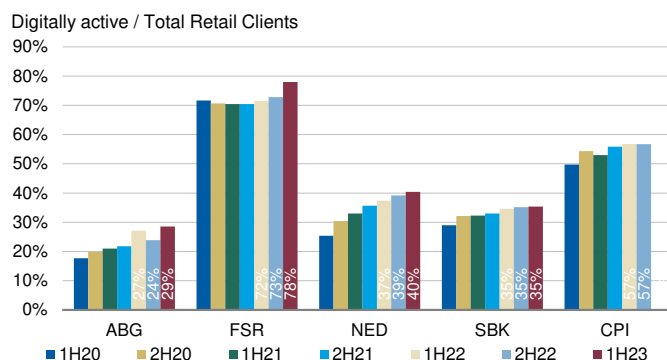
these digital banks broadening out their product offerings to improve their relevance and client primacy. Discovery Bank differs from peers, in that it's a subsidiary of a large insurer, which already has an entrenched client base. Furthermore, the Discovery's Insurance clients, tend to be more affluent and as a result require a more comprehensive product suite.

Exhibit 162: App ratings - a proxy for digital experience - show digital banks not setting themselves apart from incumbents



Source: Google Play Store, Apple App Store, Morgan Stanley Research

Exhibit 163: Incumbent banks proportion of retail clients that are digitally active



Source: Company data, Morgan Stanley Research

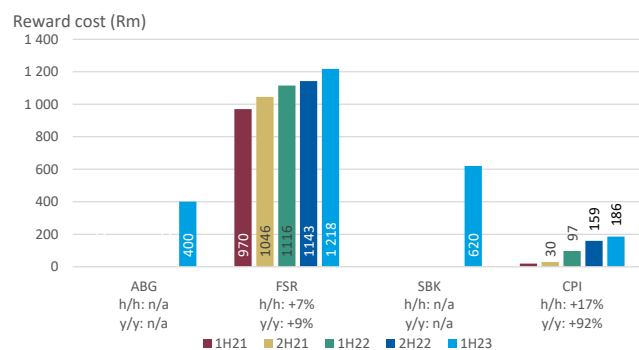
In terms of digital user experience digital banks, as a cohort, are not the leaders. While Discovery Bank profiles well, based on app ratings, Bank Zero lags some incumbent banks on user digital experience, a critical element for a consumer facing, digital only business model. We see increasing pressure on digital banks ability to compete coming from incumbent banks as the incumbents improve their digital offerings and user experience.

Levels of digital adoption among clients of some incumbent banks remains, remains low, suggesting a material part of the South Africa retail banking market is not addressable for digital banks. The incumbent banks all have digital channels, however digital adoption, while steadily increasing, is relatively low for Absa, Nedbank and Standard Bank. Even Capitec, with a youthful client demographic and simple product offering still has 43% of clients that are not digitally active.

Key nuances that matter

Digital banks typically face higher funding costs than incumbents, prompting changes to product offerings. This in turn requires digital banks to focus on higher-yielding asset strategies which, in the South Africa context, requires focusing on unsecured consumer credit. However, without a suitable proprietary data advantage, it's unclear to us how digital banks avoid adverse selection. Attracting transactional deposits would improve funding costs and we credit origination strategies and product offerings playing a role in improving client primacy and liability gathering. This explains Discovery's push into mortgages (improve client primacy and attract consumers salary deposits in the mid-to-affluent market).

Reward programs offer both challenger and incumbent banks the opportunity to improve client primacy. Incumbent banks in South Africa are increasingly using reward programs to drive client primacy,

Exhibit 164: Six-monthly reward spend by incumbent banks

Source: Company data, Morgan Stanley Research. Note: Reward spend is not uniformly disclosed

cross-sell and improve their data/insight capabilities. This presents a challenge to newer start-ups without the resources or scale to fund and leverage such reward programs.

US

Jeff Adelson, Betsy Graseck, Ryan Kenny

Competitive landscape

Among US digital banks, share gainers tend to be niche, focused on payments, or partner with incumbent banks. Digital banking in the US has been largely dominated by regulated banks. Yes, consumer deposits have increasingly shifted towards online players who are able to offer consumers higher interest rates for their savings, however the vast majority of these players are still banks. Big tech and FinTech players have made waves in attracting consumer deposits, but have still largely done this via a partnership model with banks, like Apple's foray into deposits and credit card via partnership with Goldman Sachs. While niche players have found success in digital banking, even those examples have tended to pursue a bank charter. This includes SOFI, which is now a regulated bank business, and with a more niche focus than most banks, offering personal loans and refinancing of student loans. And while neobanks have demonstrated success in attracting younger Gen Z consumers into their ecosystems, they in part took advantage of a flux of stimulus and savings in consumers' pockets during the pandemic, and recently have struggled to grow up with their consumers and better monetize the business via an expansion of more mature banking offerings.

Overall regulatory and policy landscape

In the US, incumbent banks dominate the ecosystem for checking accounts and savings accounts, in part driven by required regulatory oversight from the Fed, FDIC and OCC. While fintechs can try to obtain industrial loan company (ILC) charters from the FDIC to access deposit funding without becoming a bank holding company, we have yet to see a significant number of applications approved. As a result, fintechs have been most successful in non-depository activities such as payments and bank partnerships.

Banking service scope and ecosystem

Incumbent US banks have been investing significantly in their mobile and digital capabilities for many years. This includes JPMorgan's US\$15.3B annual tech budget, the highest spend in our coverage. Fragmentation within the US market means market share is up for grabs, making each banks' app, digital connectivity and brand more critical than ever. This has resulted in bank industry tech spend outpacing inflation, with a focus on mobile capabilities, cloud migration, interoperability, customization, insights and automation.

Relevant research

[US Banks: Branch Rationalization Tool Shows Footprints Are Getting More Competitive \(11 Jan 2023\)](#)

[SoFi Technologies, Inc.: Upgrade to EW; Risk Reward Now More Balanced \(31 Oct 2023\)](#)

[SoFi Technologies, Inc.: Too Much Too Soon: Mispriced for Bank Profitability \(13 Jul 2023\)](#)

J.P.Morgan Chase & Co.: Pressing the Tech Advantage (2 Dec 2019)

India

Sumeet Kariwala, Bhavik Shah

Digital payments, digital identity and Goods Service Tax are driving India's progress in digitization of financial services. The government and central bank have been focused on rapidly formalizing and financializing India's economy. To this effect, they have initiated measures to boost financial inclusion (Jan Dhan – no frills account), built a universal biometric identification system (Aadhaar), and implemented real-time payment systems (Unified Payments Interface –UPI and Bharat QR). Another big achievement has been the strong progress on GST and the significant data trail that is being generated on a monthly basis. Combination of these initiatives have led to creation of superior tech architecture and significant generation of transaction-level data for retail customers as well as for small businesses. This tech architecture data trail is widely used by Fintech's/Banks to assess the creditworthiness of borrowers and also provide gamut of financial services. Full fledged adoption of new data sharing architecture such as OCEN and AA is underway and can further help accelerate growth in digital lending and/or other financial services.

That said, India is yet to come up with a full stack digital banking regime. NITI Aayog (a public policy think tank of Govt.) did propose licensing and regulatory regime for digital banks in India but we are yet to see any progress on this front. The report ([link here](#)) highlights that full stack digital banks is the potential solution for the persistent policy challenge of credit deepening and is the next stage of financial inclusion. What we have seen so far is big focus by traditional banks on technology digitization across spectrum of financial services including payments, bank account opening, deposits, lending, and so on. Fintech's are also trying to collaborate with banks and offer digital bank like services. For e.g. Jupiter (a neobank) has tied up with Federal Bank to offer savings accounts, Rupeek (a digital lending platform) has tied-up with various banks to provide online gold loan services, Paytm (listed Fintech) has tied-up with HDFC Bank to distribute co-branded credit card and it also has tie-ups with various non-bank lenders for providing credit (i.e. BNPL, personal loans merchant loans).

Risk Rewards

Risk Reward – Kakao Bank Corp (323410.KS)

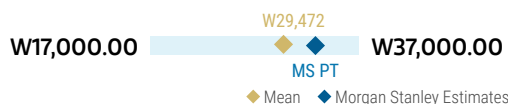
Proving Its Mettle

PRICE TARGET W32,500

Base case, economic profit valuation methodology. Our earnings estimates reflect 2023e-2024e ROE reaching 5.8% and 6.6%. Other key assumptions: cost of equity 10.1%, sustainable ROE: 14.8% (2030e), and terminal growth rate of 8.25%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



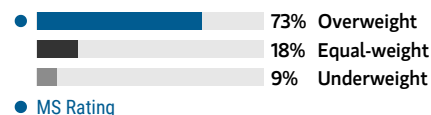
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- KKB is a significant player in the Korean banking market, with asset size of over W46tn and customer base of 21.2mn.
- KKB has the mechanics, motor, and mettle to become a top digital financial services provider in the medium/long term, but has been adversely affected by macro, regulatory and competitive factors.
- However, we are seeing some of the negative factors such as regulation and competition becoming more favorable for KKB, which should enable the bank to rekindle asset growth.
- KKB continues to work to expand its platform business and we believe this should be an important medium/long-term driver for ROE expansion.
- We think risk is now skewed more to the upside than the downside.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

W40,000

Better business conditions; KKB to gain share

Our bull case is based on more favorable macro and business conditions, which should enable KKB to realize higher loan growth and more robust net fee income vs. our base case. It implies higher ROE over 2025-30 vs. the base case (2030 ROE estimate: 16.5%). We assume 20%+ loan growth for both 2023 and 2024 also sustained in the long term.

BASE CASE

W32,500

Elements that hampered KKB's operations unwind

Our base case is based on competitive pricing, fewer regulatory restrictions, improving policy outlook (vs. commercial banks), and no REPF exposure. Our earnings estimates reflect 2023-2024 ROE reaching 5.8% and 6.6% with sustainable ROE of 14.8% being reached in 2030e and a terminal growth rate of 8.25%. We assume 20%+ sustainable loan growth, which is needed to reach optimal leverage level within five to six years.

BEAR CASE

W17,500

Macro, policy and competitive pressures persist

Our bear case is based on less favorable macro and business conditions, with policy direction being more negative and hindering digital banking (including KKB) from showing profitable growth. We assume lower loan growth, higher credit costs, and lower net fee income compared to the base case. ROE over 2025-30 is seen as lower than in the base case (2030 ROE estimate: 12%). We assume loan growth to be lower for 2023 and 2024 than our base case, while credit costs are higher.

Risk Reward – Kakao Bank Corp (323410.KS)

KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
Net Interest Margin (%) (%)	245.4	230.5	230.4	221.0
Loan Growth (%) (%)	783.7	4,147.3	2,717.9	3,169.9
Cost-to-income (%) (%)	4,277.9	3,854.8	3,784.2	3,667.1
Provisioning / Credit (%) (%)	64.6	77.2	74.0	69.8

INVESTMENT DRIVERS

- Strong loan growth taking market share from incumbents
- Platform business fees show strong growth
- Operating costs controlled effectively

GLOBAL REVENUE EXPOSURE



100% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5
MOST 3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Improvement in the regulatory and policy environment
- Expanding scope of lending
- Strong new business launches on its platform side

RISKS TO DOWNSIDE

- Tougher-than-expected regulatory/policy environment, impeding growth and/or expansion into new businesses
- Sharp downturn in the economy, leading to higher credit costs
- A strong new entrant in the market

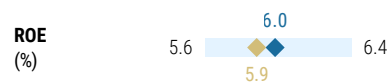
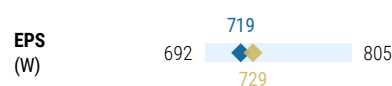
OWNERSHIP POSITIONING

Inst. Owners, % Active 80.1%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2023e



Source: Refinitiv, Morgan Stanley Research

Risk Reward – Rakuten Bank (5838.T)

Japan's No.1 digital bank; contributions to earnings from new business key

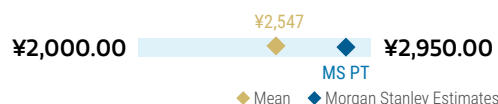
PRICE TARGET ¥2,830

F3/24e BPS ¥1,760 x1.6, EPS ¥250 x11

We use a P/B-based valuation methodology for Rakuten Bank. Leveraging its access to the Rakuten ecosphere may boost the bank's strategic value to levels other banks cannot achieve, hence our target P/B (1.6x) includes a premium to the estimated F3/25 P/B (1.5x).

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

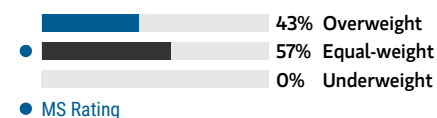
Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

■ Rakuten Bank is Japan's No.1 digital bank with core competencies in group synergy, combining finance & specialized tech. We expect the company to achieve solid growth from rising transactions as more clients use Rakuten Bank as their main account. We think the firm can achieve high ROE through an earnings structure with high operating leverage.

■ However, we think the market already recognizes Rakuten Bank's substantial growth potential and its high operating leverage relative to traditional banks. Based on relative valuations and risk/reward profiles within our banks coverage, we assign an EW rating. Full contributions to earnings from new business hold the key to further upside.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥3,590

Applied Multiple PBR2.0x

Our forecasts:

F3/24e consol. ROE 15.3%, consol. NP ¥36.9bn

F3/25e consol. ROE 17.4%, consol. NP ¥50.7bn

BASE CASE

¥2,830

Applied Multiple PBR1.6x

Our forecasts:

F3/24e consol. ROE 14.8%, consol. NP ¥35.5bn

F3/25e consol. ROE 15.1%, consol. NP ¥43.2bn

BEAR CASE

¥1,820

Applied Multiple PBR1.1x

Our forecasts:

F3/24e consol. ROE 13.5%, consol. NP ¥32.2bn

F3/25e consol. ROE 10.7%, consol. NP ¥29.6bn

Risk Reward – Rakuten Bank (5838.T)

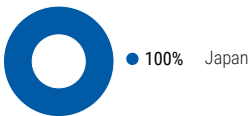
KEY EARNINGS INPUTS

Drivers	2023	2024e	2025e	2026e
Ordinary income (¥, mn)	120,445.0	138,929.8	160,931.5	178,896.6
Net Income (¥, mn)	27,692.0	35,506.2	43,198.0	50,559.3
ROE (%)	13.7	14.8	15.1	15.2

INVESTMENT DRIVERS

We expect that Rakuten Bank can achieve 3-year average ROE at 14% from F3/24 to F3/26 given the bank's synergies with the Rakuten ecosystem and an earnings structure that facilitates operating leverage effects,

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

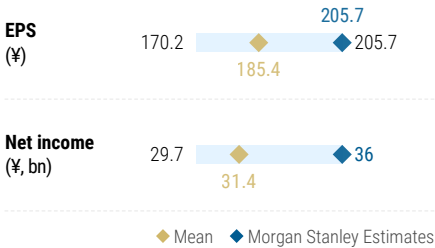
- Better yields via an improving balance of loans and monetary claims bought
- Greater than expected JGB investment opportunities
- Increasing ratio of main accounts
- Earnings contributions from advertising business

RISKS TO DOWNSIDE

- Loan balance slowdown due to competition
- The ratio of main account business may not rise
- Monetization of data business based on financial transactions data may not progress as much as expected

MS ESTIMATES VS. CONSENSUS

FY Mar 2024e



Source: Refinitiv, Morgan Stanley Research

OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

Risk Reward – SBI Sumishin Net Bank (7163.T)

One of the main digital banks; BaaS holds key to earnings growth

PRICE TARGET ¥1,920

F3/25e BPS ¥1,296 x 1.5, EPS ¥183 x 10

We use a P/B-based valuation methodology for SBI Sumishin Net Bank.

Our forecasts:

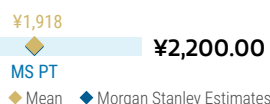
F3/24e consol. ROE 15.7%, consol. NP ¥24.4bn

F3/25e consol. ROE 15.2%, consol. NP ¥27.6bn

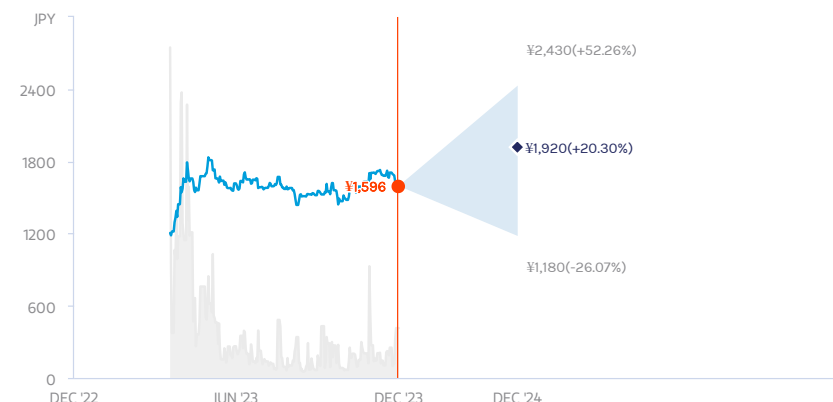
Consensus Price Target Distribution

¥1,600.00 ¥1,918 ¥2,200.00

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

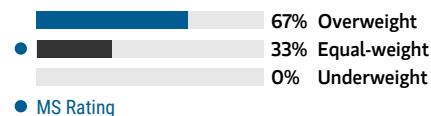
EQUAL-WEIGHT THESIS

- With the mortgage lending balance growing thanks to brand strength and price competitiveness (0.32%/year with variable interest), we think the market is already aware of the appeal of strong operating leverage and business model expansion capacity by comparison with traditional banks.

- Meanwhile, we think further share price upside would require evaluation in terms of SaaS-type metrics that draw a distinction with traditional banks, as the profit weighting of BaaS business rises.

- We consider the possibility that the bank may be outperformed by stocks that are more sensitive to yen interest rates, as its own earnings would not be much affected relatively by the removal of NIRP or YCC.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

BULL CASE

¥2,430

Applied multiple P/B 1.8x

F3/25e BPS ¥1,355 x 1.8, EPS ¥225 x 11

Our forecasts:

F3/24e consol. ROE 17.2%, consol. NP ¥27.0bn

F3/25e consol. ROE 18.1%, consol. NP ¥33.8bn

BASE CASE

¥1,920

Applied multiple P/B 1.5x

F3/25e BPS ¥1,296 x 1.5, EPS ¥183 x 10

Our forecasts:

F3/24e consol. ROE 15.7%, consol. NP ¥24.4bn

F3/25e consol. ROE 15.2%, consol. NP ¥27.6bn

BEAR CASE

¥1,180

Applied multiple P/B 1.0x

F3/25e BPS ¥1,244 x 1.0

Our forecasts:

F3/24e consol. ROE 14.3%, consol. NP ¥22.1bn

F3/25e consol. ROE 12.5%, consol. NP ¥22.1bn

Risk Reward – SBI Sumishin Net Bank (7163.T)

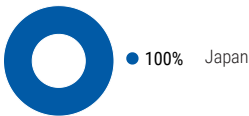
KEY EARNINGS INPUTS

Drivers	2023	2024e	2025e	2026e
Ordinary income (¥, mn)	29,390	33,706	39,430	41,957
Net Income (¥, mn)	19,932	24,405	27,597	29,366
ROE (%)	13.4	15.7	15.2	14.0

INVESTMENT DRIVERS

We assume CAGR for our 3-year forecasting period of +13.8% for consolidated net income. As the earnings structure does not see much volatility for credit costs or marketable securities income, we expect profit to increase as the loan balance grows.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger than expected mortgage loan growth, balance of card loans etc.
- Monetization of BaaS business advances more quickly than expected
- Potential dividend increase

RISKS TO DOWNSIDE

- Mortgage lending competition proves to be stiffer than expected
- Monetization of BaaS business advances more slowly than expected
- If asset control does not advance as expected

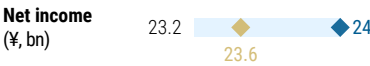
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Mar 2024e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Nu Holdings Ltd. (NU.N) Top Pick

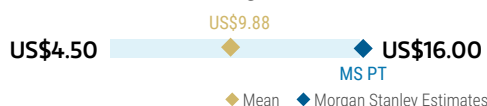
Overweight on attractive growth avenues and secular tailwinds

PRICE TARGET US\$16.00

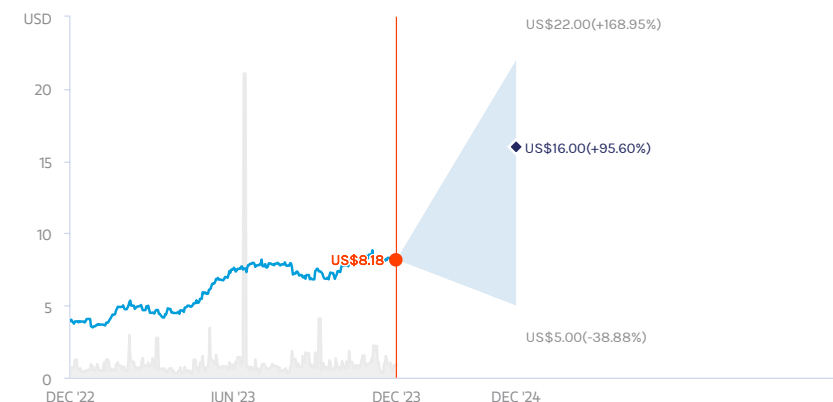
Our valuation work and price targets are based on residual income analysis. In our residual income model, the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). Our valuation model uses a discount rate of 14.2% and long-term ROE of 35%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



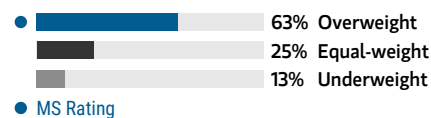
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

Nubank is well-positioned to build one of the most valuable banking franchises in Latam with superior technology, best-in-class customer satisfaction, and strong cohort performance and unit economics. The company has many attractive avenues to increase top line growth: roll out & cross-sell of new products to a huge and loyal client base, new geographies, and M&A. We think the market is massively underestimating its ability to deliver growth and profitability — particularly cross-selling personal and payroll loans in Brazil and building a successful franchise in Mexico and Colombia. We see adjusted ROE reaching ~35% in 2028 on the back of strong revenue growth and operating leverage.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Positive
Disruption: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

US\$22.00

Faster industry growth & successful cross selling

Secular industry trends surprise to the upside, digitalization and penetration of financial products grow faster than expected. Nubank can capitalize on its huge and loyal customer base by cross selling an increasing number of products, significantly improving client monetization. As a result, top line growth accelerates, while operating leverage allows additional revenues to flow down to the bottom line, supporting higher sustainable ROEs.

BASE CASE

US\$16.00

Secular tailwinds & client base expansion

Financial inclusion and credit penetration continue to improve on the back of currently low bancarization levels in Latam and ongoing digitalization trends. Nubank can successfully expand into new geographies, continuing to add new clients at a healthy pace, and launching new products in parallel. The company navigates competition effectively and gains market share. Its highly scalable business model drives attractive operating leverage and sustainable ROEs.

BEAR CASE

US\$5.00

Regulatory headwinds & tougher competition

Macro outlook deteriorates and industry growth slows down more than expected. Competition gets tougher, as new players and incumbents successfully expand digital banking initiatives. Expansion to new geographies and product launches take longer than expected to gain traction, resulting in relatively slower revenue growth. Regulatory hurdles increase pressuring profitability in the short term.

Risk Reward – Nu Holdings Ltd. (NU.N)

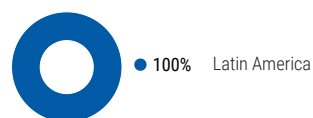
KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
Active Clients (MM)	61.2	77.8	89.6	100.0
Net Revenue (YoY, %) (%)	182.2	67.3	47.2	35.1
Gross Profit (YoY, %) (%)	126.9	105.6	62.8	43.5
Card Receivables (YoY, %) (%)	72.2	49.9	46.8	23.5
Loan to Customers (YoY, %) (%)	40.1	86.8	141.0	94.7

INVESTMENT DRIVERS

- Financial inclusion and credit penetration
- New client additions
- Roll out & cross sell of products
- Geographical expansion
- M&A
- New business verticals and products

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
---------------------	-----------------------------	---------------------	----------------------------

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Increasing penetration of financial products in LatAm
- Successful geographical expansion
- New products added to the platform
- Improving cross-sell initiatives
- M&A

RISKS TO DOWNSIDE

- Asset quality deterioration
- Regulatory headwinds
- Tougher competition could impact margins
- Execution risk of international expansion
- Potential conflicts of interest between controlling and minority holders

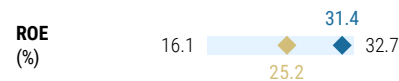
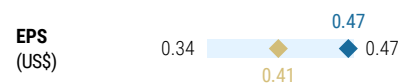
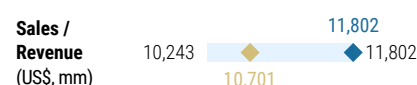
OWNERSHIP POSITIONING

Inst. Owners, % Active	93.7%	
HF Sector Long/Short Ratio	1.7x	
HF Sector Net Exposure	11.1%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2024e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – SoFi Technologies, Inc. (SOFI.O)

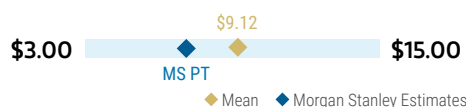
Risk Reward Now More Balanced; Equal-Weight

PRICE TARGET \$7.00

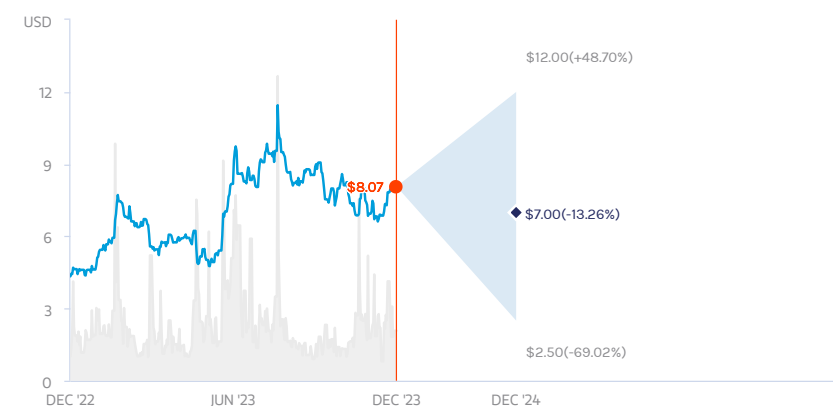
Our price target is set by triangulating 1.3x P/TBV on the bank + 5x P/Sales on Tech Platform

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



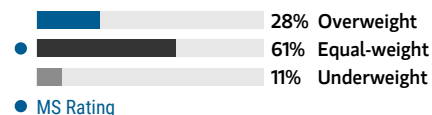
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- SOFI has been one of the strongest growth stories in consumer finance, including '22 revenue growth of 50%+ y/y; but this has been fading. Still, with recent stock weakness, now see little downside to shares at current levels.
- Our view of the bank business' longer-term profitability potential, driving to a 14% ROTCE by 2026, justifies a 1.3x P/TBV for the bank in our SOTP (driving an overall 2.0x P/TBV target valuation for SOFI including Tech Platform)
- Incrementally better '24 estimates on 3Q on NIM strength (stronger growth in loans, deposits)
- Capital raise risk likely lower for now with 1) \$475M of loan sales coming in 4Q and 2) a \$2B flow agreement over 12 months.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: *Negative*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$12.00

3x P/TBV for Bank, 6x P/Sales for Tech Platform

Fed Cuts Rates Sharply on Soft Landing, Originations Surge Post Moratorium End. Fed successful in bringing inflation down, cuts rates to 2% or lower. Student borrowers refi en masse post moratorium with lower rates. Faster customer acquisition on a shift from traditional banks to SOFI. Bolstered by cross sell to new Technisys acquisition as FI incumbents adopt core banking platform

BASE CASE

\$7.00

1.3x P/TBV for Bank, 5x P/Sales for Tech Platform

2024 revenue growth slows below consensus on slowing fee income, but in the price. Expiration of student loan moratorium enables a return to growth in student loan refi in 2H23. Capital cushions continue to moderate, but risk-based regulatory capital ratio stays above 12% as loan sales ramp in 4Q23/2024. Lending revenues income contract in 2024 on higher rates, rising NCOs, pull-to-par dynamic, and slowing loan growth. We see revenue growth slowing from ~50% in 2022 to 20%/17% in 2024/25.

BEAR CASE

\$2.50

1x P/TBV for Bank, 2x P/Sales for Tech Platform

Financial Services Fails to Fly; Student Loan Moratorium Keeps on Going. Competition slows member growth, keeping a lid on growth. SOFI fails to achieve GAAP net income profitability. Continued ramp in loan growth, sluggish loan sales quickly erode capital levels, potentially driving capital raise. Tech Platform revenue growth remains sluggish. Student refi volumes slow amid still elevated rates. Gain on sale income turns negative as credit losses ramp sharply, pull-to-par impacts.

Risk Reward – SoFi Technologies, Inc. (SOFI.O)

KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
Revenue Growth (%)	52.5	33.7	20.0	16.6
Contribution Margin (%)	34.0	42.3	42.0	42.5
Adjusted EBITDA Margin (%)	9.3	19.2	21.0	24.0
EPS	(0.40)	(0.13)	0.02	0.14

CATALYST CALENDAR

Date	Event
29 Jan 2024 - 02 Feb 2024	Q4 2023 SoFi Technologies Inc Earnings Release
29 Apr 2024 - 03 May 2024	Q1 2024 SoFi Technologies Inc Earnings Release

Source: Refinitiv, Morgan Stanley

INVESTMENT DRIVERS

- Student volumes ramp in 2023, personal loan volumes continue taking share
- Loan fee income shrinks in 2024
- But NII coming in stronger
- Capital ratio moderates to 12% in '24; but staying above requires meaningful resumption of loan sales in 4Q23/FY24

RISKS TO PT/RATING

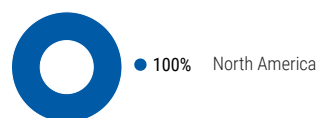
RISKS TO UPSIDE

- Tech Platform growth re-accelerates in 4Q23 into 2024
- Student originations continue to surge post moratorium
- Expenses decline sharply, driving bank business ROTCE quickly above 20%
- Loan sale execution ramps, lowering risk of capital erosion/raise

RISKS TO DOWNSIDE

- Slower loan fee income growth on elevated rates, credit deterioration, pull-to-par
- Competition intensifies, expenses high
- Student refi slows
- Regulatory

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	5/5 MOST	3 Month Horizon
-------------	---------------------	-------------	--------------------

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

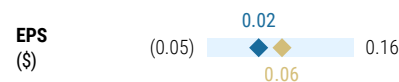
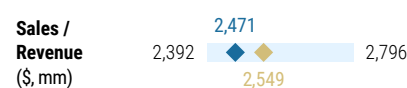
OWNERSHIP POSITIONING

Inst. Owners, % Active	45.6%	
HF Sector Long/Short Ratio	1.7x	
HF Sector Net Exposure	11.1%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2024e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Bank Jago Tbk PT (ARTO.JK)

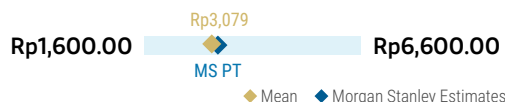
Digital banking pioneer

PRICE TARGET Rp3,208

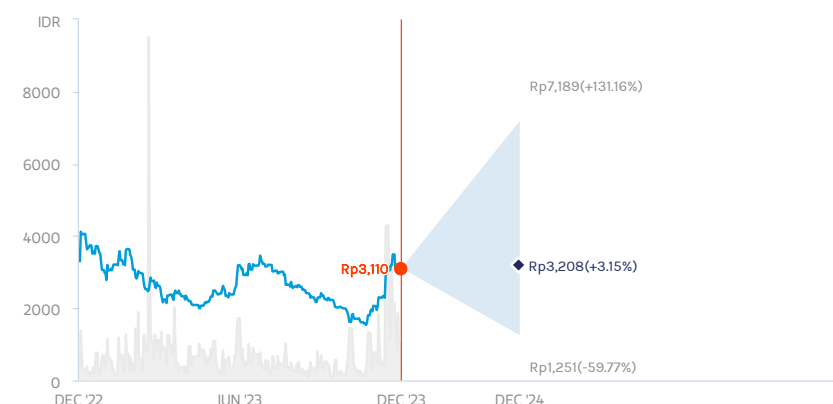
Gordon Growth model over a 20-year horizon of financial forecasts, probability-weighted 10% bull, 50% base, 40% bear. Skew reflects the impact of higher rates on this growth stock in our bear case. Key base case assumptions: cost of equity of 13.4% (risk premium 4.4%, risk-free rate 6.5%, beta 1.6), sustainable ROE 18.0%, sustainable growth 12.6%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

BULL CASE

Rp7,189

12x bull case 2024e BVPS

We assume loan growth of 5ppts and CIR is 5ppt lower than our base case. We expect RoE to rise to 25% in the medium term and we assume a sustainable growth rate of 12.6%.

BASE CASE

Rp3,977

7.3x base case 2024e BVPS

We see Jago continuing to increase its customer base, and generating loan growth of 37% and NIM at 9.5% in 2023. We expect RoE to rise to 20% in the medium term and we assume a sustainable growth rate of 12.6%.

BEAR CASE

Rp1,251

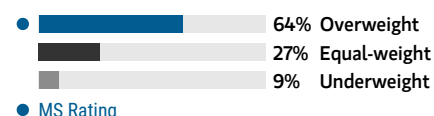
4.0x bear case 2024e BVPS

We assume loan growth is 10% lower than in our base case, and NIM falls to 9% in the medium term. We assume a sustainable growth rate of 12.6%.

OVERWEIGHT THESIS

- ARTO maintains its position as a pioneer of digital banking in Indonesia.
- Despite initial adjustment to credit underwriting, ARTO remains profitable and we expect 2022-2025e net profit CAGR of 219%.
- Valuations have more than priced in credit risks and expectations have come down a lot.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption: *Positive*

View descriptions of Risk Rewards Themes [here](#)

Risk Reward – Bank Jago Tbk PT (ARTO.JK)

KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
Loan growth (%)	75.6	37.6	38.3	37.8
NIM (%)	10.4	9.5	10.5	10.0
Fee growth (%)	552.8	150.0	100.0	70.0
Cost to income ratio (%)	71.7	69.4	60.2	53.0
Loan loss charge as % of gross loans (%)	5.3	4.0	4.8	4.8

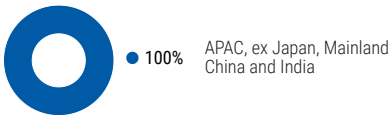
CATALYST CALENDAR

Date	Event	Source: Refinitiv, Morgan Stanley
24 May 2024 - 28 May 2024	Bank Jago Tbk PT Annual Shareholders Meeting	

INVESTMENT DRIVERS

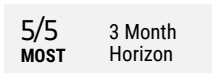
- Loan growth
- Credit costs
- NIM

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS



Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Partnership with other large Indonesian tech startups to drive faster earnings growth
- Faster-than-expected public adoption of digital banking
- Lower-than-expected credit costs amid successful credit scoring automation

RISKS TO DOWNSIDE

- Uncertainties in global rates environment
- Execution risk
- Faster-than-expected competition
- Unexpected regulatory headwinds

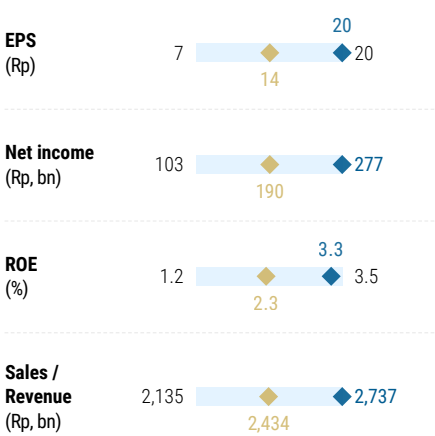
OWNERSHIP POSITIONING



Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2024e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Inter & Co Inc. (INTR.O)

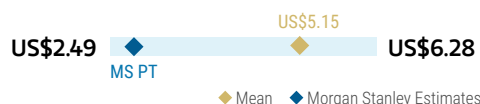
We are skeptical about the path to profitability, reiterate Underweight

PRICE TARGET US\$2.80

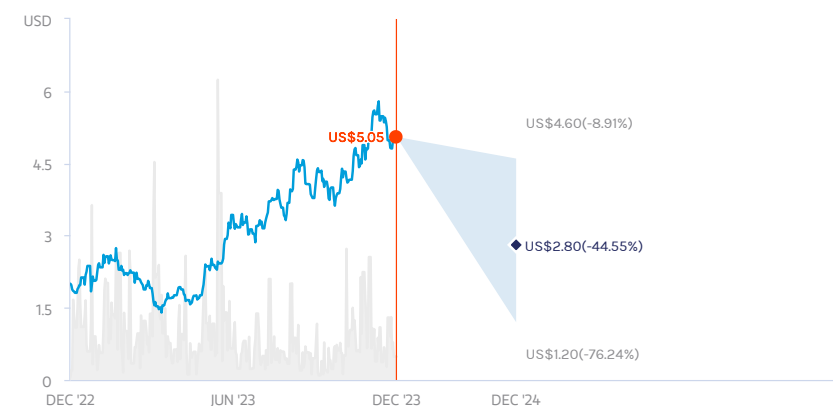
Our valuation work and price targets are based on residual income analysis. In our residual income model, the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). Our valuation uses a discount rate of 14.3%, and long-term ROE of 6.8%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



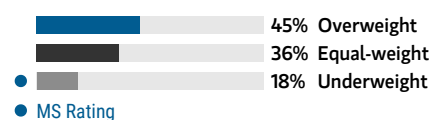
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

Inter & Co is undergoing a transformation that could result in faster growth and higher profitability. We believe it is too soon to tell if the efforts will prove successful. There are several key challenges that may hamper the bank's aggressive growth plans, including: difficulties around cross selling and client monetization, rollout of new products and geographic expansion where management has limited expertise, low profitability, and a tough competitive environment. Valuation seems high for a bank with long-term ROE below cost of equity.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Negative
Disruption: Positive
Self-help: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

US\$4.60

Adoption & operating leverage improve

Structural reforms, accommodative financial conditions and stabilization of global trends results in better than expected economic recovery. Loan growth is strong and delinquencies stay under control. NIM improves due to shift in mix, while fees benefit from cross selling initiatives. Operating leverage improves. ROE improves above the cost of equity.

BASE CASE

US\$2.80

Slow ramp-up, low ROE persists

The economy stabilizes and recovers at a modest pace, and so does investment, unemployment, and credit growth. However, client monetization remains limited, as competition remains tough. Credit growth is weak, but asset quality stays under control. Expenses continue to grow at a fast pace, limiting efficiency gains. ROE improves but stays below cost of equity.

BEAR CASE

US\$1.20

Execution challenges materialize

Tough economic conditions persist and the government fails to pass structural reforms. Execution on cross selling, monetization, and geographic expansion is weaker than expected. Credit growth is weak and asset quality deteriorates sharply. Poor operating leverage could be an issue as revenues come under pressure. Low single digit ROE persist.

Risk Reward – Inter & Co Inc. (INTR.O)

KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
NIM (%)	6.5	7.2	7.1	6.8
Loan Growth (YoY) (%)	29.3	25.6	15.0	15.3
Fee Income Growth (YoY) (%)	87.4	24.2	24.7	22.9
Total Revenue Growth (YoY) (%)	58.7	32.8	20.4	12.8
Personnel & Administrative Expenses Growth (YoY) (%)	39.4	5.0	17.1	11.9

INVESTMENT DRIVERS

- Interest rates
- Loan growth
- Fee income growth
- Cross-selling & monetization of clients
- New client sign-up
- Digital strategy
- Delinquency & provisions

GLOBAL REVENUE EXPOSURE



● 100% Latin America

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Structural reforms result in stronger economic growth
- Improving cross-sell initiatives
- Efficiency gains
- Successful geographic expansion

RISKS TO DOWNSIDE

- Economic slowdown impact delinquency & loan growth
- Cost overruns & rising delinquency due to rapid BS growth
- Regulatory headwinds
- Tougher competition
- Relatively low earnings visibility
- Execution risk of international expansion
- Limited market cap & trading liquidity

OWNERSHIP POSITIONING

Inst. Owners, % Active	94.2%	
HF Sector Long/Short Ratio	1.7x	
HF Sector Net Exposure	11.1%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

Sales / Revenue
(US\$, mm)

Note: There are not sufficient brokers supplying consensus data for this metric

Net income
(US\$, mm)

Note: There are not sufficient brokers supplying consensus data for this metric

EPS
(US\$)

Note: There are not sufficient brokers supplying consensus data for this metric

DPS
(US\$)

Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [Microsoft PowerPoint - Exhibit Links.pptx](#)
4. View explanation of 'Indicator of Change' methodology - [ESG_Indicator_of_Change_External_Link.pdf](#)
5. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
6. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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Bank, Saudi National Bank, Shinhan Financial Group, Shizuoka Financial Group, Societe Generale, SoFi Technologies, Inc., Standard Bank, Standard Chartered Bank, StoneCo Ltd., Sumitomo Mitsui FG, Sumitomo Mitsui Trust Holdings, UBS Group AG, UNICAJA BANCO SA, UniCredit S.p.A., Virgin Money UK PLC, XP Inc.

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Global Stock Ratings Distribution

(as of November 30, 2023)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1352	37%	273	43%	20%	605	39%
Equal-weight/Hold	1667	46%	303	47%	18%	708	46%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	591	16%	64	10%	11%	221	14%
Total	3,613		640			1535	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Kakao Bank Corp (323410.KS) - As of 12/11/23 in KRW
Industry : S. Korea Financials



Stock Rating History: 12/1/18 : NA/A; 3/7/19 : NA/I; 6/12/20 : NA/C; 12/11/20 : NA/I; 1/18/21 : NA/A; 10/12/21 : E/A; 5/16/23 : O/A; 7/14/23 : O/I

Price Target History: 10/12/21 : 65000; 2/28/22 : 55000; 7/4/22 : 36000; 11/2/22 : 22000; 5/16/23 : 32000; 8/4/23 : 35000; 12/8/23 : 32500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) --- Stock Price (Covered by Current Analyst) ---
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (12/12/2023)
Heewon Choi		
DB Insurance Co Ltd (005830.KS)	E (05/15/2023)	W87,700
Hanwha Life Insurance (088350.KS)	U (07/14/2023)	W2,680
Samsung Fire & Marine (000810.KS)	E (03/22/2023)	W255,000
Samsung Life Insurance (032830.KS)	O (10/19/2023)	W70,300
Joon Seok		
Hana Financial Group (086790.KS)	E (05/16/2023)	W42,100
Industrial Bank of Korea (024110.KS)	U (07/04/2022)	W11,890
Kakao Bank Corp (323410.KS)	O (05/16/2023)	W26,100
KB Financial Group (105560.KS)	O (02/23/2017)	W52,100
Samsung Card (029780.KS)	U (11/05/2021)	W32,600
Shinhan Financial Group (055550.KS)	E (05/16/2023)	W37,150

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Brazil Financial Institutions

COMPANY (TICKER)	RATING (AS OF)	PRICE* (12/11/2023)
Jorge Echevarria		
BB Seguridade Participacoes S.A. (BBSE3.SA)	U (06/25/2023)	R\$31.18
Caixa Seguridade Participacoes S.A. (CXSE3.SA)	O (06/10/2021)	R\$11.90
Porto Seguro (PSSA3.SA)	O (11/02/2022)	R\$28.13
Jorge Kuri		
B3 SA BRASIL BOLSA BALCAO (B3SA3.SA)	E (07/21/2017)	R\$13.93
Banco Bradesco (BBD.N)	O (08/07/2019)	US\$3.32
Banco BTG Pactual SA (BPAC11.SA)	O (03/28/2017)	R\$34.52
Banco do Brasil (BBAS3.SA)	O (03/07/2018)	R\$54.44
Banco Pan SA (BPAN4.SA)	O (11/29/2022)	R\$7.75
Banco Santander Brasil (BSBR.N)	O (08/07/2019)	US\$6.22
Cielo S.A. (CIEL3.SA)	E (12/18/2020)	R\$4.41
Inter & Co Inc. (INTR.O)	U (06/13/2018)	US\$5.05
Itau Unibanco Holding S.A. (ITUB.N)	E (05/19/2023)	US\$6.42
Nu Holdings Ltd. (NU.N)	O (01/03/2022)	US\$8.18
PagSeguro Digital (PAGS.N)	E (12/06/2022)	US\$10.50
StoneCo Ltd. (STNE.O)	E (12/06/2022)	US\$15.68
XP Inc (XP.O)	O (01/06/2020)	US\$23.32

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Banks

COMPANY (TICKER)	RATING (AS OF)	PRICE* (12/12/2023)
Mia Nagasaka		
Aozora Bank (8304.T)	U (02/28/2018)	¥3,068
Chiba Bank (8331.T)	E (03/27/2020)	¥1,128
Concordia Financial Group (7186.T)	E (06/26/2020)	¥708
Fukuoka Financial Group (8354.T)	E (03/27/2020)	¥3,469
Japan Post Bank (7182.T)	O (12/07/2023)	¥1,490
Mizuho Financial Group (8411.T)	E (11/30/2017)	¥2,471
Rakuten Bank (5838.T)	E (05/24/2023)	¥2,373
Resona Holdings (8308.T)	O (09/27/2021)	¥783
SBI Sumishin Net Bank (7163.T)	E (11/13/2023)	¥1,556
Shizuoka Financial Group (5831.T)	E (06/26/2020)	¥1,223

Sumitomo Mitsui FG (8316.T)	O (12/07/2012)	¥7,275
Sumitomo Mitsui Trust Holdings (8309.T)	O (12/08/2023)	¥5,544

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: EEMEA - Banks & Fintech

COMPANY (TICKER)	RATING (AS OF)	PRICE* (12/12/2023)
James Starke, CFA		
Absa Group Ltd (ABGJ.J)	O (11/15/2021)	ZAc 15,571
Capitec Bank Holdings Ltd (CPIJ.J)	E (10/11/2023)	ZAc 190,049
FirstRand Limited (FSRJ.J)	O (07/30/2021)	ZAc 6,559
Nedbank Group Ltd (NEDJ.J)	E (08/30/2023)	ZAc 19,960
Standard Bank (SBKJ.J)	E (07/30/2021)	ZAc 19,103
Nida Iqbal		
Abu Dhabi Commercial Bank PJSC (ADCB.AD)	O (08/13/2023)	AED 8.30
Alinma Bank SJSC (1150.SE)	U (06/12/2023)	SAR 36.20
Alpha Bank SA (ACBr.AT)	O (10/23/2023)	€1.50
Al Rajhi Bank (1120.SE)	O (11/14/2023)	SAR 80.10
Arab National Bank (1080.SE)	U (06/12/2023)	SAR 23.84
Banque Saudi Fransi (1050.SE)	E (09/28/2022)	SAR 37.85
Emirates NBD Bank PJSC (ENBD.DU)	O (04/12/2022)	AED 17.25
Eurobank Ergasias SA (EURBr.AT)	E (07/10/2023)	€1.61
First Abu Dhabi Bank PJSC (FAB.AD)	U (02/08/2022)	AED 13.34
National Bank of Greece SA (NBGr.AT)	O (07/10/2023)	€6.17
OTP Bank (OTPB.BU)	O (06/12/2023)	HUF 15,025.00
Pekao SA (PEO.WA)	O (11/21/2022)	PLN 146.05
Piraeus Bank SA (BOPr.AT)	O (11/21/2022)	€3.17
PKO BP (PKO.WA)	E (11/21/2022)	PLN 49.23
Riyad Bank (1010.SE)	E (11/14/2023)	SAR 27.70
Saudi Awwal Bank (1060.SE)	O (04/12/2022)	SAR 33.95
Saudi National Bank (1180.SE)	E (08/13/2023)	SAR 35.90

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INDUSTRY COVERAGE: Banks

COMPANY (TICKER)	RATING (AS OF)	PRICE* (12/12/2023)
Alvaro Serrano		
AIB Group plc (A5G.F)	U (06/06/2023)	€3.71
Banco Sabadell (SABE.MC)	E (02/15/2022)	€1.23
Bankinter (BKT.MC)	E (09/15/2022)	€6.01
Bank of Ireland (BIRG.I)	E (09/18/2014)	€8.19
Barclays Bank (BARC.L)	O (09/26/2023)	142p
BBVA (BBVA.MC)	E (09/05/2023)	€8.42
CaixaBank SA (CABK.MC)	O (04/07/2021)	€3.87
HSBC (HSBA.L)	E (08/11/2020)	619p
Lloyds Banking Group (LLOY.L)	O (11/28/2023)	46p
NatWest Group (NWG.L)	E (10/02/2023)	217p
Santander (SAN.MC)	E (11/12/2021)	€3.88
UNICAJA BANCO SA (UNI.MC)	U (01/15/2021)	€0.94
Virgin Money UK PLC (VMUK.L)	U (11/02/2021)	156p
Giulia Aurora Miotto, CFA		
ABN AMRO Bank NV (ABNd.AS)	E (02/10/2022)	€13.11
BNP Paribas (BNPP.PA)	E (09/15/2022)	€60.82

Credit Agricole S.A. (CAGR.PA)	U (04/13/2022)	€12.66
Deutsche Bank (DBKGn.DE)	E (09/22/2023)	€11.93
ING Groep (INGA.AS)	O (09/15/2022)	€13.62
KBC Group NV (KBC.BR)	E (02/15/2021)	€55.96
Societe Generale (SOGN.PA)	O (08/23/2023)	€23.71
UBS Group AG (UBSG.S)	E (10/17/2023)	SFr 25.00
Nick Lord		
Standard Chartered Bank (STAN.L)	E (03/17/2020)	639p
Pamela Zuluaga		
Banco BPM (BAMI.MI)	O (10/04/2023)	€5.09
Intesa SanPaolo S.p.A. (ISP.MI)	E (10/04/2023)	€2.68
UniCredit S.p.A. (CRDI.MI)	O (07/03/2018)	€25.04
Vishal Shah		
Commerzbank (CBKG.DE)	O (11/10/2021)	€10.84

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